

Misclassification in memory modification in *App^{NL-G-F}* knock-in mouse model of Alzheimer's disease

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This **valuable** study employs a formalized computational model of learning to assess memory deficits in Alzheimer's Disease with the goal of developing an early diagnosis tool. Using an established mouse model of the disease, the authors studied multiple behavioral tasks and ages with the goal of showing similarities in behavioral deficits across tasks. Using the model, the authors indicate specific deficits in memory (overgeneralization and over differentiation) in mice with the transgene for the disease. The evidence presented is **solid**, yet certain concerns remain regarding the interpretation of the results of the modeling.

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Abstract

Alzheimer's disease (AD), the leading cause of dementia, could potentially be mitigated through early detection and interventions. However, it remains challenging to assess subtle cognitive changes in the early AD continuum. Computational modeling is a promising approach to explain a generative process underlying subtle behavioral changes with a number of putative variables. Nonetheless, internal models of the patient remain underexplored in AD. Determining the states of an internal model between measurable pathological states and behavioral phenotypes would advance explanations about the generative process in earlier disease stages beyond assessing behavior alone. In this study, we assumed the latent cause model as an internal model and estimated internal states defined by the model parameters being in conjunction with measurable behavioral phenotypes. The 6- and 12-month-old *App^{NL-G-F}* knock-in AD model mice and the age-matched control mice underwent memory modification learning, which consisted of classical fear conditioning, extinction, and reinstatement. The results showed that *App^{NL-G-F}* mice exhibited a lower

extent of reinstatement of fear memory. Computational modeling revealed that the deficit in the *App^{NL-G-F}* mice would be due to their internal states being biased toward overgeneralization or overdifferentiation of their observations, and consequently the competing memories were not retained. This deficit was replicated in another type of memory modification learning in the reversal Barnes maze task. Following reversal learning, *App^{NL-G-F}* mice, given spatial cues, failed to infer coexisting memories for two goal locations during the trial. We concluded that the altered internal states of *App^{NL-G-F}* mice illustrated their misclassification in the memory modification process. This novel approach highlights the potential of investigating internal states to precisely assess cognitive changes in early AD and multidimensionally evaluate how early interventions may work.

Introduction

Alzheimer's disease (AD) is the leading cause of dementia, which severely disturbs daily lives in the aging population and causes a tremendous social burden (Nandi et al., 2022 [↗](#)). Neuropathological hallmarks, including amyloid- β (A β) plaques and neurofibrillary tangles (NFT), define AD biologically, and the advance in biomarkers have enabled early diagnosis (Jack Jr. et al., 2018 [↗](#), 2024 [↗](#)). Abnormal biomarkers can appear years before symptoms arise, a stage referred to as preclinical AD (Sperling et al., 2011 [↗](#), 2014 [↗](#)). As the disease progresses, the cognitive symptom gets severe, and its degree defines mild cognitive impairment (MCI) and the onset of AD dementia. Cognitive decline has been reported in a wide range of domains, including memory, language, visuospatial function, and executive function, leading to behavioral change as early as a decade before dementia (Amieva et al., 2008 [↗](#); Knopman et al., 2021 [↗](#)). The early stage of AD provides a time window to intervene in disease progression, yet there is an explanatory gap and inconclusive consensus on what particular behavioral phenotype can be directly linked to neuropathology in early AD. Given the present circumstance, the subtle and heterogeneous cognitive change before the onset of dementia poses a challenge to assess clinical outcomes with traditional measurement (Jutten et al., 2023 [↗](#)).

To precisely assess clinical symptoms and effects of early intervention for AD pathology, determining universal behavioral phenotypes as well as understanding how such behaviors change from asymptomatic to symptomatic is needed. Memory impairment is the predominant first symptom of AD (J. Barnes et al., 2015 [↗](#)). The symptoms are commonly identified as a deficit in recalling previously learned or recognized items (Grober et al., 2018 [↗](#)). Previous studies suggested that the performance of prioritized recall, in which memories are recalled in order of assigned value, is a sensitive measure in the early stages of AD (for review, see Knowlton & Castel, 2022 [↗](#)). It remains elusive how memory impairments may arise from biased information processing that could affect the encoding, maintaining, or retrieval process of memory function. Using genetic mouse models is a common approach to dissecting the effects of pathological changes on cognitive symptoms measured in behavioral tasks while controlling for potential confounding factors (Webster et al., 2014 [↗](#); Zhong et al., 2024 [↗](#)). However, the symptoms at the early stage of AD are relatively mild in traditional measurements, and phenotypes may be inconsistent among studies even when using the same AD mouse model (Jankowsky & Zheng, 2017 [↗](#)). As behavioral phenotypes in conjunction with brain pathology state may not be captured by current assessments robustly, it is crucial to seek more sensitive and comprehensive approaches.

Recently proposed computational models provide generative processes underlying observed behavior, potentially bridging the gap between behavioral phenotype and AD pathology. Indeed, such models not only replicated the memory task performance of AD patients in different disease stages (M. D. Lee et al., 2020 [↗](#); M. D. Lee & Stark, 2023 [↗](#); Pooley et al., 2011 [↗](#)) but also were valuable in detecting cognitive decline in preclinical AD patients (Bock et al., 2021 [↗](#); Vanderlip et al., 2024 [↗](#)). However, previous approaches did not suppose an internal model of the world to

predict the future from current observation given prior knowledge. Internal models with sufficient explanatory power can replicate various behavioral phenotypes in specific tasks through a common generative process. Estimating an internal model in the early stages of AD patients would elucidate how AD alters their internal states and thereby their behaviors along with disease progression, which is informative for precise diagnosis and assessment of early intervention effects. To our knowledge, only a few studies tried to estimate internal states in amnesic MCI and AD patients and reported that a certain retention rate underlying motor learning (Sutter et al., 2024 [↗](#)) but not learning rate in associative learning (Wessa et al., 2016 [↗](#)) differed from healthy control. This circumstance demands us to investigate internal models illustrating not only learning but also its interaction with memory with higher explanatory power.

In this study, we assumed the latent cause model as an internal model explaining the memory modification process to solve a problem such that a novel observation should be classified into previously acquired memory or novel memory (Gershman et al., 2017 [↗](#)). We hypothesized that the internal state of the latent cause model is altered along with disease progression from the preclinical stage of AD. To test this, we used the *App*^{NL-G-F} knock-in AD mouse model (Saito et al., 2014 [↗](#)). This model mice carry the Swedish KM670/671NL, Iberian I716F, and Arctic E693G mutations on the gene of amyloid- β precursor protein (APP) and display A β plaque accumulation and neuroinflammation across age, reproducing neuropathological change in preclinical AD without artifacts caused by APP overexpression (Saito et al., 2014 [↗](#); Sasaguri et al., 2017 [↗](#)). In this study, the 6- and 12-month-old *App*^{NL-G-F} knock-in mice and the age-matched control mice were engaged in a memory modification task consisting of Pavlovian fear conditioning, extinction, and reinstatement. Internal states across trials were estimated for each mouse, simultaneously with determining a set of model parameters such that the simulated behavior under the latent cause model given the parameters fits enough to the observed behavior. Because we found a deficit in the *App*^{NL-G-F} mice as overgeneralization or overdifferentiation of their observations into certain memories indicated by parameters in the latent cause model, we further confirmed whether this deficit could be replicated in another kind of memory modification task, the reversal Barnes maze paradigm. In the latent cause framework, we discussed how the internal states in the *App*^{NL-G-F} mice diverged from those in the control mice during the memory modification process.

Results

The latent cause model, an internal model, explains memory modification processes in the classical fear conditioning paradigm

The latent cause model proposed by Gershman et al. (2010 [↗](#)) is an internal model such that an agent infers a latent cause as a source of the co-occurrence of events in the environment. It was later adapted to model the memory modification process, a problem to infer previously acquired latent causes or a novel latent cause for every observation in the Pavlovian fear conditioning paradigm in rodents and humans (Gershman et al., 2017 [↗](#)). In the latent cause model, an agent learning an association between a conditioned stimulus (CS) and an unconditioned stimulus (US) has an internal model such that stimuli $\mathbf{x}$ (i.e., d -dimensional vector involving the CS or context) and outcome r (i.e., US) at time t are generated at an associative strength $\mathbf{w}$ (i.e., d -dimensional vector) from k -th latent cause z_{tk} of total K latent causes by a likelihood $p(\mathbf{x} | z_{tk})$ and $p(r | z_{tk})$, respectively (Fig. 1A [↗](#)). $p(\mathbf{x} | z_{tk})$ and $p(r | z_{tk})$ follow Gaussian distributions in which each mean is calculated from previous observations, while each variance (σ_x^2 and σ_r^2) was *a priori* defined as a hyperparameter. Since latent causes are unobservable, the agent given an observation infers which latent cause is likely to generate current observation by Bayes' rule. The prior of z follows the Chinese restaurant process, that is

$$P(z_t = k | \mathbf{z}_{1:t-1}) \propto \begin{cases} \sum_{t' < t} \mathcal{K}(\tau(t) - \tau(t')) \mathbb{I}[z_{t'} = k], & \text{if } k \leq K \text{ (i.e., } k \text{ is an old cause)} \\ \alpha, & \text{otherwise (i.e., } k \text{ is a new cause)} \end{cases}$$

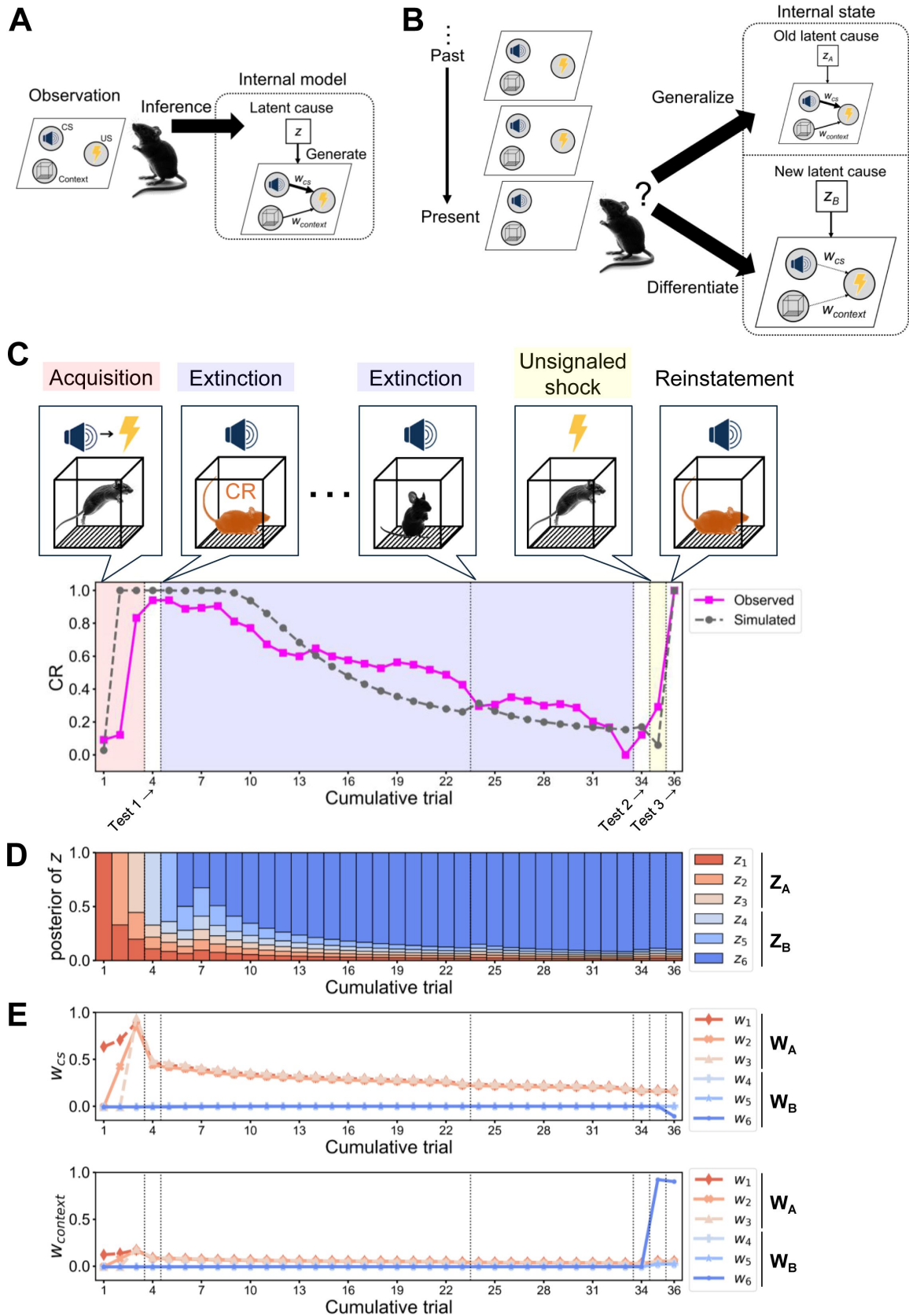


Figure 1.

Internal state in the latent cause model in the Pavlovian fear conditioning, extinction, and reinstatement.

(A) Schematic diagram of latent cause framework in a simple Pavlovian fear conditioning paradigm. An experimental mouse observes a tone, a context, and an electrical shock, which are set down in a parallelogram. Among the stimuli, the tone is designed as the conditioned stimulus (CS), and the electrical shock is designed as the unconditioned stimulus (US) with additional consideration of context effect. The mouse has an internal model such that a latent cause z generates the observation where the tone and the context induce the shock at an associative weight w_{CS} and $w_{context}$, respectively. The latent cause is a latent variable represented by a white square. The posterior probability of z is represented by a black arrow from z to the observation. The stimuli (i.e., tone, context, and shock) are observed variables represented in shaded circles, and the associative weight w is represented by a black arrow between the two. Since latent causes are unobservable, the mouse infers which latent cause is more likely from posterior distribution over latent causes given the observation following Bayes' rule.

(B) Schematic diagram of internal state posited in the latent cause model demonstrating the memory modification process in the fear conditioning paradigm. The mouse has acquired z_A through two observations of the CS accompanied by the US in the same context. Now, the mouse observes the tone alone, then infers a new latent cause z_B generating the CS and context without US as w_{CS} and $w_{context}$ equal to 0. The internal state at the time consists of z_A with largely decreased posterior probability and thereby slightly decreased w_{CS} and $w_{context}$, in addition to z_B . The mouse in this panel is likely to differentiate the current observation from z_A , as it assigns a higher probability for z_B . The posterior probability of z_A and z_B is represented by the relative size of their diagram. If the mouse generalizes z_A to the current observation without inferring z_B , it would largely decrease w_{CS} and $w_{context}$ of z_A to minimize prediction error. **(C)** Observed and simulated conditioned response (CR) during reinstatement in fear conditioning. In the acquisition phase, a CS was accompanied by a US, inducing associative fear memory in the animal, as it exhibits a higher CR to the CS. In the extinction phase, the CS was presented without the US, inducing extinction memory with decreased CR. After an unsignaled shock without the preceding CS, the CR increases again, and this is called reinstatement. The observed CR was sampled from a 2-month-old male wild-type C57BL/6j mouse (magenta line plot with square markers), where the vertical and horizontal axis indicate CR and cumulative trial, respectively. The latent cause model parameters were estimated so as to minimize the prediction error between the observed and the simulated CR given the parameter (gray line with square markers). The estimated parameter value: $\alpha = 2.4$, $g = 0.316$, $\eta = 0.64$, max. no. of iteration = 9, $w_0 = -0.006$, $\sigma_r^2 = 4.50$, $\sigma_x^2 = 4.39$, $\theta = 0.03$, $\lambda = 0.02$, $K = 6$. Trials with red, blue, and yellow backgrounds correspond to the acquisition, extinction, and unsignaled shock, where the first and second extinction consists of 19 and 10 trials, respectively (see Material and Methods). Three trials with white backgrounds are tests 1, 2, and 3 to evaluate if conditioning, extinction, and reinstatement are established. The vertical dashed lines separate the phases. **(D)** The evolution of the posterior probability and **(E)** the associative weight of each latent cause across the reinstatement experiment, where w_{CS} (top panel) and $w_{context}$ (bottom panel) were computed. **(D and E)** Each latent cause is indicated by a unique subscript number and color, w_{CS} and $w_{context}$. The latent causes acquired during the acquisition phase and their associative weight are termed z_A and w_A , respectively. Meanwhile, latent causes acquired during the extinction phase and their associative weight are termed z_B and w_B , respectively.

Prior of z_{tk} refers to the frequency that z_k was inferred so far, where $\mathbb{I}[\cdot] = 1$ if its argument is true; otherwise, 0. This prior is biased by a temporal kernel $\mathcal{K}$, as an analogy of the power law of the forgetting curve,

$$\mathcal{K}(\tau(t) - \tau(t')) = (\tau(t) - \tau(t'))^{-g}$$

to exponentially decay the probability that z_k inferred at previous trial t' is inferred again at current trial t by a temporal scaling factor g if the time interval between trial t' and t increases. A new latent cause is inferred at a probability governed by concentration parameter α . Higher α or lower likelihood of the current observation biases the agent to infer a new latent cause given an

observation (i.e., differentiation) rather than to infer an old one (i.e., generalization) (**Fig. 1B** [↗](#)). Although the latent cause model allows for having an infinite number of latent causes, agents with appropriate α and g favor to infer a smaller number of latent causes for observations instead of inferring a unique latent cause for every observation.

According to the latent cause model, the agent solves two computational problems during Pavlovian fear conditioning learning. One is to compute the posterior probability of each latent cause given its associative weight between stimuli $\mathbf{x}$ and shock r with a history of observations, and the other is to compute the associative weight of each latent cause given its posterior. Both computations are achieved through the expectation-maximization (EM) algorithm, as the former and the latter computation correspond to E-step and M-step, respectively. In the E-step, the posterior of each latent cause is computed from its prior and the likelihood of current observation given the latent cause following Bayes' rule, consequently determining the most likely latent cause at the current trial as the one with the maximal posterior, then moves to the M-step. In the M-step, the agent predicts shock by a linear combination of stimuli and associative weight in each latent cause. The weight is updated based on the Rescorla-Wagner rule ([Rescorla & Wagner, 1972](#) [↗](#)) to minimize the prediction error between observed and predicted shock. The amount of the weight change depends on learning rate η and prediction error, which is biased by the posterior probability of the latent causes. The amount of weight change will be smaller even in larger prediction errors if the latent cause is unlikely. The agent estimates weights for all latent causes and then returns to E-step. This procedure is iterated a preset number of times, and thereby, the posterior probability over latent causes and their associative weights are determined. Finally, the agent concludes the expected shock in the current trial by a linear combination of observed stimuli and an expectation of weight, which is weighted by the posterior distribution over latent causes. The conditioned response (CR) for the expected shock is determined as an integral of Gaussian distribution (mean is the expected shock, variance is λ) above threshold θ , where λ and θ are hyperparameters. Thus, these parameters of the latent cause model determine an internal state of the agent leading to particular behavioral outcomes given an observation.

Reinstatement exemplifies the memory modification process and indeed is well explained by the latent cause model. Reinstatement occurs when agents who have learned and then extinguished an association between a CS and a US exhibit CR again after being exposed to the US alone following the extinction. The latent cause model explains the reinstatement of fear memory with the changes of agent's internal state as follows. First, an agent repeatedly observed a CS accompanied with a US in a Pavlovian fear conditioning procedure (**Fig. 1C** [↗](#), "Acquisition"). Because these observations are novel for the agent, it inferred a set of latent causes $\mathbf{z}_A$ (z_1, z_2 , and z_3 in **Fig. 1D** [↗](#)) where the CS is associated with the US by associative weight $\mathbf{w}_A$ (w_1, w_2 , and w_3 in **Fig. 1E** [↗](#)) during the acquisition. Considering the effect of the context, $\mathbf{w}$ contains both w_{CS} and $w_{context}$ (**Fig. 1E** [↗](#)). After the acquisition, the agent inferred $\mathbf{z}_A$ and then predicted the upcoming US, leading to increased CR when it observed the CS (**Fig. 1C** [↗](#), "Test 1" after "Acquisition"). Second, the agent repeatedly observed the CS alone in an extinction procedure. Early in the extinction, the agent would infer $\mathbf{z}_A$ and then exhibit higher CR to the CS, while the US was absent (**Fig. 1C** [↗](#), left "Extinction"). This state increases prediction error, then prompts the agent either to decrease $\mathbf{w}_A$ toward 0 or to infer a novel set of latent causes $\mathbf{z}_B$ where the CS is not accompanied with the US, that is, associative strength $\mathbf{w}_B = 0$ (**Fig. 1B** [↗](#)); both processes are run in the mouse in **Fig. 1CDE** [↗](#). Note that the sets A and B were introduced arbitrarily for explanation here. Later in the extinction, $\mathbf{z}_B$ rather than $\mathbf{z}_A$ is inferred, leading to the decreased CR (**Fig. 1C** [↗](#), right "Extinction" or "Test 2"). Third, when the agent observes the US alone after the extinction (**Fig. 1C** [↗](#), "Unsignaled shock"), the probability of $\mathbf{z}_A$ or associative weight of some latent causes increases, hence it predicts the US and thereby the CR increases again, that is reinstatement (**Fig. 1C** [↗](#), "Reinstatement" at "Test 3"); in the mouse in **Fig. 1CDE** [↗](#), the unsignaled shock increases not only the posterior probability of $\mathbf{z}_A$ that has maintained higher w_{CS} but also $w_{context}$ in $\mathbf{w}_6$ interpreted as a modification in the association between context and US in z_6 . Thus, it is rational to expect that certain parameters of the latent cause model are in conjunction with particular

behaviors in the reinstatement paradigm (see Appendix 2 for the relationship between parameters, CR, and internal states). As demonstrated in [Gershman & Hartley \(2015\)](#), the likelihood that $\alpha > 0$ in human participants correlates with the degree of their spontaneous recovery, that is a return of the CR after a long hiatus following an extinction procedure. Furthermore, a recent study has shown that the latent cause model explains maladaptive decision-making in post-traumatic stress disorder patients as a perseverated inference of previously acquired latent causes even under situations where they should infer novel latent causes ([Norbury et al., 2022](#)). By estimating such parameters from the behaviors of each experimental mouse, how the internal state of the AD group diverges from that of the control group could be determined.

***App*^{NL-G-F} mice showed unimpaired associative fear and extinction learning but a lower extent of reinstatement**

To examine the age and *App*^{NL-G-F} knock-in effects on the reinstatement, all cohorts of control and homozygous *App*^{NL-G-F} mice were tested at the age of either 6 months or 12 months in the reinstatement paradigm based on auditory-cued fear conditioning (**Fig. 2A** and Table S1). According to previous studies, the *App*^{NL-G-F} mice displayed an age-dependent A β deposition from 4 months old and memory impairments could be detected at 6 months of age ([Masuda et al., 2016](#); [Saito et al., 2014](#); [Sakakibara et al., 2018](#)). The wide spread of A β accumulation was confirmed in this study, and results were provided in **Figure 2-figure supplement 1**. Given that age affects the A β accumulation caused by *App*^{NL-G-F} knock-in, a one-way ANCOVA was used to evaluate the *App*^{NL-G-F} knock-in effects on behavioral measures controlling for age.

In the acquisition phase of fear conditioning, the freezing rate gradually increased with the number of CS-US pairings (**Fig. 2B**, leftmost red panel). In the test 1 phase, 24 h after the acquisition, a significant genotype effect was observed in freezing rates during CS presentation [$F(1,95) = 9.679, p = 0.002$], where the *App*^{NL-G-F} mice showed a higher freezing rate than in the control (**Fig. 2C** and Table S2), indicating the associative fear memory was established in *App*^{NL-G-F} mice. As the increasing trials of CS alone presented, both groups showed a gradually decreased freezing rate in the extinction phase (**Fig. 2B**, blue panel). In the test 2 phase, 24 h after the last trial of the extinction, there was a significant age effect in freezing rates during CS presentation [$F(1,95) = 4.698, p = 0.033$], where 12-month-old mice showed a higher freezing rate than in the 6-month-old mice (**Fig. 2D** and Table S2). In the test 3 phase, 24 h after given an unsignaled shock, mice showed an elevated freezing rate, and no significant genotype or age effect was detected (**Fig. 2B, 2E** and Table S2). To evaluate the effect of the context on the CR, we measured the preCS freezing rate during 1 min before the tone presentation (**Fig. 2-figure supplement 2**). Generally, the preCS freezing rate was lower than CS freezing rate and gradually reduced but kept positive across trials after acquisition (**Fig. 2-figure supplement 2A and 2B**), suggesting the context has a certain effect on CR. When we compared the preCS freezing rate (**Figure 2-figure supplement 2E**) and the CS freezing rate (**Fig. 2E**) in test 3 within group by the paired samples *t*-tests, the CS freezing rate was significantly higher than the preCS freezing rate in all groups: 6-month-old control, $t(23) = -6.344, p < 0.001, d = -1.295$; 6-month-old *App*^{NL-G-F}, $t(24) = -4.679, p < 0.001, d = -0.936$; 12-month-old control, $t(23) = -4.512, p < 0.001, d = -0.921$; 12-month-old *App*^{NL-G-F}, $t(24) = -2.408, p = 0.024, d = -0.482$. Although the associative strength between the context and the US would be increased by the unsignaled shock, these results suggest that reinstatement was not solely explained by the CR for context, but also reactivation acquisition memory. To evaluate the magnitude of how each mouse can discriminate between different phases, we calculated the discrimination index (DI) using freezing rates during the CS presentation. If the mouse shows the same freezing rate at the two test phases, the value of DI will be 0.5, while if the mouse shows a higher freezing rate in one phase compared to the other, DI will be far from 0.5. The results showed that the DIs between test 2 and test 1 were significantly lower than 0.5 in all groups (**Fig. 2F** and Table S3), indicating the successful establishment of extinction memory. The genotype effect was detected [$F(1,95) = 5.013, p = 0.027$] (**Fig. 2F** and

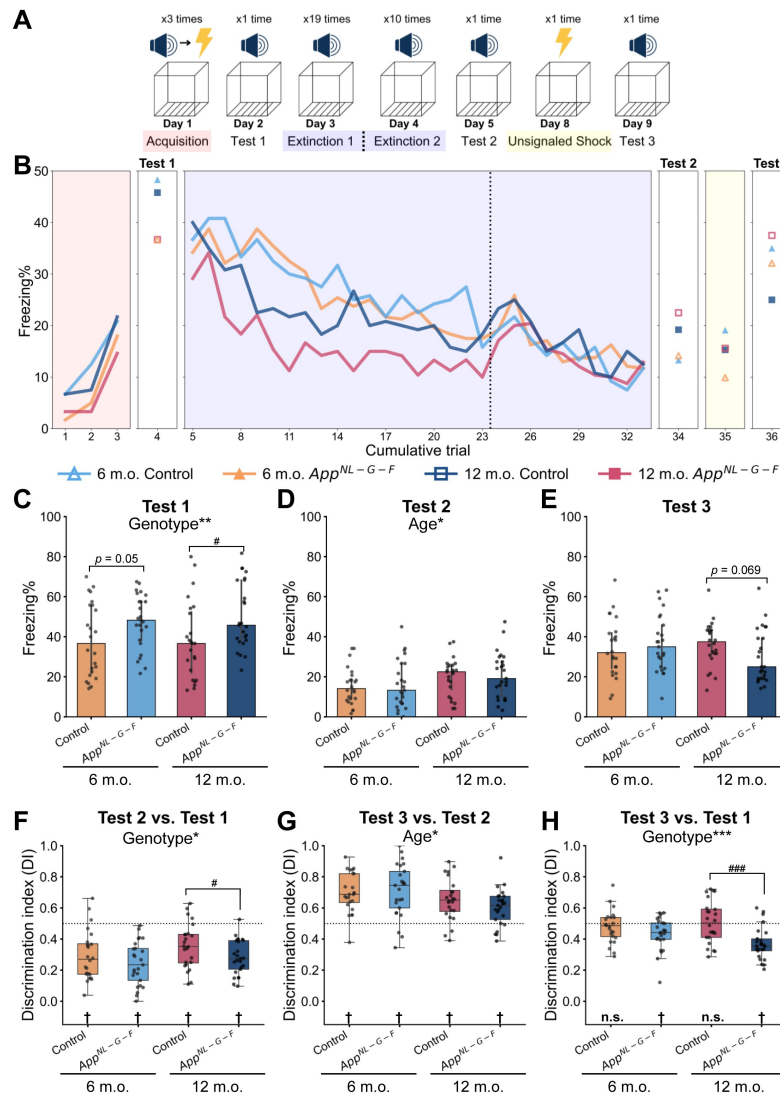


Figure 2.

App^{NL-G-F} mice exhibited successful associative learning and extinction but a lower extent of reinstatement.

(A) Schematic diagram of reinstatement paradigm in this study. In the acquisition phase, the CS was accompanied with the US 3 times. In the following phase, either the CS or the US was presented at certain times. The context was the same throughout the experiment. **(B)** Freezing rate during CS presentation across reinstatement paradigm. The markers and lines show the median freezing rate of each group in each trial. Red, blue, and yellow backgrounds represent acquisition, extinction, and unsignaled shock in **(A)**. The dashed vertical line separates the extinction 1 and extinction 2 phases. Freezing rate during CS presentation in test 1 **(C)**, test 2 **(D)**, and test 3 **(E)**. Discrimination index (DI) between test 2 and test 1 **(F)**, between test 3 and test 2 **(G)**, and between test 3 and test 1 **(H)** calculated from freezing rate during CS presentation. **(C to E)** The data are shown as median with interquartile range. **p* < 0.05, ***p* < 0.01, and ****p* < 0.001 by one-way ANCOVA with age as covariate; #*p* < 0.05, ##*p* < 0.01, and ###*p* < 0.001 by Student's *t*-test comparing control and *App^{NL-G-F}* mice within the same age. Detailed statistical results are provided in Table S2. **(F to H)** The dashed horizontal line indicates DI = 0.5, which means no discrimination between the two phases. †*p* < 0.05 by one-sample Student's *t*-test, and the alternative hypothesis specifies that the mean differs from 0.5; **p* < 0.05, ***p* < 0.01, and ****p* < 0.001 by one-way ANCOVA with age as a covariate; #*p* < 0.05, ##*p* < 0.01, and ###*p* < 0.001 by Student's *t*-test comparing control and *App^{NL-G-F}* mice within the same age. Detailed statistical results are provided in Tables S3 and S4. **(B to H)** Colors indicate different groups: orange represents 6-month-old control (*n* = 24), light blue represents 6-month-old *App^{NL-G-F}* mice (*n* = 25), pink represents 12-month-old control (*n* = 24), and dark blue represents 12-month-old *App^{NL-G-F}* mice (*n* = 25). Each black dot represents one animal.

Table S4), where App^{NL-G-F} mice showed lower values due to higher freezing in the test 1 phase. These results suggested that there was no markable fear and extinction learning deficit in App^{NL-G-F} mice at the behavioral level. The DIs between the test 3 and 2 phases in four groups were significantly higher than 0.5, indicating that the reinstatement was successfully induced after the unsignaled shock (Fig. 2G and Table S3). Six-month-old mice showed higher DIs, as a significant age effect was detected [$F(1,95) = 7.480, p = 0.007$] (Fig. 2G and Table S4). The DIs between the test 3 and 1 phases did not deviate from 0.5 for both 6- and 12-month-old control mice, suggesting that they displayed comparable freezing rates in tests 1 and 3 (Fig. 2H and Table S3). In contrast, the DIs between test 3 and 1 phases were significantly lower than 0.5 in 6-month-old App^{NL-G-F} mice [$t(24) = -3.245, p = 0.003$] and 12-month-old App^{NL-G-F} mice [$t(24) = -6.165, p < 0.001$] (Fig. 2H and Table S3). Moreover, a significant genotype effect was detected [$F(1,95) = 15.393, p < 0.001$] (Fig. 2H and Table S4), indicating App^{NL-G-F} mice exhibited a lower extent of reinstatement. Thus, these results suggest that mice retrieved fear memory in the test 1 phase and retrieved extinction memory in the test 2 phase, respectively. The fear memory and the extinction memory compete with each other: the US is present in the former, while it is absent in the latter. In the test 3 phase, after they observed the unsignaled shock, mice had to decide which memory was more relevant given the CS. The same level of freezing rate in tests 1 and 3 observed in control mice suggests that they could retain both fear and extinction memory and thereby infer the fear memory more preferentially. In contrast, App^{NL-G-F} mice displayed a lower freezing rate in test 3 compared to test 1, suggesting that they might still infer extinction memory even after the unsignaled shock or a completely new memory, while the initial fear memory might be suppressed or eliminated.

The internal state underlying reinstatement simulated with the latent cause model differs between the ages and *APP* genotype

To seek the internal state of each mouse generating the behaviors in the reinstatement paradigm, we estimated the parameters of the latent cause model, minimizing prediction errors between the observed CR and simulated CR in the latent cause model given a set of parameters (see also Material and Methods). Figure 3 demonstrates the traces of observed CR, simulated CR, and the changes in the internal state in each group. We initially confirmed that the DIs in the simulated CRs in each group well replicated those in the observed CRs, except that the DI between test 3 and test 1 in the 6-month-old App^{NL-G-F} mice did not significantly deviate from 0.5 in simulated results (c.f. Fig. 2F to 2H and Fig. 3-figure supplement 1A to 1C). To check if the latent cause model has a certain bias to simulate CR in either group, we compared the residual (i.e., observed CR minus simulated CR) between groups. The fit was similar between control and App^{NL-G-F} mice groups in the test trials, except test 3 in the 12-month-old group (Figure 3-figure supplement 1D and 1E). In test 3, the residual was significantly higher in the 12-month-old control mice than App^{NL-G-F} mice, indicating the model underestimated the reinstatement in the control mice. These results suggest that the latent cause model fits our data with little systematic bias such as an overestimation of CR for the control group in the reinstatement, supporting the validity of the comparisons in estimated parameters between groups.

In the 6-month-old group, two latent causes (z_1 and z_2) were generated in the acquisition phase with increasing associative weight (rows 2 to 4 in Fig. 3A, 3B). During the extinction phase, the same latent causes were inferred, and the corresponding associative weights declined, indicating the devaluation of previously acquired fear memory (rows 2 to 4 in Fig. 3A, 3B). Notably, the w_{cs} in w_2 in control mice remained at a higher value compared to that in App^{NL-G-F} mice. At trial 35 given an unsignaled shock, $w_{context}$ in w_1 was elevated in control mice (row 4 in Fig. 3A), while both $w_{context}$ in w_1 and w_2 were elevated in App^{NL-G-F} mice (row 4 in Fig. 3B). At trial 36 (test 3 phase), in control mice, higher values in $w_{context}$ in w_1 and w_{cs} in w_2 contributed to increased CR (Fig. 3A), whereas $w_{context}$ in w_1 and w_2 were the main components for reinstatement in App^{NL-G-F} mice (Fig. 3B). This discrepancy suggests that the expectation of the US is different between 6-month-old control and App^{NL-G-F} mice.

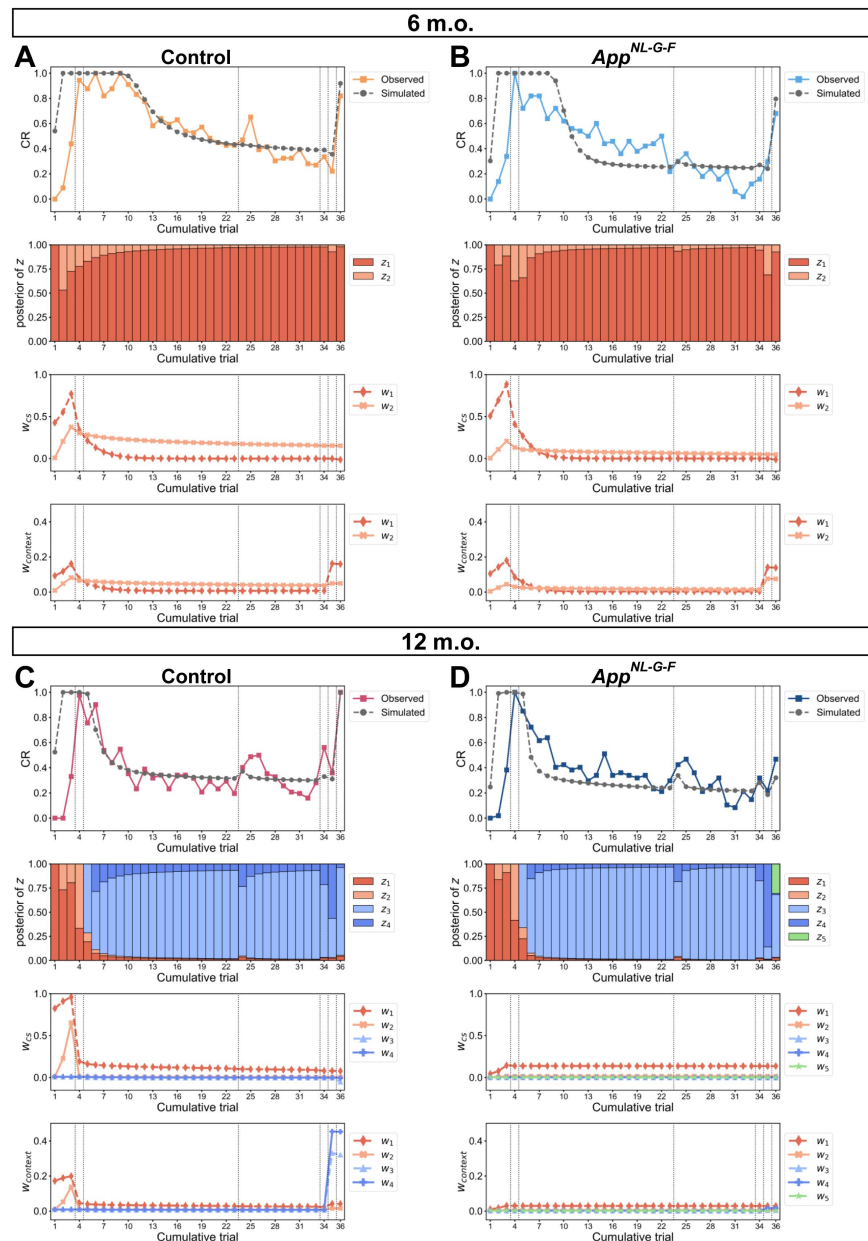


Figure 3.

The divergence of internal states between control and *App*^{NL-G-F} mice differed in age.

(A) Simulation of reinstatement in the 6-month-old control mice given a set of estimated parameters of the latent cause model. The estimated parameter value: $\alpha = 2.4$, $g = 0.05$, $\eta = 0.42$, max. no. of iteration = 2, $w_0 = 0.009$, $\sigma_r^2 = 2.90$, $\sigma_x^2 = 0.49$, $\theta = 0.008$, $\lambda = 0.018$, $K = 10$. (B) Simulation of reinstatement in 6-month-old *App*^{NL-G-F} mice. The estimated parameter value: $\alpha = 1.1$, $g = 0.61$, $\eta = 0.51$, max. no. of iteration = 2, $w_0 = 0.004$, $\sigma_r^2 = 1.51$, $\sigma_x^2 = 0.32$, $\theta = 0.015$, $\lambda = 0.019$, $K = 30$. (C) Simulation in 12-month-old control mice. The estimated parameter value: $\alpha = 1.1$, $g = 0.92$, $\eta = 0.82$, max. no. of iteration = 5, $w_0 = 0.009$, $\sigma_r^2 = 1.88$, $\sigma_x^2 = 0.51$, $\theta = 0.010$, $\lambda = 0.017$, $K = 4$. (D) Simulation in 12-month-old *App*^{NL-G-F} mice. The estimated parameter value for: $\alpha = 1.0$, $g = 1.10$, $\eta = 0.04$, max. no. of iteration = 2, $w_0 = 0.0025$, $\sigma_r^2 = 1.51$, $\sigma_x^2 = 0.18$, $\theta = 0.011$, $\lambda = 0.011$, $K = 36$.

(A to D) The first row shows the trace of observed CR and simulated CR. The observed CR is the median freezing rate during the CS presentation over the mice within each group; the observed CR of each group was divided by its maximum over all trials. The second row shows the posterior probability of each latent cause in each trial. The third and fourth rows show the associative weight of tone to shock (w_{CS}) and that of context to shock ($w_{context}$) in each trial. Each marker and color corresponds to a latent cause up to 5. Each latent cause is represented by the same color as that in the second row and contains w_{CS} and $w_{context}$. The vertical dashed lines indicate the boundaries of phases.

The transition of internal state in the 12-month-old group was similar to that in the 6-month-old group until the test 1 phase. Unlike the 6-month-old group, new latent causes (z_3 and z_4) were inferred during the extinction phase in the 12-month-old group (rows 2 to 4 in **Fig. 3C, 3D**). At trial 35 given an unsignaled shock, the control mice inferred the latent causes z_3 and z_4 acquired in the extinction phase so that their weights $w_{context}$ in $\mathbf{w}_3$ and $\mathbf{w}_4$ were updated (row 4 in **Fig. 3C**). In contrast, there was virtually no update of weights in App^{NL-G-F} mice (row 4 in **Fig. 3D**). At trial 36 (test 3 phase), the elevated CR in control mice was attributed to a successful update of associative weight after the unsignaled shock (**Fig. 3C**), whereas the CR in App^{NL-G-F} mice did not increase due to a redundant latent cause z_5 , which was newly inferred and reduced the posterior probability of latent causes having higher associative weight (**Fig. 3D**).

The unsuccessful state inference at reinstatement was caused by misclassification in App^{NL-G-F} mice

Due to the different evolution of latent causes between ages (**Fig. 3**), we separately discuss the contribution of differences in estimated parameters within the same age and their correlations with DI between test 3 (after unsignaled shock) and test 1 (after acquisition) (**Fig. 2H**). We first investigated the individual internal state differences in the 12-month-old group where the impairment was apparent at the behavioral level (**Fig. 2H**). App^{NL-G-F} mice had significantly lower α (Mann-Whitney $U = 100$, $p = 0.002$) and lower σ_x^2 (Mann-Whitney $U = 88.5$, $p = 0.008$) than control mice (**Fig. 4A**). α and σ_x^2 were significantly correlated each other (**Fig. 4-figure supplement 1**). Both were also positively correlated with the DI (**Fig. 4B**), and the trend was preserved after adjusting for the genotype effect (**Fig. 4-figure supplement 2**). While the maximal number of inferable latent causes (K) was significantly higher in App^{NL-G-F} mice than in control mice (Table S7), the total number of acquired latent causes was comparable between the groups (K_{total} in Table S9). K was negatively correlated with α and σ_x^2 (**Fig. 4-figure supplement 1**).

We assume that the effect of K on the memory modification process together with α and σ_x^2 could come to the surface in extremely artificial conditions, but not in natural conditions as in the empirical data (see Appendix 2 – table 6). Statistical results for remaining parameters are shown in Table S7 and S8. It should be noted that these estimated parameters and internal state might not be unique but one of the possible solutions.

We further characterized how the latent causes acquired in different phases contribute to the distinct internal states between control and App^{NL-G-F} mice. Lower α and lower σ_x^2 would bias App^{NL-G-F} mice toward overgeneralization as they favor inferring a small number of latent causes when facing the same cue. Indeed, the number of latent causes initially inferred at the acquisition phase was significantly lower in App^{NL-G-F} mice, where most of them inferred two latent causes, and more than half of the control mice inferred three latent causes (**Fig. 4C** and Table S9). Furthermore, when they initially observed the CS in the absence of the US in test 1, the posterior probabilities of acquisition latent causes in test 1 were significantly higher in the App^{NL-G-F} mice than in the control mice, leading to the higher CRs in test 1 in the simulation (**Fig. 4-figure supplement 3**), consistent with the observed CR (**Fig. 2C**).

According to the latent cause model, lower σ_x^2 would bias App^{NL-G-F} mice toward overdifferentiation unless the cue exactly matches to previously observed ones. In other words, they prefer to infer new potential causes when similar observations are presented instead of interpreting them generated from the same distribution. The likelihood of current stimuli given latent causes was calculated from a Gaussian distribution with the mean of stimulus values under the latent cause so far and fixed variance σ_x^2 . Repeated observation of the CS alone during the extinction would sufficiently increase the likelihood of stimuli given the latent causes inferred during the extinction in both App^{NL-G-F} and control mice. This attenuates the need to infer new latent causes and the number of latent causes acquired at the extinction phase was comparable

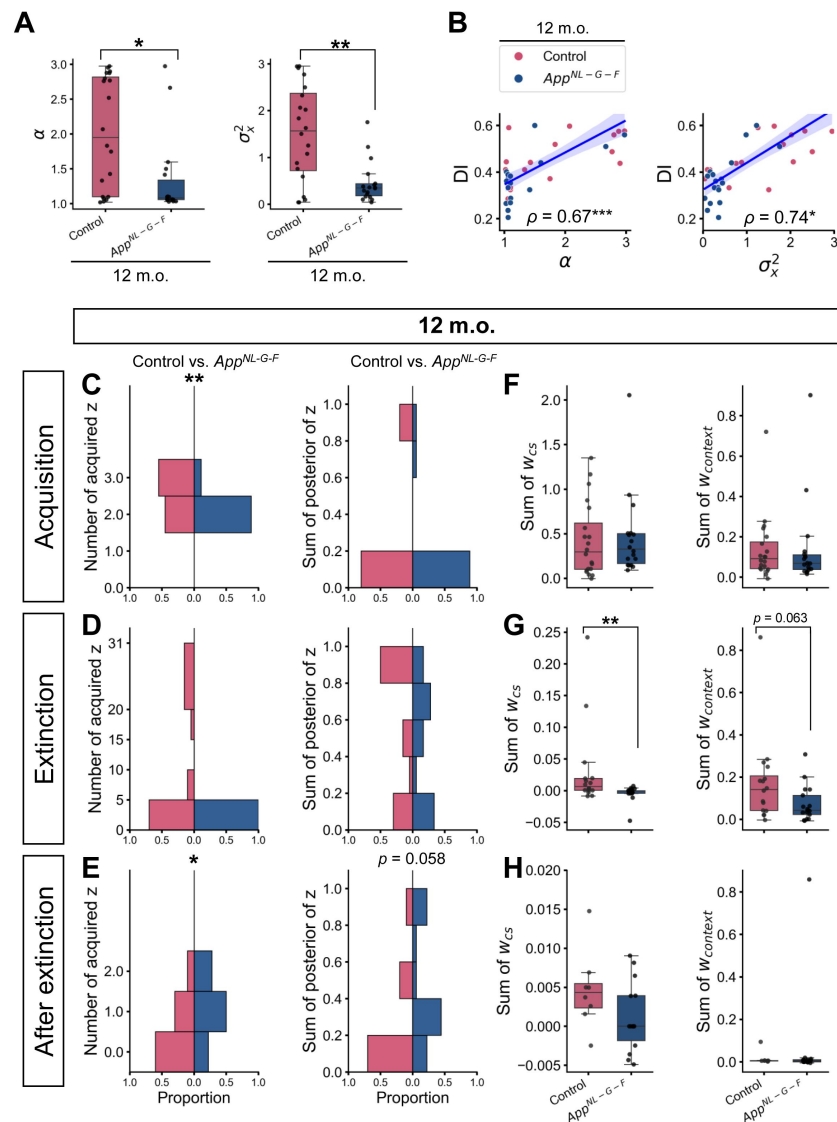


Figure 4.

Individual parameter estimation and internal state in the 12-month-old group.

(A) The estimated latent cause model parameter. **(B)** Correlation between discrimination index (DI) and estimated parameter values in the 12-month-old group. The count of latent causes initially inferred during the acquisition trials **(C, left)**, extinction trials (i.e., test 1, extinction, test 2) **(D, left)**, and trials after extinction (i.e., the unsignaled shock and test 3) **(E, left)**, with the sum of posterior probabilities **(C, D, E, right)**, and the sum of associative weights at test 3 in these latent causes **(F, G, H)**.

(A) Each black dot represents one animal. * $p < 0.05$, and ** $p < 0.01$ by the Mann-Whitney U test. **(B)** The y-axis shows the DI between test 3 and test 1. Spearman's correlation coefficient (ρ) was labeled with significance, where * $p < 0.05$ and *** $p < 0.001$. The blue line represents the linear regression model fit, and the shaded area indicates the confidence interval. Each dot represents one animal. **(C to E)** In the first column, the histogram of the number of latent causes (z) acquired in each phase is shown. The maximum number of latent causes that can be inferred is 3, 31, and 2 in panels C, D, and E. In the second column, the histogram of the sum of the posterior probabilities of the latent causes is shown. The horizontal axis indicates the proportion of the value in each group. **(F to H)** In the first and second columns, the sum of w_{CS} and $w_{context}$ of latent causes are shown in the boxplot, respectively. Note that the initial value of associative weight could take non-zero values, though those were comparable between groups (Table S7). Each black dot represents one animal. * $p < 0.05$, and ** $p < 0.01$ by the Mann-Whitney U test comparing control and App^{NL-G-F} mice, and p -values greater than 0.1 were not labeled on the plot. **(A to H)** Pink represents 12-month-old control ($n = 20$), and dark blue represents 12-month-old App^{NL-G-F} mice ($n = 18$). Detailed statistical results are provided in Tables S7, S8, and S9.

between groups (**Fig. 4D** and Table S9). However, the CS was absent in the unsignaled shock and then presented again in test 3. These observations would decrease the likelihood of stimuli given past latent causes, and consequently increase the probability to infer the new latent causes. This volatility would be more obvious in *App^{NL-G-F}* mice with lower σ_x^2 . As a result, the number of latent causes acquired after the extinction (i.e., the unsignaled shock and test 3) as well as the sum of posterior probabilities of them were greater in *App^{NL-G-F}* mice than those in the control mice, despite that the mice received either the US or the CS alone that were the same with those they observed so far (**Fig. 4E** and Table S9).

Such internal states in *App^{NL-G-F}* mice would diverge the update of associative weight from those in the control mice after extinction (**Fig. 4F, 4G** and Table S9). Both *App^{NL-G-F}* and control mice would still infer the extinction latent causes at the unsignaled shock with higher posterior probabilities. Since the update of the associative weight of each latent cause is modulated by both prediction error and posterior probability of the latent cause, the associative weights in the extinction latent causes were increased by the unsignaled shock in the control mice (**Fig. 3C**, **4G** and Table S9). In *App^{NL-G-F}* mice, however, they inferred new latent causes with low associative weight after the extinction (**Fig. 4E, 4H** and Table S9), which decreased the posterior probabilities of other latent causes and impeded update of their weight (**Fig. 3D**, **4G**). As the initial weight of new latent causes showed a trend toward significance that control mice showed a higher value (Table S7), this would explain the significant higher in w_{CS} in extinction latent cause (**Fig. 4G**). This distinct internal state therefore led to the lower CR and DI in test 3 in *App^{NL-G-F}* mice (**Fig. 2E, 2H**). These results suggest that the α and σ_x^2 would differentiate the internal state and thereby behavioral phenotypes between *App^{NL-G-F}* and control mice. In the Chinese restaurant process, the probability that a new latent cause is inferred decreases with trials, preventing that too many latent causes are inferred (Gershman & Blei, 2012). This could be one of the reasons why the overgeneralization by lower α was less effective than the overdifferentiation by lower σ_x^2 in test 3 in *App^{NL-G-F}* mice.

As shown previously in **Fig. 2** and **Fig. 2-figure supplement 1**, A β aggregation has already widely spread in the brain of 6-month-old *App^{NL-G-F}* mice, but no clear behavioral impairments were detected by the conventional analysis. Similarly to the 12-month-old group, we investigated the individual internal state differences in the 6-month-old group and tested whether the discrepancy in α and σ_x^2 emerged earlier without behavioral impairment. We found that *App^{NL-G-F}* mice had significantly lower α (Mann-Whitney $U = 38$, $p = 0.012$), and it was significantly correlated with DI (**Fig. 5A, 5B** and Table S10, S11). The σ_x^2 was comparable between control and *App^{NL-G-F}* mice, but its significant correlation with DI was found (**Fig. 5A, 5B** and Table S10, S11). Significant positive correlation between α and σ_x^2 was also found in the 6-month-old group (**Fig. 5-figure supplement 1**). Unlike in the 12-month-old group, the significant correlation between α and DI, as well as that between σ_x^2 and DI was not preserved within subgroups (**Fig. 5-figure supplement 2**), suggesting genotype would be confounder for α , σ_x^2 , and DI. The hyperparameter θ , a threshold to emit CR for stimulus inputs, was significantly higher in *App^{NL-G-F}* mice than in control mice. Since the λ was comparable between control and *App^{NL-G-F}* mice (Table S10), the CR in *App^{NL-G-F}* mice could rapidly decrease with expected US during the extinction phase, even if their CR at the test 1 was higher than those in the control. The effect of higher θ was subtle when the expected US was far from its value, and therefore had little contribution to the DI between test 3 and test 1, which is in line with the correlation result in Table S11. Statistical results for the remaining parameters are shown in Tables S10 and S11.

Similar to the 12-month-old group, the influence of α was observed in the acquisition phase, where most *App^{NL-G-F}* mice inferred two latent causes, while more than half of control mice inferred three latent causes (**Fig. 5C**). In addition, lower α in *App^{NL-G-F}* mice affect the posterior probabilities of acquisition latent cause and CR in test 1 phase (**Fig. 5-figure supplement 3**). No significant difference in the internal states was found between control and *App^{NL-G-F}* mice (**Fig. 5C to 5H** and Table S12).

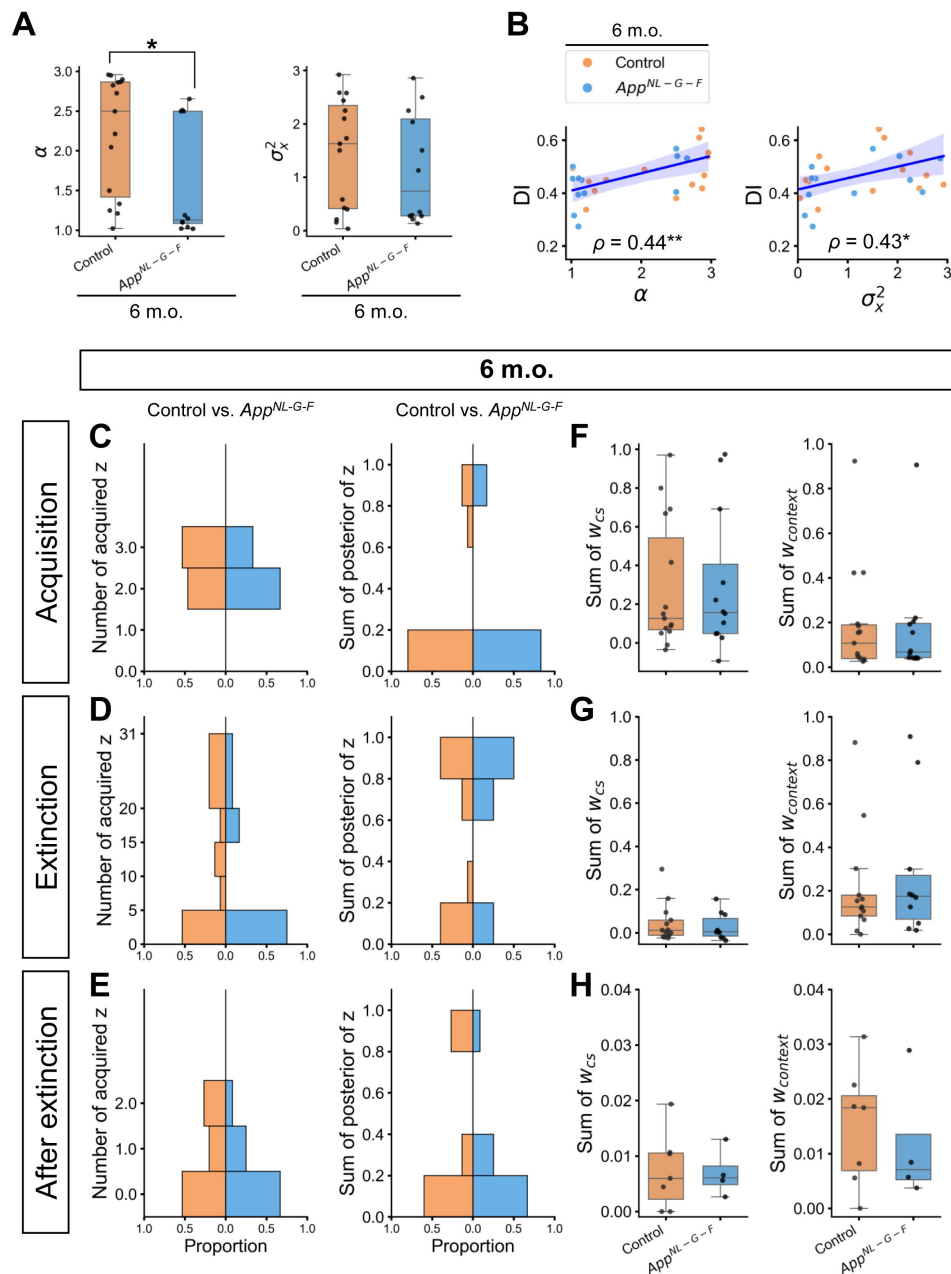


Figure 5.

Individual parameter estimation and internal state in the 6-month-old group.

(A) The estimated parameters of the latent cause model. (B) Correlation between discrimination index (DI) and estimated parameter values in the 6-month-old group. The count of latent causes initially inferred during the acquisition trials (C, left), extinction trials (i.e., test 1, extinction, test 2) (D, left), and trials after extinction (i.e., the unsignaled shock and test 3) (E, left), with the sum of posterior probabilities (C, D, E, right), and the sum of associative weights at test 3 in these latent causes (F, G, H).

(A) Each black dot represents one animal. $*p < 0.05$ by the Mann-Whitney U test. (B) The y-axis shows the DI between test 3 and test 1. Spearman's correlation coefficient (ρ) was labeled with significance, where $*p < 0.05$ and $**p < 0.01$. The blue line represents the linear regression model fit, and the shaded area indicates the confidence interval. Each dot represents one animal. (C to H) The configuration of the figure is the same as in Figure 4. All the p -values of the Mann-Whitney U test comparing control and App^{NL-G-F} mice were greater than 0.05 and were not labeled on the plot. (A to H) Orange represents 6-month-old control ($n = 15$), and light blue represents 6-month-old App^{NL-G-F} mice ($n = 12$). Detailed statistical results are provided in Tables S10, S11, and S12.

In summary, α and σ_x^2 are the main contributors to differential internal states and memory modification processes between control and *App^{NL-G-F}* mice. Regardless of age, *App^{NL-G-F}* mice have a defect in forming a new memory even if the same cue comes with a new outcome as in test 1, suggesting overgeneralization associated with lower α (Fig. 4A, 5A). With a more severe phenotype, 12-month-old *App^{NL-G-F}* mice were biased to classify similar cues into different causes, which is associated with lower variance of the stimulus, σ_x^2 (Fig. 4A). This overdifferentiation would eventuate the lower reinstatement of fear memory (Fig. 2H). Thus, these results suggest that *App^{NL-G-F}* mice failed to retain competing memories because of the misclassification of observation into memories, even though each memory accounts for past and present observations.

***App^{NL-G-F}* mice failed to infer coexisting memories in the reversal Barnes maze task**

To explore the explanatory power of the latent cause framework formalizing mechanisms of associative learning and memory modification, cohorts 4, 5, 6, and 7 were subjected to a reversal Barnes maze task (C. A. Barnes, 1979) two weeks after the reinstatement experiment (Table S1). Barnes maze is a commonly used behavioral task to examine spatial learning and memory in Alzheimer's disease model mice (Webster et al., 2014). Associative learning is fundamental in spatial learning (Leising & Blaisdell, 2009; Pearce, 2009). Although we did not make any specific assumptions of what kind of associations were learned in the Barnes maze, the trial-and-error updates of these associations involving sensory preconditioning or secondary conditioning could underlie the learning performance. The first stage contains a 6-day training and 1-day probe test to establish the first spatial memory of the target hole (Fig. 6A). During the training phase, the mice explored a circular field and were reinforced to find an escape box under a target hole as any one of 12 holes equally spaced around the perimeter of the field (Fig. 6A). In the 12-month-old group, the number of errors, latency, and travel distance from start to goal decreased across days in the training phase, suggesting successful initial learning (Fig. 6-figure supplement 1A, and Table S13). A significant interaction between day and genotype [$F(5, 170) = 2.447, p = 0.036$] was observed in the latency where control mice took a longer time to reach the target hole on training days 4, 5, and 6 (Fig. 6-figure supplement 1A, and Table S13). In order to estimate what kind of strategies mice used to solve the task relying on their memory, we performed algorithm-based strategy analysis (Suzuki & Imayoshi, 2017; Tachiki et al., 2023). A moving trajectory in each trial was categorized into one of five spatial strategies following the definition (Fig. 6-figure supplement 2). As in the conventional analysis above, the usages of the strategies were virtually comparable between *App^{NL-G-F}* and control mice (Fig. 6B and Table S14).

In the probe test phase, the mice explored the field for 5 min without the escape box, which generated a certain prediction error (Fig. 6A). While the mice retrieved the spatial memory of the target hole in spite of the novel observation, they would stay around the target hole. Consequently, *App^{NL-G-F}* mice spent significantly less time around the target hole than control mice did in the probe test 1, as a significant interaction between genotype and holes [$F(11, 374) = 4.22, p < 0.001$] was detected (Fig. 6C and Table S15). These results suggest that *App^{NL-G-F}* mice could successfully form a spatial memory of the target hole, while they did not retrieve the spatial memory of the target hole as strongly as control mice when they observed the absence of the escape box during the probe test.

In the first reversal learning phase, the position of the target hole with the escape box was moved to the opposite from day 9 to 11, then returned to the original position from day 12 to day 14 (Fig. 6A). Through the first reversal learning, we expected that the second spatial memory of the target hole is formed that competes with the spatial memory of the first target hole. In other words, this reversal training phase allowed the mice to acknowledge that both two holes were possible to be the target hole. The improved performance across reversal training days in the conventional analysis was similar to that of the initial training phase, suggesting intact behavioral

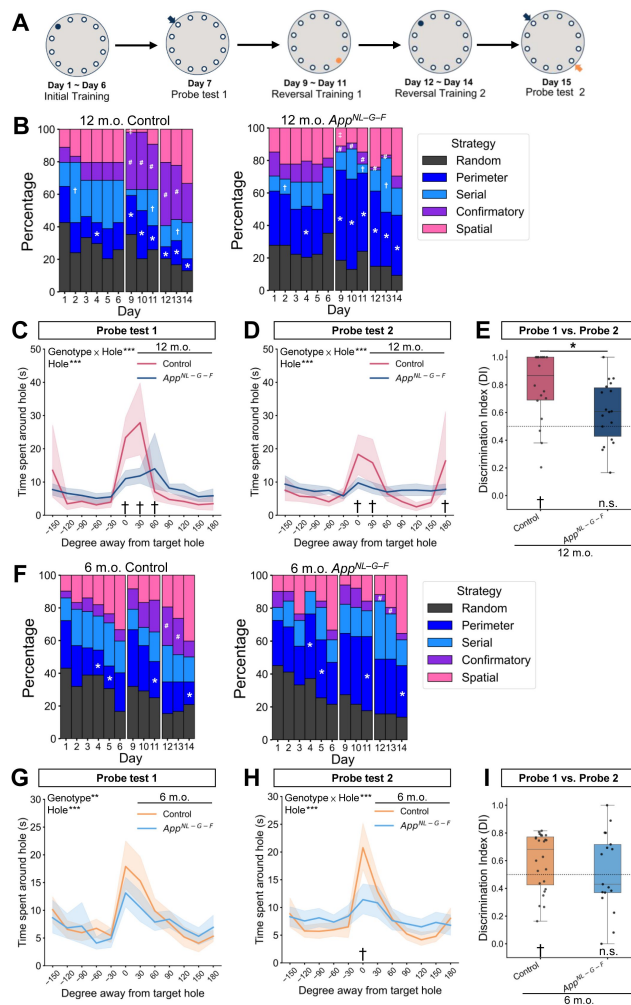


Figure 6.

App^{NL-G-F} mice failed to infer coexisting spatial memories in the reversal Barnes maze task.

(A) Schematic diagram of the reversal Barnes maze task. The largest circle represents the Barnes maze field. The filled and open small circles represent the target hole with the escape box and the remaining holes in the field, respectively. The small arrow pointing to the hole indicated the position of the target hole without an escape box in the probe test. The same color of the filled circle and arrow indicates the same position. (B) Strategy usage in initial training (days 1 to 6), the first reversal training (days 9 to 11), and the second reversal training (days 12 to 14) in the 12-month-old group. Time spent around each hole at probe test 1 (C) and probe test 2 (D) in the 12-month-old group. (E) Discrimination index (DI) of the second target hole between probe test 2 and probe test 1 in the 12-month-old group. (F) Strategy usage in initial training (days 1 to 6), the first reversal training (days 9 to 11), and the second reversal training (days 12 to 14) in the 6-month-old group. Time spent around each hole at probe test 1 (G) and probe test 2 (H) in the 6-month-old group. (I) Discrimination index (DI) of the second target hole between probe test 2 and probe test 1 in the 6-month-old group.

(B and F) $*p < 0.05$ of perimeter strategy, $^{\#}p < 0.05$ of confirmatory strategy, $^{\dagger}p < 0.05$ of serial strategy, $^{\ddagger}p < 0.05$ of spatial strategy by Wilcoxon rank-sum test comparing control and *App^{NL-G-F}* mice at the same age. (C, D, G, and H) The data are shown as mean with 95% confidence interval. $*p < 0.05$, $**p < 0.01$, and $***p < 0.001$ by mixed-design two-way [Genotype (control, *App^{NL-G-F}*) $\times$ Hole (1–12)] ANOVA; $^{\dagger}p < 0.05$ by Tukey's HSD test at the specific hole to compare control and *App^{NL-G-F}* mice at the same age. (E and I) The dashed horizontal line indicates DI = 0.5, which means no discrimination between the two phases. $^{\dagger}p < 0.05$ by the Wilcoxon signed-rank test, and the alternative hypothesis specifies that the mean differs from 0.5; $*p < 0.05$ by Mann-Whitney *U* test comparing control and *App^{NL-G-F}* mice. (C–E, G–I) Each black dot represents one animal. Colors indicate the different groups: pink represents 12-month-old control ($n = 18$), dark blue represents 12-month-old *App^{NL-G-F}* mice ($n = 18$), orange represents 6-month-old control ($n = 24$), and light blue represents 6-month-old *App^{NL-G-F}* mice ($n = 17$). Detailed statistical results are provided in Tables S14 to S16 for the 12-month-old group results and Tables S18 to S20 for the 6-month-old group results.

flexibility (Brown & Tait, 2010 [↗](#); Gawel et al., 2019 [↗](#)) (Fig. 6-figure supplement 1B [↗](#) and Table S13). Unexpectedly, 12-month-old *App^{NL-G-F}* mice reached the goal significantly faster than the control mice [Genotype effect from day 9 to 11, $F(1,68) = 10.53$, $p = 0.003$; Genotype effect from day 12 to 14, $F(1,68) = 10.27$, $p = 0.003$] (Fig. 6-figure supplement 1B [↗](#), and Table S13). This would be due to the difference in the strategies primarily taken in each group of mice. During the first reversal training (days 9 to 11), more than half of *App^{NL-G-F}* mice used the perimeter strategy, which is significantly higher than control mice (Fig. 6B [↗](#) and Table S14). The mice taking this strategy could reach the novel target hole by exploring holes sequentially without spending time around the previous target hole (Fig. 6-figure supplement 2 [↗](#)). On the other hand, the use of perimeter strategy decreased, and spatial strategy increased in control mice from days 9 to 11 (Fig. 6B [↗](#)). Moreover, the use of the confirmatory strategy was around 30% in control mice, which is significantly higher than *App^{NL-G-F}* mice (Fig. 6B [↗](#) and Table S14). This strategy is the one such that the mice visit the first and second target holes as if they relied on the two memories of the target holes where the escape box should exist (Fig. 6-figure supplement 2 [↗](#)). During the second reversal training (days 12 to 14), *App^{NL-G-F}* mice constantly used the perimeter strategy more than the control mice. Meanwhile, the control mice used the confirmatory strategy more than *App^{NL-G-F}* mice, as in the first reversal learning phase (Fig. 6B [↗](#) and Table S14). These results suggest that the search strategy diverged between *App^{NL-G-F}* mice and control mice since the reversal learning. The control mice could retrieve two competing spatial memories of the target hole during each trial. In contrast, *App^{NL-G-F}* mice were less likely to retrieve these two memories within a trial even though the use of spatial strategy was comparable with control mice on the same day.

As expected, the *App^{NL-G-F}* mice explored significantly shorter time for the second target hole as well as the first target hole, compared to the control mice (Fig. 6D [↗](#) and Table S15). A significant interaction between genotype and hole was detected [Genotype $\times$ Hole, $F(11, 374) = 3.24$, $p < 0.001$]. The exploration time was uniform over holes in *App^{NL-G-F}* mice, while control mice showed hole preferences around two target holes as if the *App^{NL-G-F}* mice and the control mice took the perimeter and the confirmatory strategy, respectively. At last, we calculated the discrimination index (DI) from the exploration time for the first and second target holes in probe test 2 and test 1. As in the reinstatement experiment, we expected that if the mice did not retrieve a spatial memory of the second target hole during probe test 2, the exploration time of this hole would be the same between probe tests 1 and 2, resulting in DI around 0.5. The DI for the second target hole was significantly higher than 0.5 in control mice, indicating that the mice successfully modified the spatial memories from probe tests 1 to 2 at a discriminable level (Fig. 6E [↗](#) and Table S16). On the contrary, the DI in *App^{NL-G-F}* mice was comparable to 0.5 and significantly lower than that of control mice, indicating that their spatial memory for the second target hole was not modified after the reversal learning (Fig. 6E [↗](#) and Table S16). We confirmed that the DIs for the first target hole were close to 0.5 and comparable between groups, suggesting that the exploration time for the first target hole remained similarly long in both probe tests (Fig. 6-figure supplement 1C [↗](#), and Table S16).

The distinct learning process observed in the 12-month-old group was also present in the 6-month-old group. The performance shown in the conventional analysis was comparable between the 6-month-old control and *App^{NL-G-F}* mice (Fig. 6-figure supplement 2A, 2B [↗](#), and Table S17). The use of perimeter strategy was over 30% throughout two reversal training phases in 6-month-old *App^{NL-G-F}* mice, while control mice used significantly higher confirmatory strategy in the second reversal training (Fig. 6F [↗](#), and Table S18). During the probe tests, the difference in exploration time around the target hole was less prominent in the 6-month-old group, while *App^{NL-G-F}* mice uniformly explored the holes as in the 12-month-old *App^{NL-G-F}* mice, especially in the probe test 2 (Fig. 6G [↗](#), 6H [↗](#), and Table S19). Unlike 12-month-old control mice, the 6-month-old control mice did not show a strong preference for the second target hole in probe test 2 (Fig. 6H [↗](#)). However, their DIs for the first and second target holes were significantly higher than 0.5, suggesting an increased confidence of spatial memory for each target hole (Fig. 6I [↗](#), Fig. 6-figure supplement

3C [↗](#), and Table S20). In App^{NL-G-F} mice, the DIs for the first and second target hole were comparable to 0.5, as those in 12-month-old App^{NL-G-F} mice, suggesting few modifications of the spatial memories (Fig. 6I [↗](#), Fig. 6-figure supplement 3C [↗](#), and Table S20).

In summary, these results suggest that the control mice retained competing spatial memories for the first and the second target holes following reversal learning. In contrast, the App^{NL-G-F} mice retained at most one spatial memory during a trial. In other words, the App^{NL-G-F} mice, given the spatial cues, struggled to infer the spatial memory of either the first or second target hole, leading to the rare usage of the confirmatory strategy. Thus, we confirmed the misclassification in the App^{NL-G-F} mice in the reversal Barnes maze paradigm. Although both DIs in the reinstatement (Fig. 2H [↗](#)) and the Barnes maze experiment (Fig. 6E, 6I [↗](#)) suggest that inferred memories in each time point would diverge between the App^{NL-G-F} mice and the control mice, no significant correlations were detected (12-month-old group: Spearman's correlation coefficient = 0.062, $p = 0.738$, $N = 32$; 6-month-old group: Spearman's correlation coefficient = -0.029, $p = 0.862$, $N = 39$), potentially due to differential effects of A β accumulation on associative and instrumental learning.

Discussion

Recent advances in disease-modifying intervention have shed light on AD treatment and prevention. Identifying the earliest manifestation of cognitive decline is critical for intervention. However, the behavioral phenotypes and altering trajectories are heterogeneous in predementia AD patients (Duara & Barker, 2022 [↗](#); Jutten et al., 2023 [↗](#)), yet standard neuropsychological tests have been optimized for detecting stereotypical symptoms near or after the onset of dementia (Snyder et al., 2014 [↗](#)). Recently proposed generative models successfully contributed to predicting nature-nurture factors conjugate with A β and behavioral outcomes from the preclinical to severe stage of AD (Hwang et al., 2023 [↗](#); Petrella et al., 2019 [↗](#); Yada & Honda, 2023 [↗](#)), while only a few studies explicitly assume an internal model of the patients (Sutter et al., 2024 [↗](#)). Assuming an internal model would broaden and deepen interpretations of various cognitive deficits in terms of common computational account (Kocagoncu et al., 2021 [↗](#)).

In line with this, our perspective is that states of an internal model would be one candidate of factors to generate a wide variety of behavioral phenotypes in AD, and hence probing the internal state would contribute not only to detecting early signs of AD but also to multidimensionally evaluating how the disease gets worse and how early intervention works. In this study, we assumed the latent cause model proposed by Gershman et al. (2017) [↗](#) as an internal model and unveiled the internal state in 6- and 12-month-old App^{NL-G-F} mice, a preclinical mouse model of AD, from their behavioral data by estimating the parameters of the latent cause model well replicating the behaviors.

The misclassification in the memory modification process underlies the impairment of App^{NL-G-F} mice in the reinstatement of conditioned fear memory

Initially, we confirmed a significant accumulation of A β increased along aging in the App^{NL-G-F} mice (Fig. 2-figure supplement 1 [↗](#)). We then tested whether the memory modification process was behaviorally impaired in the App^{NL-G-F} mice across the auditory-cued fear conditioning, extinction, and reinstatement. First, the mice learned an association between a tone as a CS and an electrical shock as a US, as they exhibited increased CR in test 1 (Fig. 2B, 2C [↗](#)). Next, they observed the CS alone repeatedly through the extinction, and then they exhibited decreased CR in test 2 (Fig. 2B, 2D [↗](#)). After they observed the US alone, the mice exhibited increased CR again in test 3, suggesting that the reinstatement was established (Fig. 2B, 2E [↗](#)). However, the App^{NL-G-F} mice exhibited a lower extent of reinstatement compared to the control (Fig. 2H [↗](#)). The DI

between test 1 and 3 in the *App*^{NL-G-F} mice was significantly lower than 0.5, while that in the control was comparable with 0.5 (Fig. 2H), suggesting that the *App*^{NL-G-F} mice would have lower expectations of upcoming shock in the reinstatement than immediately after the acquisition. These results were the first to report the deficits in the reinstatement in the *App*^{NL-G-F} mice, suggesting that A β accumulation induces the defect in the memory modification process after the extinction.

Although our results support that the A β accumulation in *App*^{NL-G-F} mice would have little effect on associative fear learning and extinction, the results have been mixed in studies using fear conditioning tasks. Two studies found normal contextual or auditory-cued fear conditioning responses up to 18 months old (Kundu et al., 2021; Sakakibara et al., 2018), which aligns with our results. On the contrary, another two studies reported that *App*^{NL-G-F} mice have defects in both contextual and cued fear conditioning from the age of five months (Emre et al., 2022; Mehla et al., 2019). The inconsistency suggests the impact of A β may be complicated and cannot be simply detected by common behavioral tasks, highlighting the need for novel approaches to probe in-depth behavioral changes.

Recent advances in computational psychiatry offer a wealth of internal models to interpret a patient's internal states (Hauser et al., 2022; Huys et al., 2016; Montague et al., 2012). For instance, Norbury et al. (2022) simulated a memory modification process in severe post-traumatic stress disorder (PTSD) patients by the latent cause model and found that they were biased to infer old memories rather than to acquire new memories for novel observations, as they have lower concentration parameter α . In this study, we found that *App*^{NL-G-F} mice exhibited lower α both at 6 months and 12 months of age (Fig. 4A, 5A). As the A β accumulation increased with age (Fig. 2-figure supplement 1), lower σ_x^2 was observed in 12-month-old *App*^{NL-G-F} mice (Fig. 4A). As α and σ_x^2 significantly correlated with DI between test 3 and 1 (Fig. 4B, 5B), we considered both as critical parameters that shaped the diverged internal states underlying reinstatement between control and *App*^{NL-G-F} mice. Although we have considered alternative internal models, the latent cause model offered better explanatory power to our data (see Appendix 1), therefore we stood on the latent cause model to explain reinstatement.

App^{NL-G-F} mice with lower α favored to generalize the observation as if it arose from the old cause instead of differentiating it as a new memory. Indeed, their posterior probabilities of acquisition latent causes were higher than those of the control mice when a novel observation was given in test 1 (Fig. 4-figure supplement 3B), contributing to the higher CR in *App*^{NL-G-F} mice (Fig. 2C, Fig. 4-figure supplement 3A). Such overgeneralization associated with lower α may, in part, explain the deficits in discriminating highly similar visual stimuli reported in AD mouse model (Ding et al., 2023; Saifullah et al., 2020; Zhu et al., 2017), preclinical AD patients (Leal et al., 2019; H. Lee et al., 2020), and MCI patients (Ally et al., 2013; Belliart-Guérin & Planche, 2023; Laczó et al., 2021; Parizkova et al., 2020; Wesnes et al., 2014; Yassa et al., 2010), as they classify the lure as previous learned object.

Following the Chinese restaurant process, the impact of α on internal states decreases with trials, and meanwhile that of σ_x^2 relatively increases in our reinstatement paradigm. *App*^{NL-G-F} mice in 12-month-old with lower σ_x^2 favored to differentiate similar observations into unique memories, especially in later phases, due to the decreased posterior probabilities of existing latent causes. Indeed, *App*^{NL-G-F} mice had a larger number of latent causes acquired after the extinction with posterior probabilities (Fig. 4E), resulting in a limited update of the associative weight of the existing latent causes by unsignaled shock (Fig. 4D), and thereby lower CR in test 3 (Fig. 2E, 2H). These results may explain the impaired ability to transfer previously learned association rules could be due to misclassified them as separated knowledge in preclinical autosomal dominant AD mutation carriers and mild AD patients (Bódi et al., 2009; Petok et al., 2018). Such overdifferentiation associated with lower σ_x^2 might underlie delusions, a common neuropsychiatric symptom reported in AD dementia (Kumfor et al., 2022), as an extended latent

cause model could simulate the emergence of delusional thinking (Erdmann & Mathys, 2022 [↗](#)). Thus, the deficit in reinstatement of conditioned fear memory in the 12-month-old *App^{NL-G-F}* mice could be due to their overgeneralization or overdifferentiation of observations into memories, and consequently the competing memories were not retained through the memory modification process (Fig. 4 [↗](#) and 5 [↗](#)).

Our study demonstrated that estimating internal states with the parameters of the latent cause model provided additional explanations for the cognitive differences between control and *App^{NL-G-F}* mice. It is undoubted that conventional behavioral tests for AD patients or AD mouse models have provided evidence of cognitive decline and are useful for stratification in clinical trials. Nonetheless, it is also possible that certain deficits have antecedently emerged in internal states, even if behavioral deficits are not obvious as in 6-month-old *App^{NL-G-F}* mice. The computational phenotype was proposed by Montague et al. (2012 [↗](#)) and defined as parameters of a computational model being in conjunction with measurable behavioral or biological phenotypes. In this sense, α and σ_x^2 in the latent cause model might satisfy this definition in this study. Computational phenotype was established to fill explanatory gaps between biological and psychological evidence in psychiatry, which would be informative likewise in AD studies.

The misclassification prevents *App^{NL-G-F}* mice from retaining competing memories

We further tested whether the deficit in the *App^{NL-G-F}* mice observed in the reinstatement experiments can be replicated in the reversal Barnes maze paradigm, where the reversal training phases could resemble the extinction phase in the reinstatement paradigm, in that “counterexample” of the previously established memory is observed. The *App^{NL-G-F}* and control mice exhibited similar learning curves in the initial training phase regardless of age (Fig. 6-figure supplement 1A [↗](#) and 3A [↗](#)). These data align with previous studies that did not find a significant spatial learning deficit in *App^{NL-G-F}* mice up to 10 months old by using the Morris water maze, a common spatial learning task relevant to the Barnes maze (Latif-Hernandez et al., 2019 [↗](#); Saifullah et al., 2020 [↗](#); Whyte et al., 2018 [↗](#)). Although *App^{NL-G-F}* mice were previously reported to have longer latency and make more errors during training in the Barnes maze (Broadbelt et al., 2022 [↗](#); Sakakibara et al., 2018 [↗](#)), these studies haven’t described the mice’s navigation strategies to explain such deficits.

During the reversal training, *App^{NL-G-F}* mice displayed behavioral flexibility as control mice did based on the conventional analysis (Fig. 6-figure supplement 2B [↗](#) and 3B [↗](#)), similar to findings in Sakakibara et al. (2018) [↗](#). However, our strategy analysis revealed that this comparable performance was achieved in a sub-optimal way. The usage of confirmatory strategy such that mice selectively explored the first and the second target hole was significantly higher in the control mice during the reversal training phase, especially in the 12-month-old (Fig. 6B, 6F [↗](#)). After the reversal learning, the confirmatory strategy would become the optimal strategy at the first trial in each day to know today’s goal location. In contrast, the *App^{NL-G-F}* mice primarily took the perimeter search strategy such that mice explored the target hole in a “brute-force” manner (Fig. 6B, 6F [↗](#)), consistent with deficits in navigation ability found in preclinical AD patients (Allison et al., 2016 [↗](#); Coughlan et al., 2018 [↗](#)). These results suggest that the control mice solved the task by inferring the two competing spatial memories representing the two possible target holes, whereas the *App^{NL-G-F}* mice solved the task not relying on selective spatial memories but the perimeter strategy, while both mice were given the same spatial cues. Thus, our hypothesis that the *App^{NL-G-F}* mice have a deficit in classifying observed cues into two or more competing memories in the same context would be supported by the results of the reversal Barnes maze experiment.

In the perspective from the latent cause framework, we posit that the mice would learn a larger number of association rules between stimuli in the maze compared to those in the reinstatement of the fear conditioning, and either infer new memories or modify existing memories for the unexpected observations in the Barnes maze (e.g., changed location or absence of escape box) as in the reinstatement paradigm. In the reversal Barnes maze paradigm, the animals would infer that a latent cause generates the stimuli in the maze at certain associative weights in each trial and would adjust behavior by retaining competing memories. Both overgeneralization and overdifferentiation could explain the lower exploration time of the target hole in the *App^{NL-G-F}* mice. In the case of overgeneralization, the mice would overwrite the existing spatial memory of the target hole with a memory that the escape box is absent. In the case of overdifferentiation, the mice would infer a new memory such that the goal does not exist in the novel field, in addition to the old memory where the goal exists in the previous field. In both cases, the *App^{NL-G-F}* mice would not infer that the location of the goal is fixed at a particular point and failed to retain competing spatial memories of the goal, leading to relying on a less precise, non-spatial strategy to solve the task.

As the latent cause model was built on the fear conditioning paradigm and does not explicitly assume the contribution of other cognitive functions such as attention or working memory, this study provides no direct evidence as to which model parameters are in conjunction with the behavioral outcomes in the reversal Barnes maze and which parameters involve other the cognitive functions that are presumably affected by AD pathology (Finke et al., 2013 [↗](#); Kirova et al., 2015 [↗](#); Malhotra, 2019 [↗](#)). To evaluate them, further studies would be required to integrate computational models encompassing different cognitive functions, such as multiple successor representations (Madarasz & Behrens, 2019 [↗](#)) or hidden state inference (Sanders et al., 2020 [↗](#)) for spatial learning, upon more fundamental principles, such as the predictive coding in a hierarchical neural network (Kocagoncu et al., 2021 [↗](#)). We believe that such models would not only explain a wide variety of symptoms in dementia and AD in a common framework but also predict disease progression or effects of interventions for various cognitive functions.

Putative brain regions involving the altered internal states in the *App^{NL-G-F}* mice

This study did not explicitly provide evidence about neural mechanisms underlying the defect in memory classification in *App^{NL-G-F}* mice. Nonetheless, we speculated that possible neural circuits are presumably involved in the results reported here according to the findings of the previous studies.

It is well known that the amygdala, the hippocampus, and the medial prefrontal cortex are critical in regulating fear memory and its extinction (Maren et al., 2013 [↗](#)). In addition, the ventral tegmental area (VTA) plays a modulatory role with its wide projection to these brain regions (Beier et al., 2015 [↗](#); Cai & Tong, 2022 [↗](#)). The dysfunction in the abovementioned brain regions may cause the defect of reinstatement. Indeed, previous studies have shown that the inhibition of dopaminergic signal from VTA to the infralimbic cortex (Hitara-Imamura et al., 2015 [↗](#)) or inhibition of glutamatergic signal from VTA to the dorsal hippocampus (Han et al., 2020 [↗](#)) during unsignaled shock reduced the reinstatement of fear memory in mice.

Although the neural implementation of the latent cause model has not been demonstrated yet, contributions of certain neuronal populations are expected. It is well established that prediction error in classical or operant conditioning learning is computed in the VTA (Lerner et al., 2021 [↗](#)). The prediction error is signaled from the VTA to the hippocampal CA1 via the dopaminergic projections, while the VTA receives a feedback signal from the CA1 (Lisman & Grace, 2005 [↗](#)). The pattern separation and completion were implemented in the dentate gyrus (DG) and the CA3 in the hippocampus, respectively (Neunuebel & Knierim, 2014 [↗](#)). As Sanders et al. (2020) [↗](#) pointed out, one might find an analogy between the process of pattern completion/separation and the

inference of old/new latent cause, in the sense that both consist of a generalization/differentiation process to assign a current observation to either the same class with the previous observations or a novel class. The dysfunction in this circuit has long been implicated in memory problems in aging and AD (Palmer & Good, 2011 [↗](#); Wilson et al., 2006 [↗](#)). Gershman et al. (2017) [↗](#) predicted that activities of newborn neurons in the DG and neurons in the CA3 correlate with the prior probability over latent causes, while the dopaminergic projection from the VTA to the hippocampal CA1 represents the posterior probability over latent causes. In other words, prediction error signals from the VTA to the CA1 might serve to compute the likelihood of the observation given the latent causes. The feedback from the CA1 to the VTA might signal posterior probability over latent causes and then modulate the extent of associative weight update.

Different lines of evidence suggest that hippocampus functions were affected in *App^{NL-G-F}* mice, including lower neuronal firing rates in the CA1 region (Inayat et al., 2023 [↗](#)), degradation of gamma oscillation power in the CA3 region (Arroyo-García et al., 2021 [↗](#)), and disrupted place cell remapping in CA1 region (Jun et al., 2020 [↗](#)). While VTA dysfunction was not reported in *App^{NL-G-F}* mice, studies using Tg2576 mice, which overexpress mutated APP, have shown that neurodegeneration in the VTA occurred at an early age and consequent dysfunction was correlated with impairments in the hippocampus (Nobili et al., 2017 [↗](#); Spoletti et al., 2024 [↗](#)). Given the evidence above, dysfunctions caused by pathological Aβ in the hippocampus and VTA are potential candidates underlying unsuccessful latent cause inference in *App^{NL-G-F}* mice, leading to a lower extent of reinstatement. Although this study considered limited disease factors, cognitive impairments in AD patients result from a combination of neuropathological factors and cannot be ascribed solely to Aβ deposition in clinical practice. Examining whether the interaction of neuropathology can be seen in the internal state with a mouse model including tau pathology, e.g., *App^{NL-G-F}/MAPT* double knock-in mice (Hashimoto et al., 2019 [↗](#); Saito et al., 2019 [↗](#)), would improve the translational interpretation in this study.

Conclusion

Uncovering the underdeveloped symptoms in the AD continuum is critical to early detection and intervention by filling the explanatory gap between the pathological states and the cognitive decline in AD. We expect that seeking internal state as well as parameters in an internal model in conjunction with behavioral or biological phenotypes could provide in-depth explanations about behavioral outcomes in AD beyond assessing behavior alone. Our study is the first to report that the alteration in the internal state of *App^{NL-G-F}* mice biased their memory modification process toward overgeneralization or overdifferentiation.

Materials and Methods

Animals

Male *App^{NL-G-F}* homozygous knock-in mice (abbreviated as *App^{NL-G-F}* mice in this study) and age-matched control mice were subjected to behavioral tests at 6 and 12 months old. Cohort and genotype information was provided in Table S1. We decided to assign at least 20 mice for each group (6-month-old *App^{NL-G-F}* mice, 6-month-old control mice, 12-month-old *App^{NL-G-F}* mice, and 12-month-old control mice) on average when designing the reinstatement experiment (see below). Animals were housed under a 12-h light/dark cycle in a temperature-controlled environment (23–24°C temperature and 40%–50% relative humidity) with food pellets (Japan SLC) and water provided *ad libitum*. Behavioral experiments were performed during the dark cycle (8 am–8 pm). After the behavioral experiments, mice were sacrificed by cervical dislocation unless their brains were sampled. All animal procedures were performed in accordance with the Kyoto University animal care committee's regulations (approval number: Lif-K24014).

Behavioral test

Reinstatement paradigm in fear conditioning

The reinstatement of conditioned fear response after a reminder of unconditioned stimulus is observed in different species (Hermans et al., 2005 [\[1\]](#); Hitora-Imamura et al., 2015 [\[2\]](#); Monfils et al., 2009 [\[3\]](#)). In the present study, the four-chamber fear conditioning testing system (O'HARA & CO., LTD, Tokyo, Japan) was used to deliver tone and electrical foot shock, and the top view camera in the chamber recorded mouse behavior at 2 Hz, as previously described (Tachiki et al., 2023 [\[4\]](#)). The procedure contained the following phases (**Fig. 2A** [\[5\]](#)): acquisition (day 1), test 1 (day 2), extinction (day 3 and day 4), test 2 (day 5), unsignaled shock (day 8), and test 3 (day 9). The context (acrylic box) was the same throughout the test. Mice were transferred to the experimental room from the breeding room and then habituated for 30 min in their homecage in a rack until the phase started. In each phase, mice were moved between their homecage and the chamber via a transporting cage filled with woodchips. In the acquisition phase, a 60-s tone (65 dB, 10 kHz) as the conditioned stimulus (CS) was presented 300 s later since the mice were released in the chamber. The last 2 seconds of the CS were accompanied by an electrical foot shock (0.3 mA) as the unconditioned stimulus (US). This CS-US association was repeated 3 times with a 90-s interval. After the last US, the mice were left in the chamber for 140 s and then returned to the homecage. In the test phases, a 60-s tone was given 120 s after the mice entry. In the extinction phase, only a 60-s tone was given 19 times on day 3 and 10 times on day 4 with a 200-s interval. In the unsignaled shock phase, an electrical foot shock (0.7 mA) was given 10 min after the mice were placed into the chamber. After the electrical shock, the mice were left in the chamber for 5 min and then transferred back to the homecage. The chamber and grid were cleaned with 70% propanol before trials started. The freezing behavior was measured as a conditioned response (CR), and the criterion for freezing is defined as the change of detected mouse area being less than 10 pixels within 1 second. The freezing rate was calculated as the percentage of time spent freezing 1 min before (preCS) and during the tone presentation (CS). Discrimination index (DI) has been widely used as a normalized measure to evaluate the extent to which the system can distinguish between two conditions. To evaluate the magnitude of how each mouse can discriminate between different phases, DI is calculated as $CR(t) / [CR(t) + CR(t')]$, where CR indicates freezing rate during CS presentation; t and t' indicate any two test phases.

Barnes maze

Barnes maze is a commonly used behavioral task to examine spatial learning and reference memory in Alzheimer's disease model mice (Webster et al., 2014 [\[6\]](#)). The apparatus and setting were the same as previously reported (Suzuki & Imayoshi, 2017 [\[7\]](#); Tachiki et al., 2023 [\[4\]](#)) (Bio-Medica, Osaka, Japan). The maze consisted of a circular open arena (98 cm in diameter) with 12 holes equally spaced around the perimeter. A black iron escape box was placed under one of the holes as a goal during the training phase. The location of the goal stayed constant for a given mouse but was randomized across mice. Four unique visual cues were located at the outside of the arena. Behavior during the trials was recorded using a GigE Vision camera (UI-5240SE-NIR; IDS Imaging Development Systems GmbH, Obersulm, Germany) mounted on the top of the maze to estimate the mouse's position in each frame. All programs used for data acquisition, processing, saving, and synchronized device controls were written in LabVIEW 2013 (National Instruments, TX, USA). The procedure contained the following phases (**Fig. 6A** [\[8\]](#)): habituation (day 0), initial training (day 1 to day 6), probe test 1 (day 7), retraining (day 8), reversal training 1 (day 9 to day 11), reversal training 2 (day 12 to day 14) and probe test 2 (day 15). Mice were transferred to the experimental room from the breeding room and then waited for 30 min in their homecage in a rack by the phase started. In the habituation phase, mice were released to the center of the field and could freely explore the field for 5 min. After the trial, the mice were moved into the escape box for another 5 min, then back to the homecage. In the training phase, the mice had to explore

the field until they successfully got into the escape box within 10 min. Then, they were returned to the homecage and on standby for the next trial. In the probe test, the procedure was the same as the habituation phase except that the mouse was returned to the homecage directly after the field exploration. The field was cleaned with 70% ethanol thoroughly, and the paper bedding in the escape box was changed per trial to reduce the effect of possible olfactory cues in the maze.

The analysis of the Barnes maze was performed on the custom code based on our previous study (Suzuki & Imayoshi, 2017 [DOI](#)) with some modifications in strategy analysis. The code for strategy analysis is freely available online at https://github.com/suzuki-yusuke/BM_analysis.git [DOI](#). In brief, the conventional analysis included the number of errors (count to visit holes other than the goal), latency, and travel distance from start to goal in each trial in the training phase, and time spent around each hole in the probe test. The strategy analysis was performed to qualitatively evaluate what kind of exploratory strategies were used by mice based on their belief in each trial (Suzuki & Imayoshi, 2017 [DOI](#); Tachiki et al., 2023 [DOI](#)). For each mouse, the trajectory from start to goal in each trial was automatically assigned to any one of the exploration strategies following the definition described in **Figure 6-figure supplement 2** [DOI](#). In this study, confirmatory and perimeter strategies were assumed in addition to the strategies previously reported. The former is the one such that the mice visit two holes, the current or past goal located opposite in the field, as if they have two memories such that the goal exists in each of the two holes. The latter is the one such that the mice find the goal simply running along with the holes, as if they do not have memories that the goal exists in a specific hole. To evaluate how the belief in the mice changed after the reversal training, the discrimination index (DI) is calculated as $ER(t_2) / [ER(t_1) + ER(t_2)]$, where ER indicates the exploration rate; t_1 and t_2 indicate probe test 1 and probe test 2. The ER was calculated as time spent around the hole divided by the maximum exploration time across all holes in each mouse. The DI is calculated for the first target hole (hole 1) and the second target hole (hole 7).

Simulating reinstatement and estimating parameters in the latent cause model

As the latent cause model serves as a generative model, it allows us to simulate the reinstatement experiment while exploring a set of model parameters, minimizing the prediction error between traces of simulated and observed CR for each mouse. The observed trace of CR across all trials in each mouse was initially normalized. In the simulation, the reinstatement experiment was replicated as follows. The number of trials was the same with our experiment: trials 1 to 3 was the acquisition phase, trial 4 was the test 1 phase, trials 5 to 23 was the first extinction phase, trials 24 to 33 was the second extinction phase, trial 34 was the test 2 phase, trial 35 was the unsignaled shock phase, and trial 36 was the test 3 phase. All trials contained stimuli, while the acquisition and the unsignaled shock trials contained the US. The stimulus given a trial was represented as a two-dimensional vector, where the first and the second dimensions indicated the CS and the context, respectively. If the CS was presented at the phases except for the unsignaled shock, the value got 1; otherwise, 0. Because the context was always presented, the value of context was given 0.2 for all trials. The US was represented as a binary value, either 1 at the acquisition and the unsignaled shock or 0 otherwise. The temporal distance of 24 h between each phase was set to be 20, the same as in the original study (Gershman et al., 2017 [DOI](#)).

Ten model parameters to be estimated were concentration parameter (α), temporal scaling parameter (g), learning rate (η), maximum number of iterations in the EM algorithm (max. no. of iteration), initial associative weight (w_0), variance of the US (σ_r^2) and the stimulus (σ_x^2), response threshold (θ), response gain (λ), and maximum number of latent causes (K), where g is not explicitly described in the original model (Gershman et al., 2017 [DOI](#)) while this is implemented in their source code (<https://github.com/sjgershm/memory-modification.git> [DOI](#)). They were estimated through the slice sampling algorithm in the Markov chain Monte Carlo method (Neal, 2003). In implementation, we used the MATLAB function, *slicesample*. The initial guess and upper and lower

bounds of each parameter were listed in Appendix 2 – table 1. The initial size of the sampling window for each parameter was the difference between the upper and lower bounds of the parameter divided by 50. In each sampling step, a set of parameters was proposed, and then a CR trace was simulated in the virtual reinstatement experiment. For each parameter, one sample was drawn for every 100 steps in 200000 times sampling, then the first 1000 samples were discarded as burn-in, and eventually 1000 samples were left. The convergence of the sample chain was determined if the Gelman-Rubin statistic was smaller than 1.1. The expected value of each parameter was considered to be the maximum bin in the sample histogram, for which the bin width was statistically optimized by the method proposed by Shimazaki & Shinomoto (2007) [Shimazaki & Shinomoto \(2007\)](#). As max. no. of iteration and K are integers, each expected value was rounded toward infinity. Given the expected parameters, the likelihood of squared errors between the simulated and the observed CR trace was evaluated on a half-normal distribution ($\mu = 0$, $\sigma = 0.3$) so that increasing error decreased the likelihood. If the likelihoods of errors were smaller than the thresholds (1 for acquisition trials; 0.25 for tests 1, 2, and 3 trials; 0.5 for remaining trials) in all trials and the Gelman-Rubin statistic < 1.1 , the sampling procedure above was terminated. Otherwise, it was continued while keeping the sample chain and setting the initial guess as the last value of the sampling chain. If the procedure was not terminated within 20 iterations, the observed CR trace was classified as an anomaly. The code described above is freely available online at https://github.com/suzuki-yusuke/comp_modeling_lcm.git.

Given the estimated parameters for each mouse, their internal state and CR were simulated for every trial. The internal states consisted of latent causes, their posterior probabilities, and associative weight. The latent causes were categorized based on which phase they were initially inferred: acquisition trials, extinction trials (i.e., test 1, extinction, test 2), and trials after extinction (i.e., the unsignaled shock and test 3).

Immunohistochemistry

After the designated behavioral test, mice were anesthetized by intraperitoneal injection with a cocktail of 0.3 mg/kg medetomidine (Nippon Zenyaku Kogyo, Koriyama, Japan), 4.0 mg/kg midazolam (Sandoz, Minato, Japan), and 5.0 mg/kg butorphanol (Meiji Seika Pharma, Chuo, Japan) and transcardially perfused phosphate buffer saline (PBS) followed by 4% paraformaldehyde (PFA). Brain samples were preserved in 4% PFA overnight, and the solution was replaced with 30% sucrose in PBS. Then, the sample was embedded in the OCT compound (Tissue TEK, Sakura finetek Japan, Chuo, Japan) and sectioned at 30 μm (CM1950, Leica Biosystems, Tokyo, Japan). Every six sections were collected. Ten coronal brain sections from the anterior part (AP = +2.2 ~ +1 mm from Bregma) and the posterior part (AP = -1.5 ~ -2.5 mm from Bregma) were subjected to staining in a 12-well plate. Sections were washed in PBS with 0.3% Triton X-100 (PBST) to remove excessive OCT compound, then blocked in 5% normal donkey serum (NDS) in PBST for an hour at room temperature and incubated with primary antibodies on a shaker at 4°C overnight. The following day, sections were washed in PBST and reacted with second antibodies and DAPI for 2 hours at room temperature. Finally, the sections were washed in PBST, mounted on slides, and coverslipped with antifade reagent (FluoroMount-G®, SoutherBiotech, Cat#0100-01). Primary antibodies, including mouse anti- β -Amyloid (1:1000, Biolegend, Cat#803001, RRID:AB_2564653), rabbit anti-Iba1 (1:1000, FUJIFILM Wako Pure Chemical Corporation, Cat#019-19741, RRID:AB_839504), and rat anti-GFAP (1:1000, Invitrogen, Cat#13-0300, RRID:AB_2532994), were prepared in PBST with 1% NDS. Second antibodies against mouse Alexa Fluor-568nm (Invitrogen, Cat#A10037, RRID:AB_11180865), rabbit Alexa Fluor-647nm (Invitrogen, Cat#A31573, RRID:AB_2536183), and rat Alexa Fluor-488nm (Invitrogen, Cat#A21208, RRID:AB_2535794) were used in a dilution of 1:500 in PBST with 1% NDS. Fluorescent immunohistochemistry slides were imaged on a fluorescence microscope (BZ-X800, Keyence, Japan) with a 10x objective, and the stitching function was used to cover the entire section. Images were stitched in built-in analysis software (Keyence, Japan). The A β plaque area of each brain was quantified under QUINT workflow (Yates et al., 2019 [Yates et al., 2019](#)). First, a series of brain section images were registered to Allen Brain Atlas CCFv3 on QuickNII (Puchades et al., 2019 [Puchades et al., 2019](#)) and fine-adjusted on VisuAlign (<https://www.nitrc.org/projects/visualign>). Next, the A β

staining channel images were segmented in Ilastik (Berg et al., 2019 [DOI](#)). Finally, the integration of registration and segmented results was performed in Nutil (Groeneboom et al., 2020 [DOI](#)) to quantify A β plaque in selected brain regions. The A β plaque area (%) is calculated as the percentage of positive A β staining area covering the whole coronal brain section.

Statistical analysis

For the fear conditioning, extinction, and reinstatement results, samples with freezing rates in test phases exceeding the 1.5 times interquartile range of the group were treated as outliers and excluded for further statistical analysis. The one-way ANCOVA was performed to analyze genotype effects on the freezing rate and DI, with age as a covariate. The Student's *t*-test was performed to compare the freezing rate and the DI between control and *App*^{NL-G-F} mice within the same age. To test whether the DI differs from 0.5, the one-sample *t*-test was performed within the group. To compare the estimated parameters and properties of latent causes between control and *App*^{NL-G-F} mice within the same age, the Mann-Whitney *U* test was performed, as the Shapiro-Wilk normality test reported a violation of normality assumption in the data. The Spearman's rank correlation coefficient between the estimated model parameters and the DIs between test 3 and test 1 phase were calculated.

For the reversal Barnes maze results, a mixed-designed two-way [Genotype (control, *App*^{NL-G-F}) $\times$ Day (1-6, 9-11, or 12-14)] ANOVA was performed for the number of errors, the latency, and the travel distance in the initial and reversal training. A mixed-designed two-way [Genotype (control, *App*^{NL-G-F}) $\times$ Hole (1-12)] ANOVA was performed for the time spent around each hole in the probe test 1 and 2. If a significant interaction between genotype and hole was detected, Tukey's HSD test at a specific hole between control and *App*^{NL-G-F} was performed. The Wilcoxon rank sum test was performed to compare the use of each strategy on each training day between control and *App*^{NL-G-F} within the same age. To test whether the DI between probe tests 1 and 2 differs from 0.5, the Wilcoxon signed-rank test was performed within the group, as the Shapiro-Wilk normality test reported a violation of the normality assumption in the data. The Mann-Whitney *U* test was performed to compare the DI between control and *App*^{NL-G-F} mice within the same age.

All statistical analyses were done in JASP 0.18.3.0 (JASP team, 2024) and MATLAB R2022b (MathWorks Inc., MA, USA). The significance level for all statistical tests was set at 0.05. The power analysis table was provided in Table S21 and Table S22 by using G*Power 3.1 (Faul et al., 2009). Not all of the multiple comparisons were corrected for, and the sample sizes written in the figure legends do not include outliers detected through the exclusion procedure mentioned above.

Figure supplements

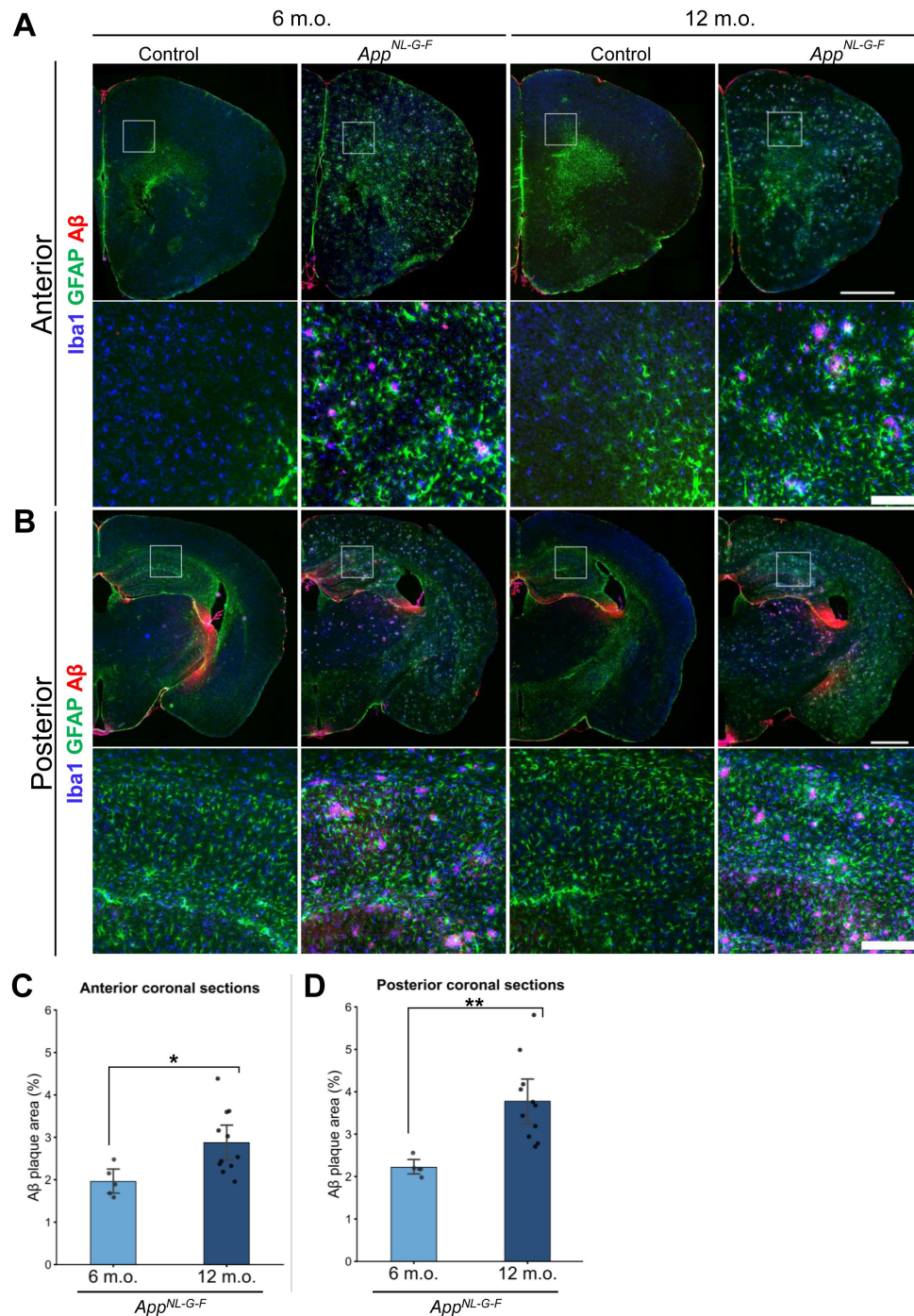


Figure 2 – figure supplement 1.

Representative image of immunofluorescence staining using Iba1 (blue), GFAP (green), and Aβ (red) antibodies on anterior

(A) and posterior (B) coronal brain sections. Quantification results of Aβ plaque area in anterior (C) and posterior (D) coronal brain section of *App^{NL-G-F}* mice.

(A and B) The scale bar on the images in the first row represents 1 mm. The images in the second row of the panel are the magnified view of the inlet in the same column, where the scale bar on the images represents 150 μm in panel A and 250 μm in panel B. (C and D) The data are shown as mean with 95% confidence interval. Each black dot represents one animal. $n = 5$ in 6-month-old *App^{NL-G-F}* mice; $n = 10$ in 12-month-old *App^{NL-G-F}* mice. Student's t -test was performed to compare 6- and 12-month-old *App^{NL-G-F}* mice. The results were $t(14)=2.555$, $p = 0.023^*$ in (C) and $t(14)=3.536$, $p = 0.003^{**}$ in (D).

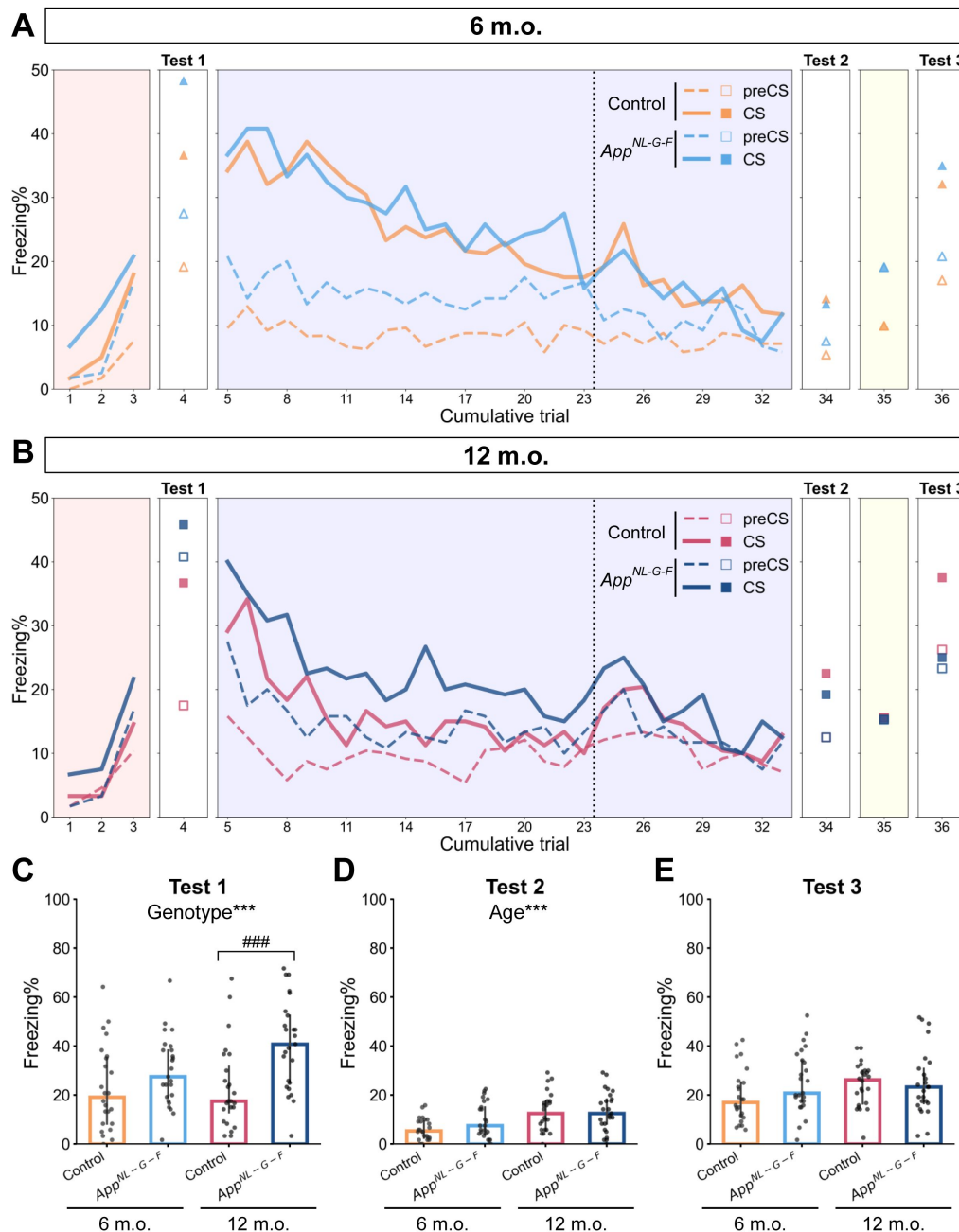


Figure 2 – figure supplement 2.

Freezing rate before tone presentation in the reinstatement paradigm

Freezing rate before and during tone presentation across reinstatement paradigm in the 6-month-old group (**A**) and 12-month-old group (**B**). The markers and lines show the median of each group in each trial. Dashed and solid lines indicate the freezing rate for 1 min before the CS presentation (preCS) and that for 1 min during the tone presentation (CS), respectively. Red, blue, and yellow backgrounds represent acquisition, extinction, and unsignaled shock as in **Figure 2A**. The dashed vertical line separates the extinction 1 and extinction 2 phases. Freezing rate before CS presentation in test 1 (**C**), test 2 (**D**), and test 3 (**E**). The data are shown as median with interquartile range. Each black dot represents one animal. * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$ by one-way ANCOVA with age as covariate; # $p < 0.05$, ## $p < 0.01$, and ### $p < 0.001$ by Student's t -test comparing control and *App^{NL-G-F}* mice within the same age.

(A to E) Colors indicate different groups: orange represents 6-month-old control ($n = 24$), light blue represents 6-month-old *App^{NL-G-F}* mice ($n = 25$), pink represents 12-month-old control ($n = 24$), and dark blue represents 12-month-old *App^{NL-G-F}* mice ($n = 25$).

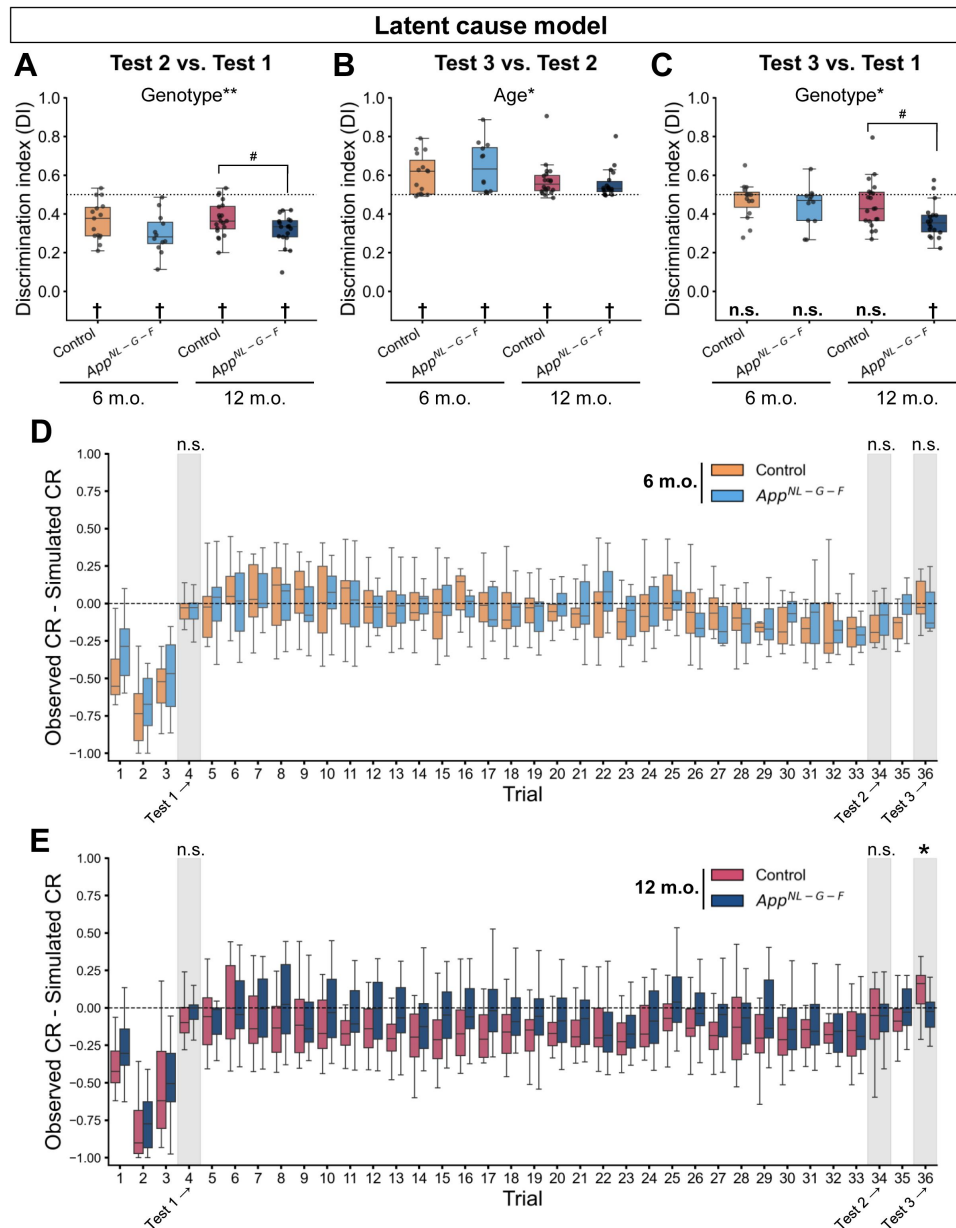


Figure 3 – figure supplement 1.

Replication of DI in the test phases and prediction error across trials in the latent cause model.

Discrimination index (DI) between test 2 and test 1 (**A**), between test 3 and test 2 (**B**), and between test 3 and test 1 (**C**) calculated from simulated CR. The signed prediction error between observed and simulated CR in the 6-month-old group (**D**) and 12-month-old group (**E**).

(**A to C**) Each black dot represents one animal. The dashed horizontal line indicates $DI = 0.5$, which means no discrimination between the two phases. $^{\dagger}p < 0.05$ by one-sample Student's t -test, and the alternative hypothesis specifies that the mean differs from 0.5; $*p < 0.05$, and $**p < 0.01$ by one-way ANCOVA with age as a covariate; $\#p < 0.05$ by Student's t -test comparing control and *App^{NL-G-F}* mice within the same age. Detailed statistical results are provided in Tables S5 and S6. (**D and E**) The gray areas indicate the test phases. $*p < 0.05$ by Mann-Whitney U test comparing control and *App^{NL-G-F}* mice within the same age. Statistical results are as follows: $U = 93.5$, $p = 0.883$ for test 1, $U = 110$, $p = 0.347$ for test 2, $U = 67$, $p = 0.277$ for test 3 in the 6-month-old group; $U = 229$, $p = 0.156$ for test 1, $U = 167$, $p = 0.718$ for test 2, $U = 101$, $p = 0.02$ for test 3 in the 12-month-old group. (**A to E**) Colors in the boxplot indicate different groups: orange represents 6-month-old control ($n = 15$), light blue represents 6-month-old *App^{NL-G-F}* mice ($n = 12$), pink represents 12-month-old control ($n = 20$), and dark blue represents 12-month-old *App^{NL-G-F}* mice ($n = 18$).

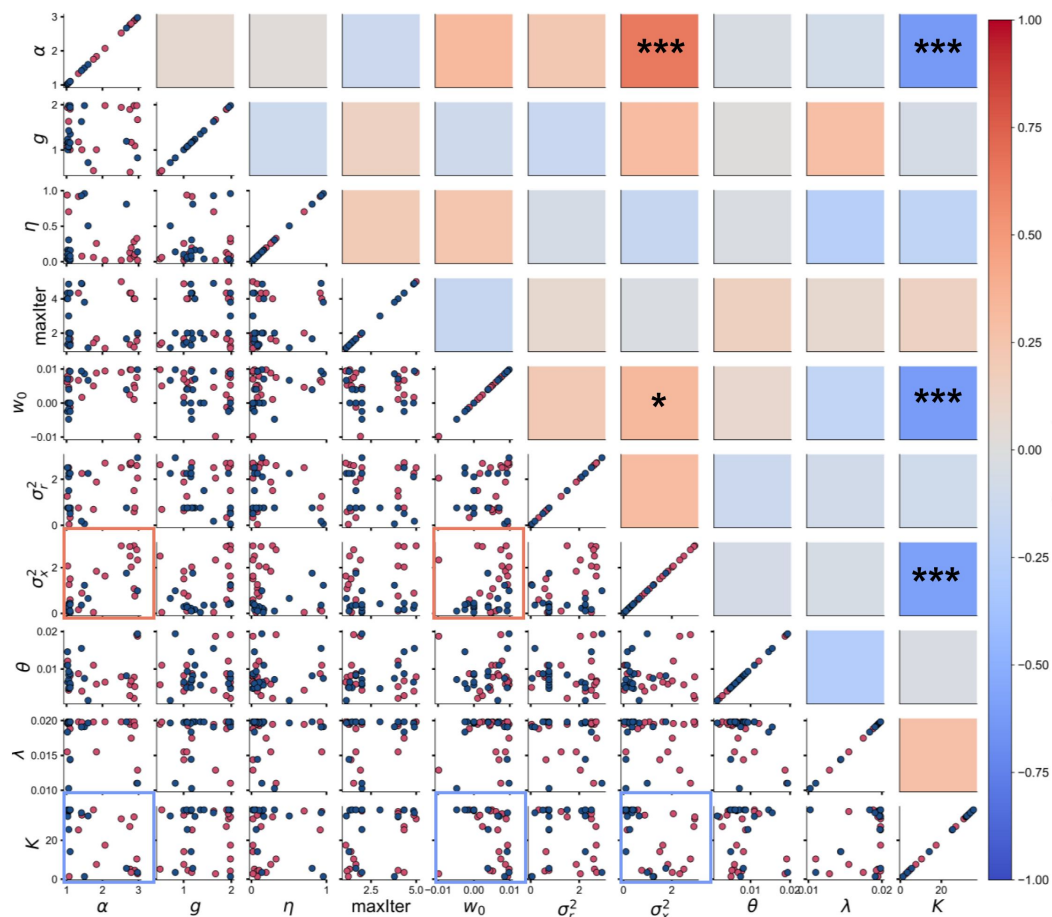


Figure 4 – figure supplement 1.

Correlation matrix of estimated parameters in the 12-month-old group.

The lower triangle shows the scatter plots of pairwise combinations of two parameters where the colors of the dot in the scatterplot indicate groups: pink represents 12-month-old control mice ($n = 20$), and dark blue represents 12-month-old *App*^{NL-G-F} mice ($n = 18$). The square outline highlights parameter pairs with statistically significant correlations. The upper triangle shows the heatmap of Spearman's correlation coefficient (ρ), where colors indicate the strength and direction of the correlation, as shown in the color bar. Significant correlations are annotated in the heatmap: * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$.

Figure 4 – figure supplement 2.

Correlations between DI and estimated parameters within the 12-month-old control group

(A) and *App^{NL-G-F}* group (B).

(A to B) Spearman's correlation coefficient (ρ) was labeled with significance, where $*p < 0.05$, $**p < 0.01$, and $***p < 0.001$.

The blue line represents the linear regression model fit, and the shaded area indicates the confidence interval. Each dot represents one animal and its color indicates groups: pink represents 12-month-old control ($n = 20$), and dark blue represents 12-month-old *App^{NL-G-F}* mice ($n = 18$).

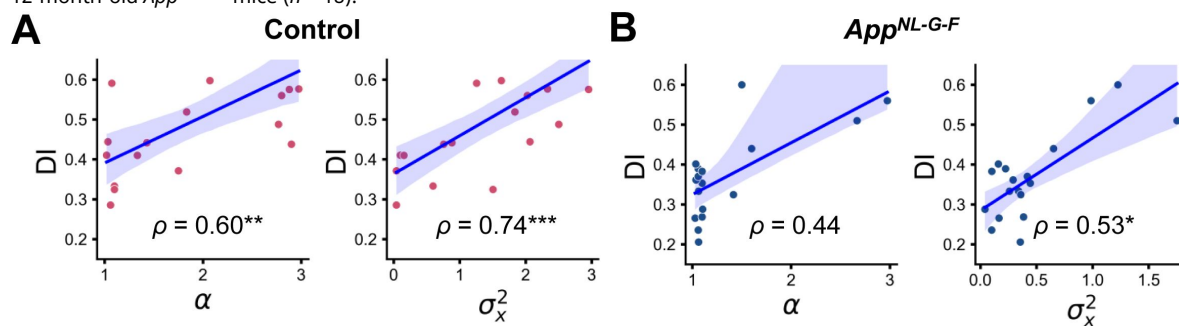


Figure 4 – figure supplement 3.

Simulated CR (A), and posterior probabilities (B), associative weights (C) of latent causes initially inferred during the acquisition trials in the 6-month-old group in test 1 phase.

(A) The data are shown as median with interquartile range. Each black dot represents one animal. Control (Mdn = 0.84, IQR = 0.44) vs. *App^{NL-G-F}* (Mdn = 1, IQR = 0.13), Mann-Whitney $U = 247$, $p = 0.041$. (B) The horizontal axis of the histogram indicates the proportion of the value in each group. Control (Mdn = 0.60, IQR = 0.10) vs. *App^{NL-G-F}* (Mdn = 1, IQR = 0), Mann-Whitney $U = 255$, $p = 0.008$. (C) Each dot represents one animal. Control (Mdn = 0.308, IQR = 0.482) vs. *App^{NL-G-F}* (Mdn = 0.373, IQR = 0.401), Mann-Whitney $U = 197$, $p = 0.633$ for w_{CS} ; Control (Mdn = 0.075, IQR = 0.088) vs. *App^{NL-G-F}* (Mdn = 0.077, IQR = 0.091), Mann-Whitney $U = 179$, $p = 0.988$ for $w_{context}$. (A to C) Colors in the bar plot or boxplot indicate groups: pink represents 12-month-old control ($n = 20$), and dark blue represents 12-month-old *App^{NL-G-F}* mice ($n = 18$).

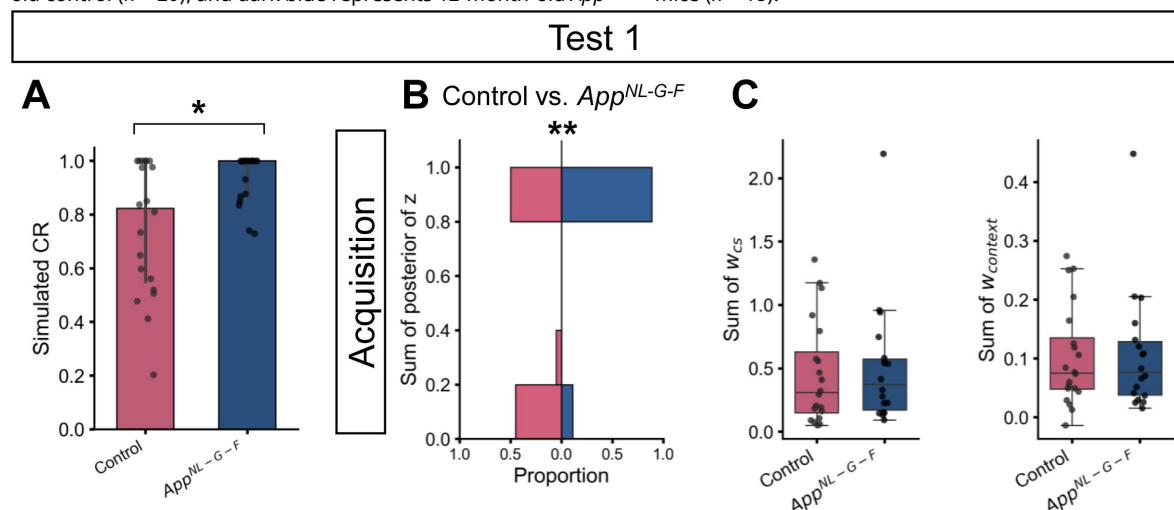


Figure 5 – figure supplement 1.

Correlation matrix of estimated parameters in the 6-month-old group.

The figure format is the same as in [Figure 4-figure supplement 1](#), except that the colors of the dots in the scatterplot indicate groups: orange represents 6-month-old control ($n = 15$), and light blue represents 6-month-old *App^{NL-G-F}* mice ($n = 12$). Significant correlations are annotated in the heatmap: $*p < 0.05$, $*p < 0.01$, and $***p < 0.001$.

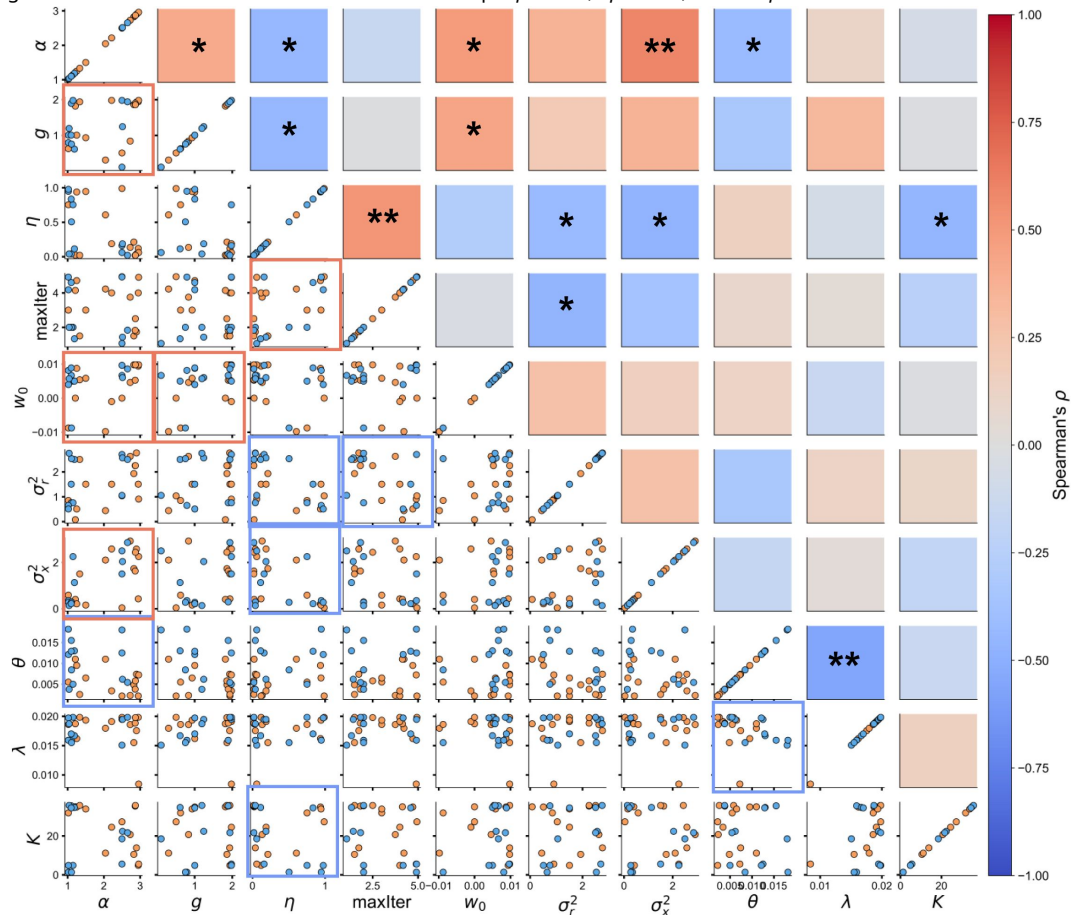
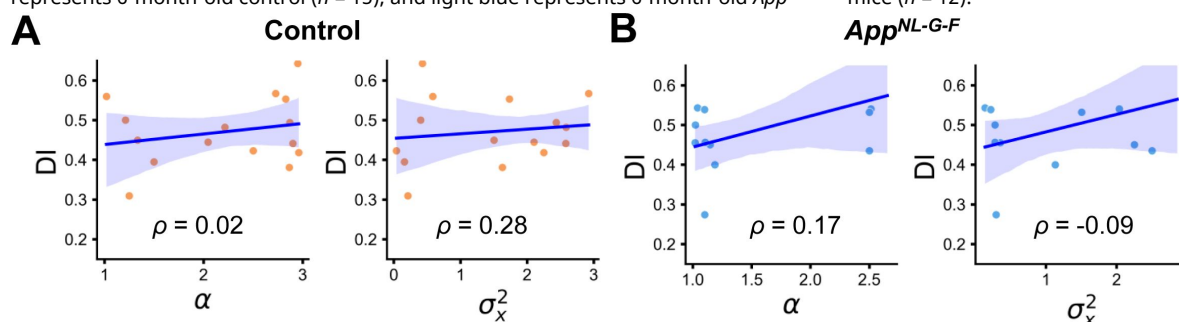


Figure 5 – figure supplement 2.

Correlation between DI and selected parameters in 6-month-old control (A) and *App^{NL-G-F}* mice (B).

The figure format is the same as in [Figure 4-figure supplement 2](#), the colors of the dots indicate groups: orange represents 6-month-old control ($n = 15$), and light blue represents 6-month-old *App^{NL-G-F}* mice ($n = 12$).



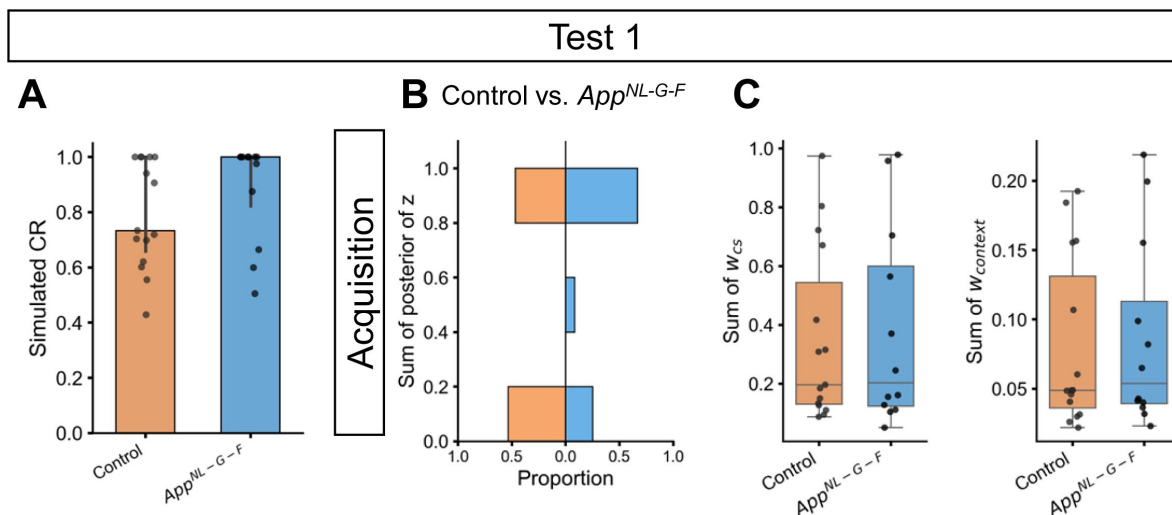


Figure 5 – figure supplement 3.

Simulated CR (A), and posterior probabilities (B), associative weights (C) of latent causes initially inferred during the acquisition trials in the 6-month-old group in test 1 phase.

The figure format is the same as in [Figure 4–figure supplement 3](#). **(A)** The data are shown as median with interquartile range. Each black dot represents one animal. Control (Mdn = 0.73, IQR = 0.34) vs. *App^{NL-G-F}* (Mdn = 1, IQR = 0.18), Mann-Whitney $U = 110$, $p = 0.325$. **(B)** The horizontal axis of the histogram indicates the proportion of the value in each group. Control (Mdn = 0.08, IQR = 0.10) vs. *App^{NL-G-F}* (Mdn = 1, IQR = 0.62), Mann-Whitney $U = 112$, $p = 0.249$. **(C)** Each dot represents one animal. Control (Mdn = 0.196, IQR = 0.414) vs. *App^{NL-G-F}* (Mdn = 0.203, IQR = 0.476), Mann-Whitney $U = 91$, $p = 0.981$ for w_{CS} ; Control (Mdn = 0.049, IQR = 0.095) vs. *App^{NL-G-F}* (Mdn = 0.054, IQR = 0.074), Mann-Whitney $U = 94$, $p = 0.867$ for $w_{context}$. **(A to C)** Colors in the bar plot or boxplot indicate groups: orange represents 6-month-old control ($n = 15$), and light blue represents 6-month-old *App^{NL-G-F}* mice ($n = 12$).

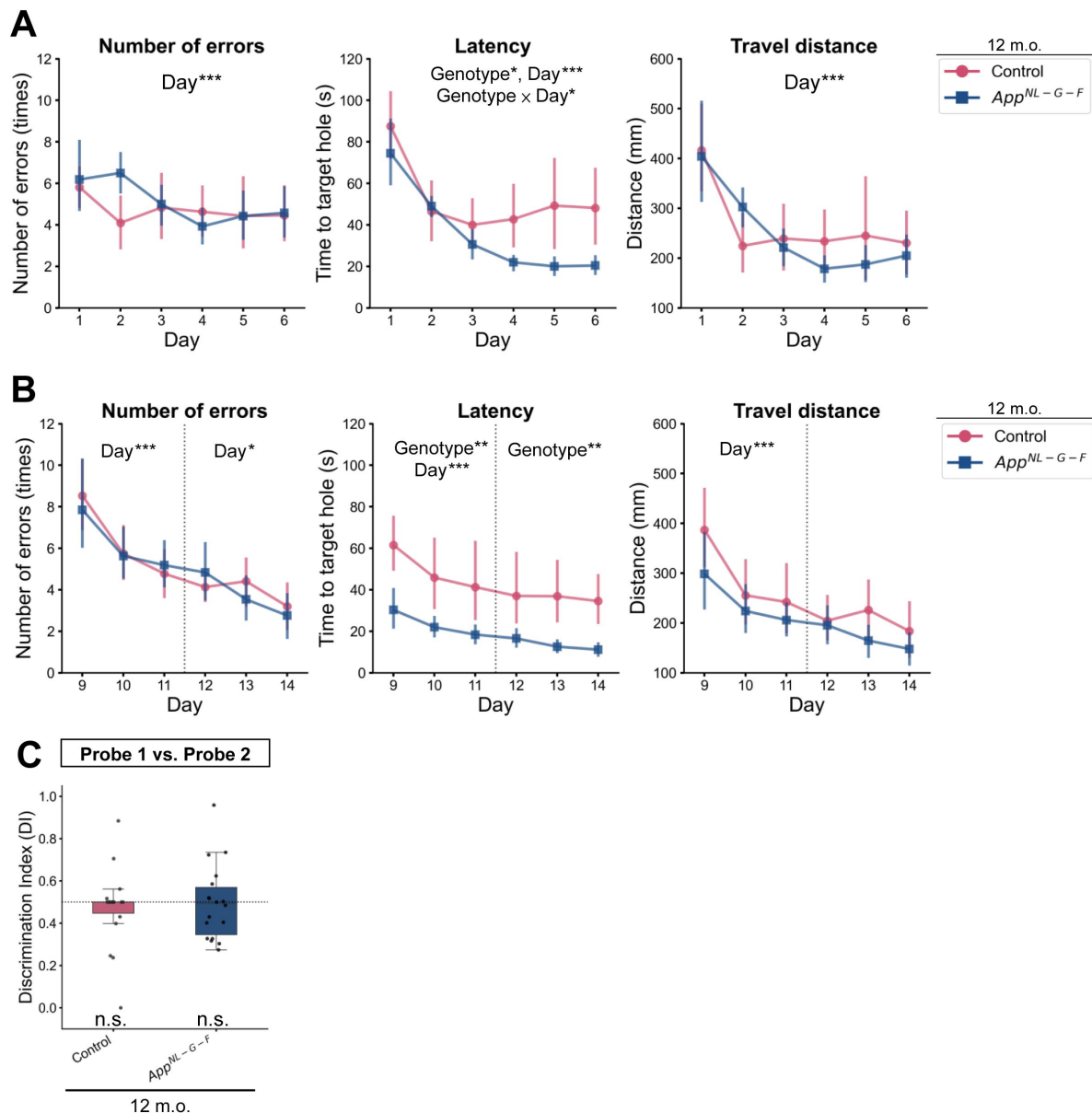


Figure 6 – figure supplement 1.

Conventional analysis for initial

(A) and reversal (B) training day in the Barnes maze task and discrimination index of the first target hole between probe test 2 and probe test 1 (C) in the 12-month-old group. Colors indicate different groups: pink represents 12-month-old control ($n = 18$), and dark blue represents 12-month-old *App^{NL-G-F}* mice ($n = 18$).

(A and B) The left, middle, and right panels show the number of errors, latency, and travel distance during training days. The data are shown as mean with 95% confidence interval. * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$ by mixed-design two-way [Genotype (control, *App^{NL-G-F}*) × Day (1–6)] ANOVA. Detailed statistical results are provided in Table S13. (C) Each black dot represents one animal. The dashed horizontal line indicates DI = 0.5, which means no discrimination between the two phases. † $p < 0.05$ by the Wilcoxon signed-rank test, and the alternative hypothesis specifies that the mean differs from 0.5; * $p < 0.05$ by Mann-Whitney U test comparing control and *App^{NL-G-F}* mice. Detailed statistical results are provided in Table S16.

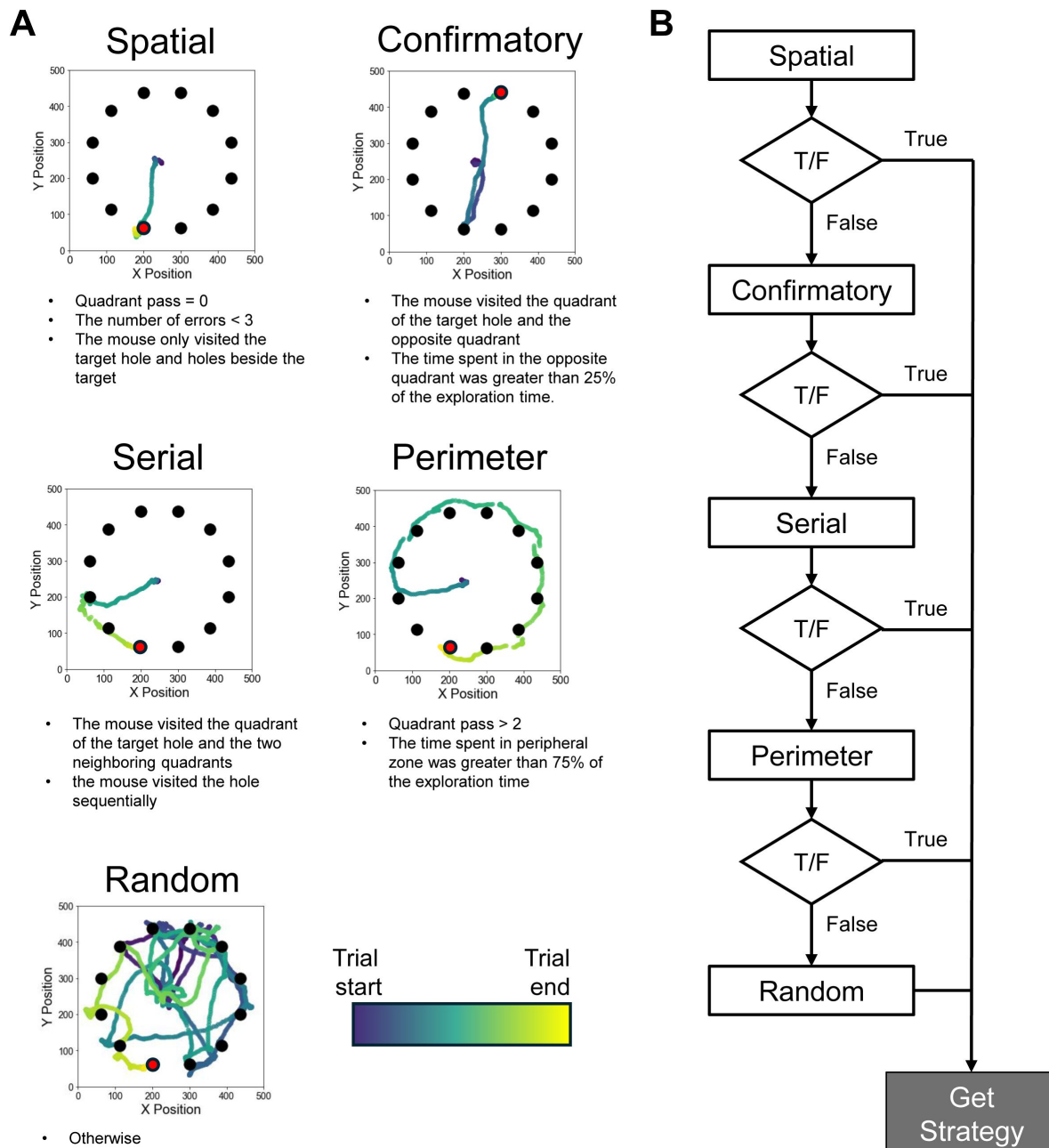


Figure 6 – figure supplement 2.

(A) Animal trajectory examples and definition of each strategy. The red solid circle indicates the target hole, and the black solid circle indicates other holes in the field. The dotted line represents the animal's trajectory, and the color reflects the time in the trial. **(B)** The algorithm of strategy analysis.

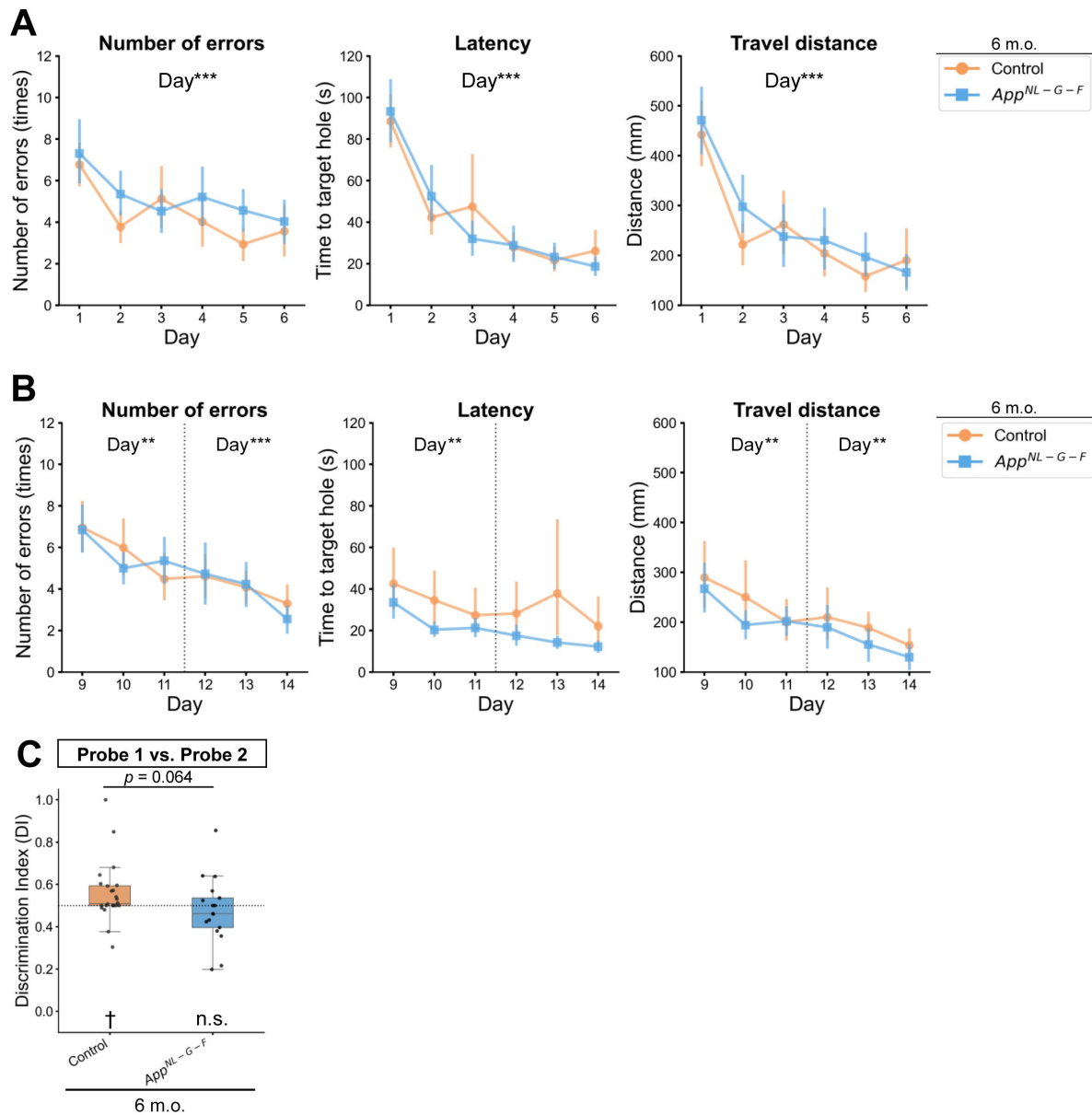


Figure 6 - figure supplement 3.

Conventional analysis for initial

(A) and reversal (B) training day in the Barnes maze task and discrimination index of the first target hole between probe test 2 and probe test 1 (C) in the 6-month-old group. The configuration of the figure is the same as in **Figure 6-figure supplement 1**. Colors indicate the different groups: orange represents 6-month-old control ($n = 24$), and light blue represents 6-month-old App^{NL-G-F} mice ($n = 17$). Detailed statistical results are provided in Table S17 for panels A and B; Table S20 for panel C.

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Additional information

Author Contributions

M.H.: Writing – original draft, Conceptualization, Formal analysis, Investigation, Methodology, Visualization

Y.S.: Writing – review & editing, Conceptualization, Formal analysis, Methodology, Software, Supervision

H.S.: Writing – review & editing, Resource

T.S.: Writing – review & editing, Resource

T.C.S.: Resource

I.I.: Writing – review & editing, Funding acquisition, Resource, Supervision

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Additional files

Appendix1 [↗](#)

Appendix2 [↗](#)

Supplementary tables [↗](#)

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Reviewer #1 (Public review):

I applaud the authors' for providing a thorough response to my comments from the first round of review. The authors' have addressed the points I raised on the interpretation of the behavioral results as well as the validation of the model (fit to the data) by conducting new analyses, acknowledging the limitations where required and providing important counterpoints. As a result of this process, the manuscript has considerably improved. I have no further comments and recommend this manuscript for publication.

<https://doi.org/10.7554/eLife.105347.2.sa3>

Reviewer #2 (Public review):**Summary:**

This manuscript proposes that the use of a latent cause model for assessment of memory-based tasks may provide improved early detection in Alzheimer's Disease as well as more differentiated mapping of behavior to underlying causes. To test the validity of this model, the authors use a previously described knock-in mouse model of AD and subject the mice to several behaviors to determine whether the latent cause model may provide informative predictions regarding changes in the observed behaviors. They include a well-established fear learning paradigm in which distinct memories are believed to compete for control of behavior. More specifically, it's been observed that animals undergoing fear learning and subsequent fear extinction develop two separate memories for the acquisition phase and the extinction phase, such that the extinction does not simply 'erase' the previously acquired memory. Many models of learning require the addition of a separate context or state to be added during the extinction phase and are typically modeled by assuming the existence of a new state at the time of extinction. The Niv research group, Gershman et al. 2017, have shown that the use of a latent cause model applied to this behavior can elegantly predict the formation of latent states based on a Bayesian approach, and that these latent states can facilitate the persistence of the acquisition and extinction memory independently. The authors of this manuscript leverage this approach to test whether deficits in production of the internal states, or the inference and learning of those states, may be disrupted in knock-in mice that show both a build-up of amyloid-beta plaques and a deterioration in memory as the mice age.

Strengths:

I think the authors' proposal to leverage the latent cause model and test whether it can lead to improved assessments in an animal model of AD is a promising approach for bridging the gap between clinical and basic research. The authors use a promising mouse model and apply this to a paradigm in which the behavior and neurobiology are relatively well understood - an ideal situation for assessing how a disease state may impact both the neurobiology and behavior. The latent cause model has the potential to better connect observed behavior to underlying causes and may pave a road for improved mapping of changes in behavior to neurobiological mechanisms in diseases such as AD. The authors also compare the latent cause model to the Rescorla-Wagner model and a latent state model allowing for better assessment of the latent cause model as a strong model for assessing reinstatement.

Weaknesses:

I have several substantial concerns which I've detailed below. These include important details on how the behavior was analyzed, how the model was used to assess the behavior, and the interpretations that have been made based on the model.

(1) There is substantial data to suggest that during fear learning in mice separate memories develop for the acquisition and extinction phases, with the acquisition memory becoming more strongly retrieved during spontaneous recovery and reinstatement. The Gershman paper, cited by the authors, shows how the latent causal model can predict this shift in latent causes by allowing for the priors to decay over time, thereby increasing the posterior of the acquisition memory at the time of spontaneous recovery. In this manuscript, the authors suggest a similar mechanism of action for reinstatement, yet the model does not appear to return to the acquisition memory after reinstatement, at least based on the simulation and examples shown in figures 1 and 3. More specifically, in figure 1, the authors indicate that the posterior probability of the latent cause, z_A (the putative acquisition memory), increases, partially leading to reinstatement. This does not appear to be the case as test 3 (day 36) appears to have similar posterior probabilities for z_A as well as similar weights for the CS as compared to the last days of extinction. Rather, the model appears to mainly modify the weights in the most recent latent cause, z_B - the putative the 'extinction state', during reinstatement. The authors suggest that previous experimental data have indicated that spontaneous recovery or reinstatement effects are due to an interaction of the acquisition and extinction memory. These studies have shown that conditioned responding at a later time point after extinction is likely due to a balance between the acquisition memory and the extinction memory, and that this balance can shift towards the acquisition memory naturally during spontaneous recovery, or through artificial activation of the acquisition memory or inhibition of the extinction memory (see Lacagnina et al. for example). Here the authors show that the same latent cause learned during extinction, z_B , appears to dominate during the learning phase of reinstatement, with rapid learning to the context - the weight for the context goes up substantially on day 35 - in z_B . This latent cause, z_B , dominates at the reinstatement test, and due to the increased associative strength between the context and shock, there is a strong CR. For the simulation shown in figure 1, it's not clear why a latent cause model is necessary for this behavior. This leads to the next point.

(2) The authors compared the latent cause model to the Rescorla-Wagner model. This is very commendable, particularly since the latent cause model builds upon the RW model, so it can serve as an ideal test for whether a more simplified model can adequately predict the behavior. The authors show that the RW model cannot successfully predict the increased CR during reinstatement (Appendix figure 1). Yet there are some issues with the way the authors have implemented this comparison:

(2A) The RW model is a simplified version of the latent cause model and so should be treated as a nested model when testing, or at a minimum, the number of parameters should be taken into account when comparing the models using a method such as the Bayesian Information Criterion, BIC.

(2B) The RW model provides the associative strength between stimuli and does not necessarily require a linear relationship between V and the CR. This is the case in the original RW model as well as in the LCM. To allow for better comparison between the models, the authors should be modeling the CR in the same manner (using the same probit function) in both models. In fact, there are many instances in which a sigmoid has been applied to RW associative strengths to predict CRs. I would recommend modeling CRs in the RW as if there is just one latent cause. Or perhaps run the analysis for the LCM with just one latent cause - this would effectively reduce the LCM to RW and keep any other assumptions identical across the models.

(2C) In the paper, the model fits for the alphas in the RW model are the same across the groups. Were the alphas for the two models kept as free variables? This is an important question as it gets back to the first point raised. Because the modeling of the reinstatement behavior with the LCM appears to be mainly driven by latent cause z_B , the extinction memory, it may be possible to replicate the pattern of results without requiring a latent cause model. For example, the 12-month-old App NL-G-F mice behavior may have a deficit in learning about the context. Within the RW model, if the alpha for context is set to zero for

those mice, but kept higher for the other groups, say $\alpha_{\text{context}} = 0.8$, the authors could potentially observe the same pattern of discrimination indices in figure 2G and 2H at test. Because the authors don't explicitly state which parameters might be driving the change in the DI, the authors should show in some way that their results cannot simply be due to poor contextual learning in the 12 month old App NL-G-F mice, as this can presumably be predicted by the RW model. The authors' model fits using RW don't show this, but this is because they don't consider this possibility that the alpha for context might be disrupted in the 12-month-old App NL-G-F mice. Of course, using the RW model with these alphas won't lead to as nice of fits of the behavior across acquisition, extinction, and reinstatement as the authors' LCM, the number of parameters are substantially reduced in the RW model. Yet the important pattern of the DI would be replicated with the RW model (if I'm not mistaken), which is the important test for assessment of reinstatement.

(3) As stated by the authors in the introduction, the advantage of the fear learning approach is that the memory is modified across the acquisition-extinction-reinstatement phases. Although perhaps not explicitly stated by the authors, the post-reinstatement test (test 3) is the crucial test for whether there is reactivation of a previously stored memory, with the general argument being that the reinvigorated response to the CS can't simply be explained by relearning the CS-US pairing, because re-exposure the US alone leads to increase response to the CS at test. Of course there are several explanations for why this may occur, particularly when also considering the context as a stimulus. This is what I understood to be the justification for the use of a model, such as the latent cause model, that may better capture and compare these possibilities within a single framework. As such, it is critical to look at the level of responding to both the context alone and to the CS. It appears that the authors only look at the percent freezing during the CS, and it is not clear whether this is due to the contextual-US learning during the US re-exposure or to increased responding to the CS - presumably caused by reactivation of the acquisition memory. The authors do perform a comparison between the preCS and CS period, but it is not clear whether this is taken into account in the LCM. For example, the instance of the model shown in figure 1 indicates that the 'extinction cause', or cause z6, develops a strong weight for the context during the reinstatement phase of presenting the shock alone. This state then leads to increased freezing during the final CS probe test as shown in the figure. If they haven't already, I think the authors must somehow incorporate these different phases (CS vs ITI) into their model, particularly since this type of memory retrieval that depends on assessing latent states is specifically why the authors justified using the latent causal model. In more precise terms, it's not clear whether the authors incorporate a preCS/ITI period each day the cue is presented as a vector of just the context in addition to the CS period in which the vector contains both the context and the CS. Based on the description, it seemed to me that they only model the CRs during the CS period on days when the CS is presented, and thereby the context is only ever modeled on its own (as just the context by itself in the vector) on extinction days when the CS is not presented. If they are modeling both timepoints each day that the CS is presented, then I would recommend explicitly stating this in the methods section.

(4) The authors fit the model using all data points across acquisition and learning. As one of the other reviewers has highlighted, it appears that there is a high chance for overfitting the data with the LCM. Of course, this would result in much better fits than models with substantially fewer free parameters, such as the RW model. As mentioned above, the authors should use a method that takes into account the number of parameters, such as the BIC.

(5) The authors have stated that they do not think the Barnes maze task can be modeled with the LCM. Whether or not this is the case, if the authors do not model this data with the LCM, the Barnes maze data doesn't appear valuable to the main hypothesis. The authors suggest that more sophisticated models such as the LCM may be beneficial for early detection of diseases such as Alzheimer's, so the Barnes maze data is not valuable for providing evidence of this hypothesis. Rather, the authors make an argument that the memory deficits in the

Barnes maze mimic the reinstatement effects providing support that memory is disrupted similarly in these mice. Although, the authors state that the deficits in memory retrieval are similar across the two tasks, the authors are not explicit as to the precise deficits in memory retrieval in the reinstatement task - it's a combination of overgeneralizing latent causes during acquisition, poor learning rate, over differentiation of the stimuli.

<https://doi.org/10.7554/eLife.105347.2.sa2>

Reviewer #3 (Public review):

Summary:

This paper seeks to identify underlying mechanisms contributing to memory deficits observed in Alzheimer's disease (AD) mouse models. By understanding these mechanisms, they hope to uncover insights into subtle cognitive changes early in AD to inform interventions for early-stage decline.

Strengths:

The paper provides a comprehensive exploration of memory deficits in an AD mouse model, covering early and late stages of the disease. The experimental design was robust, confirming age-dependent increases in A β plaque accumulation in the AD model mice and using multiple behavior tasks that collectively highlighted difficulties in maintaining multiple competing memory cues, with deficits most pronounced in older mice.

In the fear acquisition, extinction, and reinstatement task, AD model mice exhibited a significantly higher fear response after acquisition compared to controls, as well as a greater drop in fear response during reinstatement. These findings suggest that AD mice struggle to retain the fear memory associated with the conditioned stimulus, with the group differences being more pronounced in the older mice.

In the reversal Barnes maze task, the AD model mice displayed a tendency to explore the maze perimeter rather than the two potential target holes, indicating a failure to integrate multiple memory cues into their strategy. This contrasted with the control mice, which used the more confirmatory strategy of focusing on the two target holes. Despite this, the AD mice were quicker to reach the target hole, suggesting that their impairments were specific to memory retrieval rather than basic task performance.

The authors strengthened their findings by analyzing their data with a leading computational model, which describes how animals balance competing memories. They found that AD mice showed somewhat of a contradiction: a tendency to both treat trials as more alike than they are (lower α) and similar stimuli as more distinct than they are (lower σ_x) compared to controls.

Weaknesses:

While conceptually solid, the model struggles to fit the data and to support the key hypothesis about AD mice's inability to retain competing memories. These issues are evident in Figure 3:

(1) The model misses trends in the data, including the gradual learning of fear in all groups during acquisition, the absence of a fear response at the start of the experiment, and the faster return of fear during reinstatement compared to the gradual learning of fear during acquisition. It also underestimates the increase in fear at the start of day 2 of extinction, particularly in controls.

(2) The model explains the higher fear response in controls during reinstatement largely through a stronger association to the context formed during the unsignaled shock phase,

rather than to any memory of the conditioned stimulus from acquisition (as seen in Figure 3C). In the experiment, however, this memory does seem to be important for explaining the higher fear response in controls during reinstatement (as seen in Author Response Figure 3). The model does show a necessary condition for memory retrieval, which is that controls rely more on the latent causes from acquisition. But this alone is not sufficient, since the associations within that cause may have been overwritten during extinction. The Rescorla-Wagner model illustrates this point: it too uses the latent cause from acquisition (as it only ever uses a single cause across phases) but does not retain the original stimulus-shock memory, updating and overwriting it continuously. Similarly, the latent cause model may reuse a cause from acquisition without preserving its original stimulus-shock association.

These issues lead to potential overinterpretation of the model parameters. The differences in α and σ_x are being used to make claims about cognitive processes (e.g., overgeneralization vs. over differentiation), but the model itself does not appear to capture these processes accurately.

The authors could benefit from a model that better matches the data and captures the retention and retrieval of fear memories across phases. While they explored alternatives, including the Rescorla-Wagner model and a latent state model, these showed no meaningful improvement in fit. This highlights a broader issue: these models are well-motivated but may not fully capture observed behavior.

Conclusion:

Overall, the data support the authors' hypothesis that AD model mice struggle to retain competing memories, with the effect becoming more pronounced with age. While I believe the right computational model could highlight these differences, the current models fall short in doing so.

<https://doi.org/10.7554/eLife.105347.2.sa1>

Author response:

The following is the authors' response to the current reviews.

Public Reviews:

Reviewer #1 (Public review):

I applaud the authors' for providing a thorough response to my comments from the first round of review. The authors' have addressed the points I raised on the interpretation of the behavioral results as well as the validation of the model (fit to the data) by conducting new analyses, acknowledging the limitations where required and providing important counterpoints. As a result of this process, the manuscript has considerably improved. I have no further comments and recommend this manuscript for publication.

We are pleased that our revisions have addressed all the concerns raised by Reviewer #1.

Reviewer #2 (Public review):

Summary:

This manuscript proposes that the use of a latent cause model for assessment of memory-based tasks may provide improved early detection in Alzheimer's Disease as well as more differentiated mapping of behavior to underlying causes. To test the validity of this model, the authors use a previously described knock-in mouse model of AD and

subject the mice to several behaviors to determine whether the latent cause model may provide informative predictions regarding changes in the observed behaviors. They include a well-established fear learning paradigm in which distinct memories are believed to compete for control of behavior. More specifically, it's been observed that animals undergoing fear learning and subsequent fear extinction develop two separate memories for the acquisition phase and the extinction phase, such that the extinction does not simply 'erase' the previously acquired memory. Many models of learning require the addition of a separate context or state to be added during the extinction phase and are typically modeled by assuming the existence of a new state at the time of extinction. The Niv research group, Gershman et al. 2017, have shown that the use of a latent cause model applied to this behavior can elegantly predict the formation of latent states based on a Bayesian approach, and that these latent states can facilitate the persistence of the acquisition and extinction memory independently. The authors of this manuscript leverage this approach to test whether deficits in production of the internal states, or the inference and learning of those states, may be disrupted in knock-in mice that show both a build-up of amyloid-beta plaques and a deterioration in memory as the mice age.

Strengths:

I think the authors' proposal to leverage the latent cause model and test whether it can lead to improved assessments in an animal model of AD is a promising approach for bridging the gap between clinical and basic research. The authors use a promising mouse model and apply this to a paradigm in which the behavior and neurobiology are relatively well understood - an ideal situation for assessing how a disease state may impact both the neurobiology and behavior. The latent cause model has the potential to better connect observed behavior to underlying causes and may pave a road for improved mapping of changes in behavior to neurobiological mechanisms in diseases such as AD.

The authors also compare the latent cause model to the Rescorla-Wagner model and a latent state model allowing for better assessment of the latent cause model as a strong model for assessing reinstatement.

Weaknesses:

I have several substantial concerns which I've detailed below. These include important details on how the behavior was analyzed, how the model was used to assess the behavior, and the interpretations that have been made based on the model.

(1) There is substantial data to suggest that during fear learning in mice separate memories develop for the acquisition and extinction phases, with the acquisition memory becoming more strongly retrieved during spontaneous recovery and reinstatement. The Gershman paper, cited by the authors, shows how the latent causal model can predict this shift in latent causes by allowing for the priors to decay over time, thereby increasing the posterior of the acquisition memory at the time of spontaneous recovery. In this manuscript, the authors suggest a similar mechanism of action for reinstatement, yet the model does not appear to return to the acquisition memory after reinstatement, at least based on the simulation and examples shown in figures 1 and 3. More specifically, in figure 1, the authors indicate that the posterior probability of the latent cause, z_A (the putative acquisition memory), increases, partially leading to reinstatement. This does not appear to be the case as test 3 (day 36) appears to have similar posterior probabilities for z_A as well as similar weights for the CS as compared to the last days of extinction. Rather, the model appears to mainly modify the weights in the most recent latent cause, z_B - the putative the 'extinction state', during reinstatement. The authors suggest that previous experimental data have indicated that spontaneous recovery or reinstatement

effects are due to an interaction of the acquisition and extinction memory. These studies have shown that conditioned responding at a later time point after extinction is likely due to a balance between the acquisition memory and the extinction memory, and that this balance can shift towards the acquisition memory naturally during spontaneous recovery, or through artificial activation of the acquisition memory or inhibition of the extinction memory (see Lacagnina et al. for example). Here the authors show that the same latent cause learned during extinction, z_B , appears to dominate during the learning phase of reinstatement, with rapid learning to the context - the weight for the context goes up substantially on day 35 - in z_B . This latent cause, z_B , dominates at the reinstatement test, and due to the increased associative strength between the context and shock, there is a strong CR. For the simulation shown in figure 1, it's not clear why a latent cause model is necessary for this behavior. This leads to the next point.

We would like to first clarify that our behavioral paradigm did not last for 36 days, as noted by the reviewer. Our reinstatement paradigm contained 7 phases and 36 trials in total: acquisition (3 trials), test 1 (1 trial), extinction 1 (19 trials), extinction 2 (10 trials), test 2 (1 trial), unsignaled shock (1 trial), test 3 (1 trial). The day is labeled under each phase in Figure 2A.

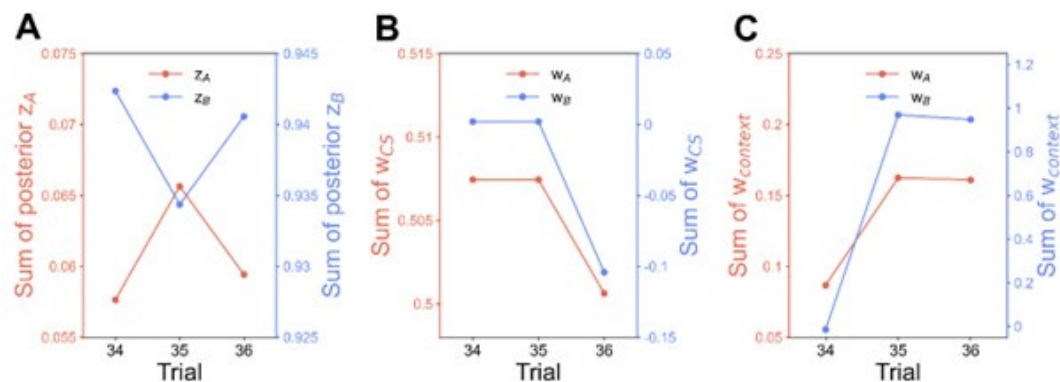
We have provided explanations on how the reinstatement is explained by the latent cause model in the first round of the review. Briefly, both acquisition and extinction latent causes contribute to the reinstatement (test 3). The former retains the acquisition fear memory, and the latter has the updated $w_{context}$ from unsignaled shock. Although the reviewer is correct that the z_B in Figure 1D makes a great contribution during the reinstatement, we would like to argue that the elevated CR from test 2 (trial 34) to test 3 (trial 36) is the result of the interaction between z_A and z_B .

We provided Author response image 1 using the same data in Figure 1D and 1E to further clarify this point. The posterior probability of z_A increased after an unsignaled shock (trial 35), which may be attributed to the return of acquisition fear memory. The posterior probability of z_A then decreased again after test 3 (trial 36) because there was no shock in this trial. Along with the weight change, the expected shock change substantially in these three trials, resulting in reinstatement. Note that the mapping of expected shock to CR in the latent cause model is controlled by parameter θ and λ . Once the expected shock exceeds the threshold θ , the CR will increase rapidly if λ is smaller.

Lastly, accepting the idea that separate memories are responsible for acquisition and extinction in the memory modification paradigm, the latent cause model (LCM) is a rational candidate modeling this idea. Please see the following reply on why a simple model like the Rescorla-Wagner (RW) model is not sufficient to fully explain the behaviors observed in this study.

Author response image 1.

The sum posterior probability (A), the sum of associative weight of CS (B), and the sum of associative weight of context (C) of acquisition and extinction latent causes in Figure 1D and 1E.



(2) The authors compared the latent cause model to the Rescorla-Wagner model. This is very commendable, particularly since the latent cause model builds upon the RW model, so it can serve as an ideal test for whether a more simplified model can adequately predict the behavior. The authors show that the RW model cannot successfully predict the increased CR during reinstatement (Appendix figure 1). Yet there are some issues with the way the authors have implemented this comparison:

(2A) The RW model is a simplified version of the latent cause model and so should be treated as a nested model when testing, or at a minimum, the number of parameters should be taken into account when comparing the models using a method such as the Bayesian Information Criterion, BIC.

We acknowledge that the number of parameters was not taken into consideration when we compared the models. We thank the reviewer for the suggestion to use the Bayesian Information Criterion (BIC). However, we did not use BIC in this study for the following reasons. We wanted a model that can explain fear conditioning, extinction and reinstatement, so our first priority is to fit the test phases. Models that simulate CRs well in non-test phases can yield lower BIC values even if they fail to capture reinstatement. When we calculate the BIC by using the half normal distribution ($\mu = 0$, $\sigma = 0.3$) as the likelihood for prediction error in each trial, the BIC of the 12-month-old control is -37.21 for the RW model (Appendix 1–figure 1C) and -11.60 for the LCM (Figure 3C). Based on this result, the RW model would be preferred, yet the LCM was penalized by the number of parameters, even though it fit better in trial 36. Because we did not think this aligned with our purpose to model reinstatement, we chose to rely on the practical criteria to determine whether the estimated parameter set is accepted or not for our purpose (see Materials and Methods). The number of accepted samples can thus roughly be seen as the model's ability to explain the data in this study. These exclusion criteria then created imbalances in accepted samples across models (Appendix 1–figure 2). In the RW model, only one or two samples met the criteria, preventing meaningful statistical comparisons of BIC within each group. Overall, though we agreed that BIC is one of the reasonable metrics in model comparison, we did not think it aligns with our purpose in this study.

(2B) The RW model provides the associative strength between stimuli and does not necessarily require a linear relationship between V and the CR. This is the case in the original RW model as well as in the LCM. To allow for better comparison between the models, the authors should be modeling the CR in the same manner (using the same probit function) in both models. In fact, there are many instances in which a sigmoid has been applied to RW associative strengths to predict CRs. I would recommend modeling CRs in the RW as if there is just one latent cause. Or perhaps run the analysis for the LCM with just one latent cause - this would effectively reduce the LCM to RW and keep any other assumptions identical across the models.

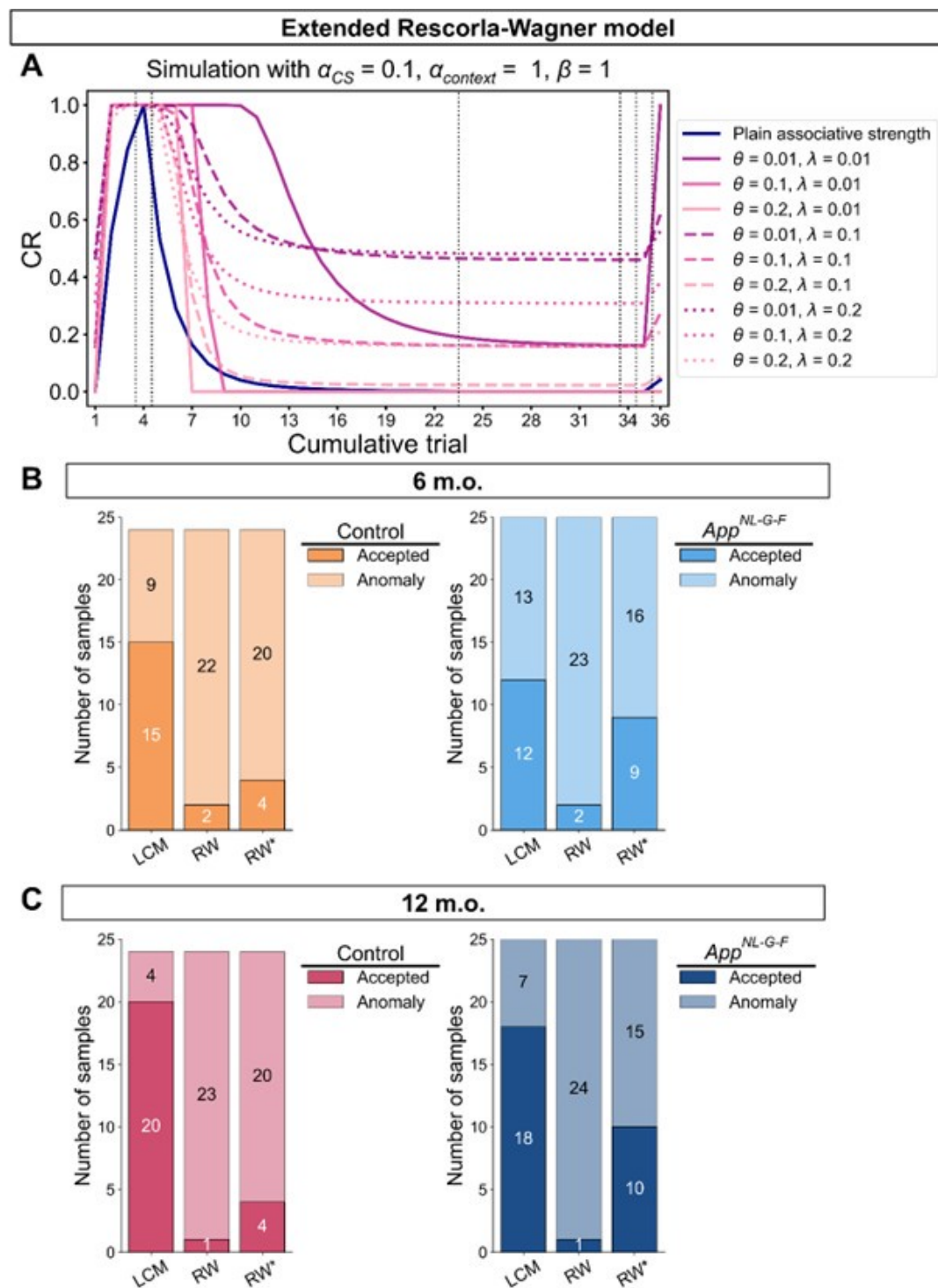
Regarding the suggestion to run the analysis using the LCM with one latent cause, we agree that this method is almost identical to the RW model, which is also mentioned in the original paper (Gershman et al., 2017). Importantly, it would also eliminate the RW model's advantage of assigning distinct learning rates to different stimuli, highlighted in the next comment (2C).

We thank the reviewer for suggesting applying the transformation of associative strength (V) to CR as in the LCM. We examined this possibility by heuristically selecting parameter values to test how such a transformation would influence the RW model (Author response image 2A). Specifically, we set $\alpha_{CS} = 0.5$, $\alpha_{context} = 1$, $\beta = 1$, and introduced the additional parameters θ and λ , as in the LCM. This parameter set is determined heuristically to address the reviewer's concern about a higher learning rate of context. The dark blue line is the plain associative strength. The remaining lines are CR curves under different combinations of θ and λ .

Consistent with the reviewer's comment, under certain parameter settings ($\theta = 0.01$, $\lambda = 0.01$), the extended RW model can reproduce higher CRs at test 3, thereby approximating the discrimination index observed in the 12-month-old control group. However, this modification changes the characteristics of CRs in other phases from those in the plain RW model. In the acquisition phase, the CRs rise more sharply. In the extinction phase, the CRs remain high when θ is small. Though changing λ can modulate the steepness, the CR curve is flat on the second day of the extinction phase, which does not reproduce the pattern in observed data (Figure 2B). These trade-offs suggest that the RW model with the sigmoid transformation does not improve fit quality and, in fact, sacrifices features that were well captured by simpler RW simulations (Appendix 1—figure 1A to 1D). To further evaluate this extended RW model (RW*), we applied the same parameter estimation method used in the LCM for individual data (see Materials and Methods). For each animal, α_{CS} , $\alpha_{context}$, β , θ , and λ were estimated with their lower and upper bounds set as previously described (see Appendix 1, Materials and Methods). The results showed that the number of accepted samples slightly increased compared to the RW model without sigmoidal transformation of CR (RW* vs. RW in Author response image 2B, 2C). However, this improvement did not surpass the LCM (RW* vs. LCM in Author response image 2B, Author response image 1C). Overall, these results suggest that while using the same method to map the expected shock to CR, the RW model does not outperform the LCM. Practically, further extension, such as adding novel terms, might improve the fitting level. We would like to note that such extensions should be carefully validated if they are reasonable and necessary for an internal model, which is beyond the scope of this study. We hope this addresses the reviewer's concerns about the implementation of the RW model.

Author response image 2.

Simulation (A) and parameter estimation (B and C) in the extended Rescorla-Wagner model.



(2C) In the paper, the model fits for the alphas in the RW model are the same across the groups. Were the alphas for the two models kept as free variables? This is an important question as it gets back to the first point raised. Because the modeling of the reinstatement behavior with the LCM appears to be mainly driven by latent cause z_B , the extinction memory, it may be possible to replicate the pattern of results without requiring a latent cause model. For example, the 12-month-old App NL-G-F mice behavior may have a deficit in learning about the context. Within the RW model, if the alpha for context is set to zero for those mice, but kept higher for the other groups, say

alpha_context = 0.8, the authors could potentially observe the same pattern of discrimination indices in figure 2G and 2H at test. Because the authors don't explicitly state which parameters might be driving the change in the DI, the authors should show in some way that their results cannot simply be due to poor contextual learning in the 12 month old App NL-G-F mice, as this can presumably be predicted by the RW model. The authors' model fits using RW don't show this, but this is because they don't consider this possibility that the alpha for context might be disrupted in the 12-month-old App NL-G-F mice. Of course, using the RW model with these alphas won't lead to as nice of fits of the behavior across acquisition, extinction, and reinstatement as the authors' LCM, the number of parameters are substantially reduced in the RW model. Yet the important pattern of the DI would be replicated with the RW model (if I'm not mistaken), which is the important test for assessment of reinstatement.

We would like to clarify that we estimated three parameters in the RW model for individuals: α_{CS} , $\alpha_{context}$, and β . Even if we did so, many samples did not satisfy our criteria (Appendix 1–figure 2). Please refer to the “Evaluation of model fit” in Appendix 1 and the legend of Appendix 1–figure 1A to 1D, where we have written the estimated parameter values.

We did not agree that paralyzing the contextual learning by setting $\alpha_{context}$ as 0 in the RW model can explain the CR curve of 12-month-old AD mice well. Specifically, the RW model cannot capture the between-day extinction dynamics (i.e., the increase in CR at the beginning of day 2 extinction) and the higher CR at test 3 relative to test 2 (i.e., DI between test 3 and test 2 is greater than 0.5). In addition, because the context input (= 0.2) was relatively lower than the CS input (= 1), and there is only a single unsignaled shock trial, even setting $\alpha_{context} = 1$ results in only a limited increase in CR (Appendix 1–figure 1A to 1D; see also Author response image 2 9). Thus, the RW model cannot replicate the reinstatement effect or the critical pattern of discrimination index, even under conditions of stronger contextual learning.

(3) As stated by the authors in the introduction, the advantage of the fear learning approach is that the memory is modified across the acquisition-extinction-reinstatement phases. Although perhaps not explicitly stated by the authors, the post-reinstatement test (test 3) is the crucial test for whether there is reactivation of a previously stored memory, with the general argument being that the reinvigorated response to the CS can't simply be explained by relearning the CS-US pairing, because re-exposure the US alone leads to increase response to the CS at test. Of course there are several explanations for why this may occur, particularly when also considering the context as a stimulus. This is what I understood to be the justification for the use of a model, such as the latent cause model, that may better capture and compare these possibilities within a single framework. As such, it is critical to look at the level of responding to both the context alone and to the CS. It appears that the authors only look at the percent freezing during the CS, and it is not clear whether this is due to the contextual-US learning during the US re-exposure or to increased responding to the CS - presumably caused by reactivation of the acquisition memory. The authors do perform a comparison between the preCS and CS period, but it is not clear whether this is taken into account in the LCM. For example, the instance of the model shown in figure 1 indicates that the 'extinction cause', or cause z6, develops a strong weight for the context during the reinstatement phase of presenting the shock alone. This state then leads to increased freezing during the final CS probe test as shown in the figure. If they haven't already, I think the authors must somehow incorporate these different phases (CS vs ITI) into their model, particularly since this type of memory retrieval that depends on assessing latent states is specifically why the authors justified using the latent causal model. In more precise terms, it's not clear whether the authors incorporate a preCS/ITI period each day the cue is presented as a vector of just the context in addition to the CS period in which the vector contains both the context and the CS. Based on the description, it seemed to me that they only model the CRs during the CS

period on days when the CS is presented, and thereby the context is only ever modeled on its own (as just the context by itself in the vector) on extinction days when the CS is not presented. If they are modeling both timepoints each day that the CS is presented, then I would recommend explicitly stating this in the methods section.

In this study, we did not model the preCS freezing rate, and we thank the reviewer for the suggestion to model preCS periods as separate context-only trials. In our view, however, this approach is not consistent with the assumptions of the LCM. Our rationale is that the available periods of context and the CS are different. We assume that observation of the context lasts from preCS to CS. If we simulate both preCS (context) and CS (context and tone), the weight of context would be updated twice. Instead, we follow the same method as described in the original code from Gershman et al. (2017) to consider the context effect. We agree that explicitly modeling preCS could provide additional insights, but we believe it would require modifying or extending the LCM. We consider this an important direction for future research, but it is outside the scope of this study.

(4) The authors fit the model using all data points across acquisition and learning. As one of the other reviewers has highlighted, it appears that there is a high chance for overfitting the data with the LCM. Of course, this would result in much better fits than models with substantially fewer free parameters, such as the RW model. As mentioned above, the authors should use a method that takes into account the number of parameters, such as the BIC.

Please refer to the reply to public review (2A) for the reason we did not take the suggestion to use BIC. In addition, we feel that we have adequately addressed the concern of overfitting in the first round of the review.

(5) The authors have stated that they do not think the Barnes maze task can be modeled with the LCM. Whether or not this is the case, if the authors do not model this data with the LCM, the Barnes maze data doesn't appear valuable to the main hypothesis. The authors suggest that more sophisticated models such as the LCM may be beneficial for early detection of diseases such as Alzheimer's, so the Barnes maze data is not valuable for providing evidence of this hypothesis. Rather, the authors make an argument that the memory deficits in the Barnes maze mimic the reinstatement effects providing support that memory is disrupted similarly in these mice. Although, the authors state that the deficits in memory retrieval are similar across the two tasks, the authors are not explicit as to the precise deficits in memory retrieval in the reinstatement task - it's a combination of overgeneralizing latent causes during acquisition, poor learning rate, over differentiation of the stimuli.

We would like to clarify that we valued the latent cause model not solely because it is more sophisticated and fits more data points, but it is an internal model that implicates the cognitive process. Please also see the reply to the recommendations to authors (3) about the reason why we did not take the suggestion to remove this data.

Reviewer #3 (Public review):

Summary:

This paper seeks to identify underlying mechanisms contributing to memory deficits observed in Alzheimer's disease (AD) mouse models. By understanding these mechanisms, they hope to uncover insights into subtle cognitive changes early in AD to inform interventions for early-stage decline.

Strengths:

The paper provides a comprehensive exploration of memory deficits in an AD mouse model, covering early and late stages of the disease. The experimental design was robust, confirming age-dependent increases in A β plaque accumulation in the AD model mice and using multiple behavior tasks that collectively highlighted difficulties in maintaining multiple competing memory cues, with deficits most pronounced in older mice.

In the fear acquisition, extinction, and reinstatement task, AD model mice exhibited a significantly higher fear response after acquisition compared to controls, as well as a greater drop in fear response during reinstatement. These findings suggest that AD mice struggle to retain the fear memory associated with the conditioned stimulus, with the group differences being more pronounced in the older mice.

In the reversal Barnes maze task, the AD model mice displayed a tendency to explore the maze perimeter rather than the two potential target holes, indicating a failure to integrate multiple memory cues into their strategy. This contrasted with the control mice, which used the more confirmatory strategy of focusing on the two target holes. Despite this, the AD mice were quicker to reach the target hole, suggesting that their impairments were specific to memory retrieval rather than basic task performance.

The authors strengthened their findings by analyzing their data with a leading computational model, which describes how animals balance competing memories. They found that AD mice showed somewhat of a contradiction: a tendency to both treat trials as more alike than they are (lower α) and similar stimuli as more distinct than they are (lower σ_x) compared to controls.

Weaknesses:

While conceptually solid, the model struggles to fit the data and to support the key hypothesis about AD mice's inability to retain competing memories. These issues are evident in Figure 3:

(1) The model misses trends in the data, including the gradual learning of fear in all groups during acquisition, the absence of a fear response at the start of the experiment, and the faster return of fear during reinstatement compared to the gradual learning of fear during acquisition. It also underestimates the increase in fear at the start of day 2 of extinction, particularly in controls.

(2) The model explains the higher fear response in controls during reinstatement largely through a stronger association to the context formed during the unsigned shock phase, rather than to any memory of the conditioned stimulus from acquisition (as seen in Figure 3C). In the experiment, however, this memory does seem to be important for explaining the higher fear response in controls during reinstatement (as seen in Author Response Figure 3). The model does show a necessary condition for memory retrieval, which is that controls rely more on the latent causes from acquisition. But this alone is not sufficient, since the associations within that cause may have been overwritten during extinction. The Rescorla-Wagner model illustrates this point: it too uses the latent cause from acquisition (as it only ever uses a single cause across phases) but does not retain the original stimulus-shock memory, updating and overwriting it continuously. Similarly, the latent cause model may reuse a cause from acquisition without preserving its original stimulus-shock association.

These issues lead to potential overinterpretation of the model parameters. The differences in α and σ_x are being used to make claims about cognitive processes (e.g., overgeneralization vs. over differentiation), but the model itself does not appear to capture these processes accurately.

The authors could benefit from a model that better matches the data and captures the retention and retrieval of fear memories across phases. While they explored alternatives, including the Rescorla-Wagner model and a latent state model, these showed no meaningful improvement in fit. This highlights a broader issue: these models are well-motivated but may not fully capture observed behavior.

Conclusion:

Overall, the data support the authors' hypothesis that AD model mice struggle to retain competing memories, with the effect becoming more pronounced with age. While I believe the right computational model could highlight these differences, the current models fall short in doing so.

We thank the reviewer for the insightful comments. For the comments (1) and (2), please refer to our previous author response to comments #26 and #27. We recognize that the models tested in this study have limitations and, as noted, do not fully capture all aspects of the observed behavioral data. We see this as an important direction for future research and value the reviewer's suggestions.

Recommendations for the authors:

Reviewer #2 (Recommendations for the authors):

I have maintained some of the main concerns included in the first round of reviews as I think they remain concerns with the new draft, even though the authors have included substantially more analysis of their data, which is appreciated. I particularly found the inclusion of the comparative modeling valuable, although I think the analysis comparing the models should be improved.

(1) This relates to point 1 in the public assessment or #16 in the response to reviewers from the authors. The authors raise the point that even a low posterior can drive behavioral expression (lines 361-365 in the response to authors), and so the acquisition latent cause may partially drive reinstatement. Yet in the stimulation shown in figure 1D, this does not seem to be the case. As I mentioned in the public response, in figure 1, the posteriors for z_A are similar on day 34 and day 36, yet only on day 36 is there a strong CR. At least in this example, it does not appear that z_A contributes to the increased responding from day 34 (test 2) to day 36 (test 3). There may be a slight increase in z_1 in figure 3C, but the dominant change from day 34 to day 36 appears to be the increase in the posterior of z_3 and the substantial increase in w_3 . The authors then cite several papers which have shown the shift in balance between what it is the putative acquisition memory and extinction memory (i.e. Lacagnina et al.). Yet I do not see how this modeling fits with most of the previous findings. For example, in the Lacagnina et al. paper, activation of the acquisition ensemble or inhibition of the extinction ensemble drives freezing, whereas the opposite pattern reduces freezing. What appears to be the pattern in the modeling in this paper is primarily learning of context in the extinction latent cause to predict the shock. As I mention in point 2C of the public review, it's not clear why this pattern of results would require a latent cause model. Would a high alpha for context and not the CS not give a similar pattern of results in the RW model? At least for giving similar results of the DIs in figure 2?

First, we would like to clarify that the x-axis in Figure 1D is labeled "Trial," not "Day." Please refer to the reply to public review (1), where we clarified the posterior probability of the latent cause from trials 34 to 36. Second, although we did not have direct neural circuit evidence in this study, we discussed the similarities between previous findings and the modeling in the first review. Briefly, our main point focuses on the interaction between

acquisition and extinction memory. In other words, responses at different times arise from distinct internal states made up of competing memories. We assume that the reviewer expects a modeling result showing nearly full recovery of acquisition memory, which aligns with previous findings where optogenetic activation of the acquisition engram can partially mimic reinstatement (Zaki et al., 2022; see also the response to comment #12 in the first round of review). We acknowledge that such a modeling result cannot be achieved with the latent cause model and see it as a potential future direction for model improvement.

Please also refer to the reply to public review (2) about how a high alpha for context in the RW model cannot explain the pattern we observed in the reinstatement paradigm.

(2) This is related to point 3 in the public comments and #13 in the response to reviewers. I raised the question of comparing the preCS/ITI period with the CS period, but my main point was why not include these periods in the LCM itself as mentioned in more detail in point 3 in the current public review. The inclusion of the comparisons the authors performed helped, but my main point was that the authors could have a better measure of wcontext if they included the preCS period as a stimulus each day (when only the context is included in the stimulus). This would provide better estimates of wcontext. As stated in the public review, perhaps the authors did this, but my understanding of the methods this was not the case, rather, it seems the authors only included the CS period for CRs within the model (at least on days when the CS was present).

Please refer to the reply to public review (3) about the reason why we did not model the preCS freezing rate.

(3) This relates to point 4 in the public review and #15 and #24 in the response to authors. The authors have several points for why the two experiments are similar and how results may be extrapolated - lines 725-733. The first point is that associative learning is fundamental in spatial learning. I'm not sure that this broad connection between the two studies is particularly insightful for why one supports the other as associative learning is putatively involved in most behavioral tasks. In the second point about reversals, why not then use a reversal paradigm that would be easier to model with LCM? This data is certainly valuable and interesting, yet I don't think it's helpful for this paper to state qualitatively the similarities in the potential ways a latent cause framework might predict behavior on the Barnes maze. I would recommend that the authors either model the behavior with LCM, remove the experiment from the paper, or change the framing of the paper that LCM might be an ideal approach for early detection of dementia or Alzheimer's disease.

We would like to clarify that our aim was not to present the LCM as an ideal tool for early detection of AD symptoms. Rather, our focus is on the broader idea of utilizing internal models and estimating individual internal states in early-stage AD. Regarding using a reversal paradigm that would be easier to model with LCM, the most straightforward approach is to use another type of paradigm for fear conditioning, then to examine the extent to which similar behavioral characteristics are observed between paradigms within subjects. However, re-exposing the same mice to such paradigms is constrained by strong carry-over effects, limiting the feasibility of this experiment. Other behavioral tasks relevant to AD that avoid shock generally involve action selection for subsequent observation (Webster et al., 2014), which falls outside the structure of LCM. Our rationale for including the Barnes maze task is that spatial memory deficit is implicated in the early stage of AD, making it relevant for translational research. While we acknowledge that exact modeling of Barnes maze behavior would require a more sophisticated model (as discussed in the first round of review), our intention to use the reversal Barnes maze paradigm is to suggest a presumable memory modification learning in a non-fear conditioning paradigm. We also discussed

whether similar deficits in memory modification could be observed across two behavioral tasks.

(4) Reviewer # mentioned that the change in pattern of behavior only shows up in the older mice questioning the clinical relevance of early detection. I do think this is a valid point and maybe should be addressed. There does seem to be a bit of a bump in the controls on day 23 that doesn't appear in the 6-month group. Perhaps this was initially a spontaneous recovery test indicated by the dotted vertical line? This vertical line does not appear to be defined in the figure 1 legend, nor in figures 2 and 3.

We would like to emphasize that the App^{NL-G-F} knock-in mouse is widely considered a model of early-stage AD, characterized by A β accumulation with little to no neurofibrillary tangle pathology or neuronal loss (see Introduction). By examining different ages, we can assess the contribution of both the amount and duration of A β accumulation as well as age-related factors. Modeling the deficit in the memory modification process in the older App^{NL-G-F} knock-in mice, we suggested a diverged internal state in early-stage AD in older age, and this does not diminish the relevance of the model for studying early cognitive changes in AD.

We would also like to clarify again that the x-axis in the figure is “Trial,” not “Day.” The vertical dashed lines in these figures indicate phase boundaries, and they were defined in the figure legend: in Figure 1C, “The vertical dashed lines separate the phases.”; in Figure 2B, “The dashed vertical line separates the extinction 1 and extinction 2 phases.”; in Figure 3, “The vertical dashed lines indicate the boundaries of phases.”

(5) Are the examples in figure 3 good examples? The example for the 12-month-old control shows a substantial increase in weights for the context during test 3, but not for the CS. Yet in the bar plots in Figure 4 G and H, this pattern seems to be different. The weights for the context appear to substantially drop in the "after extinction" period as compared to the "extinction" period. It's hard to tell the change from "extinction" to "after extinction" for the CS weights (the authors change the y-axis for the CS weights but not for the context weights from panels G to H).

We would like to clarify that in Figure 3C, the increase in weights for context is not presented during test 3 (trial 36), noted by the reviewer; rather, it is the unsignaled shock phase (trial 35).

We assumed that the reviewer might misunderstand that the labels on the left in Figure 4, “Acquisition”, “Extinction”, and “After extinction”, indicate the time point. However, the data shown in Figure 4C to 4H are all from the same time point: test 3 (trial 36). The grouping reflects the classification of latent causes based on the trial in which they were inferred. In addition, for Figures 4G and 4H, the y-axis limits were not set identically because the data range for “Sum of w_{CS} ” varied. This was done to ensure the visibility of all data points. In Figure 4, each dot represents one animal. Take Figure 3D as an example. The point in Figure 4G is the sum of w_3 and w_4 in trial 36, and the point in Figure 4H is w_5 in trial 36, note that the subscript numerals indicate latent cause index. We hope this addresses the reviewer’s question about the difference between the two figures.

The following is the authors’ response to the original reviews

Reviewer #1 (Public review):

Summary:

The authors show certain memory deficits in a mouse knock-in model of Alzheimer's Disease (AD). They show that the observed memory deficits can be explained by a

computational model, the latent cause model of associative memory. The memory tasks used include the fear memory task (CFC) and the 'reverse' Barnes maze. Research on AD is important given its known huge societal burden. Likewise, better characterization of the behavioral phenotypes of genetic mouse models of AD is also imperative to advance our understanding of the disease using these models. In this light, I applaud the authors' efforts.

Strengths:

(1) Combining computational modelling with animal behavior in genetic knock-in mouse lines is a promising approach, which will be beneficial to the field and potentially explain any discrepancies in results across studies as well as provide new predictions for future work.

(2) The authors' usage of multiple tasks and multiple ages is also important to ensure generalization across memory tasks and 'modelling' of the progression of the disease.

Weaknesses:

[#1] (1) I have some concerns regarding the interpretation of the behavioral results. Since the computational model then rests on the authors' interpretation of the behavioral results, it, in turn, makes judging the model's explanatory power difficult as well. For the CFC data, why do knock-in mice have stronger memory in test 1 (Figure 2C)? Does this mean the knock-in mice have better memory at this time point? Is this explained by the latent cause model? Are there some compensatory changes in these mice leading to better memory? The authors use a discrimination index across tests to infer a deficit in re-instatement, but this indicates a relative deficit in re-instatement from memory strength in test 1. The interpretation of these differential DIs is not straightforward. This is evident when test 1 is compared with test 2, i.e., the time point after extinction, which also shows a significant difference across groups, Figure 2F, in the same direction as the re-instatement. A clarification of all these points will help strengthen the authors' case.

We appreciate the reviewer for the critical comments. According to the latent cause framework, the strength of the memory is influenced by at least 2 parameters: associative weight between CS and US given a latent cause and posterior probability of the latent cause. The modeling results showed that a higher posterior probability of acquisition latent cause, but not higher associative weight, drove the higher test 1 CR in App^{NL-G-F} mice (Results and Discussion; Figure 4 – figure supplement 3B, 3C). In terms of posterior, we agree that App^{NL-G-F} mice have strong fear memory. On the other hand, this suggests that App^{NL-G-F} mice exhibited a tendency toward overgeneralization, favoring modification of old memories, which adversely affected the ability to retain competing memories. The strong memory in test 1 would be a compensatory effect of overgeneralization.

To estimate the magnitude of reinstatement, at least, one would have to compare CRs between test 2 (extinction) and test 3 (reinstatement), as well as those between test 1 (acquisition) and test 3. These comparisons represent the extent to which the memory at the reinstatement is far from that in the extinction, and close to that in the acquisition. Since discrimination index (DI) has been widely used as a normalized measure to evaluate the extent to which the system can distinguish between two conditions, we applied DI consistently to behavioral and simulated data in the reinstatement experiment, and the behavioral data in the reversal Barnes maze experiment, allowing us to evaluate the discriminability of an agent in these experiments. In addition, we used DI to examine its correlation with estimated parameters, enabling us to explore how individual discriminability may relate to the internal state. We have already discussed the differences in

DI between test 3 and test 1, as well as CR in test 1 between control and *App^{NL-G-F}* in the manuscript and further elaborated on this point in Line 232, 745-748.

[#2] (2) I have some concerns regarding the interpretation of the Barnes maze data as well, where there already seems to be a deficit in the memory at probe test 1 (Figure 6C). Given that there is already a deficit in memory, would not a more parsimonious explanation of the data be that general memory function in this task is impacted in these mice, rather than the authors' preferred interpretation? How does this memory weakening fit with the CFC data showing stronger memories at test 1? While I applaud the authors for using multiple memory tasks, I am left wondering if the authors tried fitting the latent cause model to the Barnes maze data as well.

While we agree that the deficits shown in probe test 1 may imply impaired memory function in *App^{NL-G-F}* mice in this task, it would be difficult to explain this solely in terms of impairments in general memory function. The learning curve and the daily strategy changes suggested that *App^{NL-G-F}* mice would have virtually intact learning ability in the initial training phase (Figure 6B, 6F, Figure 6 – figure supplement 1 and 3). For the correspondence relationship between the reinstatement and the reversal Barnes maze learning from the aspect of memory modification process, please also see our reply to comment #24. We have explained why we did not fit the latent cause model to the Barnes maze data in the provisional response.

[#3] (3) Since the authors use the behavioral data for each animal to fit the model, it is important to validate that the fits for the control vs. experimental groups are similar to the model (i.e., no significant differences in residuals). If that is the case, one can compare the differences in model results across groups (Figures 4 and 5). Some further estimates of the performance of the model across groups would help.

We have added the residual (i.e., observed CR minus simulated CR) in Figure 3 – figure supplement 1D and 1E. The fit was similar between control and *App^{NL-G-F}* mice groups in the test trials, except test 3 in the 12-month-old group. The residual was significantly higher in the 12-month-old control mice than *App^{NL-G-F}* mice, suggesting the model underestimated the reinstatement in the control, yet the DI calculated from the simulated CR replicates the behavioral data (Figure 3 – figure supplement 1A to 1C). These results suggest that the latent cause model fits our data with little systematic bias such as an overestimation of CR for the control group in the reinstatement, supporting the validity of the comparisons in estimated parameters between groups. These results and discussion have been added in the manuscript Line 269-276.

One may notice that the latent cause model overestimated the CR in acquisition trials in all groups in Figure 3 – figure supplement 1D and 1E. We have discussed this point in the reply to comment #26, 34 questioned by reviewer 3.

[#4] (4) Is there an alternative model the authors considered, which was outweighed in terms of prediction by this model?

Yes, we have further evaluated two alternative models: the Rescorla-Wagner (RW; Rescorla & Wagner, 1972) model and the latent state model (LSM; Cochran & Cisler, 2019). The RW model serves as a baseline, given its known limitations in explaining fear return after extinction. The LSM is another contemporary model that shares several concepts with the latent cause model (LCM) such as building upon the RW model, assuming a latent variable inferred by Bayes' rule, and involving a ruminative update for memory modification. We evaluated the three models in terms of the prediction accuracy and reproducibility of key behavioral

features. Please refer to the Appendix 1 for detailed methods and results for these two models.

As expected, the RW model fit well to the data till the end of extinction but failed to reproduce reinstatement (Appendix 1 – figure 1A to 1D). Due to a large prediction error in test 3, few samples met the acceptance criteria we set (Appendix 1 – figure 2 and 3A). Conversely, the LSM reproduced reinstatement, as well as gradual learning in acquisition and extinction phases, particularly in the 12month-old control (Appendix 1 – figure 1G). The number of accepted samples in the LSM was higher than in the RW model but generally lower than in the LCM (Appendix 1 – figure 2). The sum of prediction errors over all trials in the LSM was comparable to that in the LCM in the 6-month-old group (Appendix 1 – figure 4A), it was significantly lower in the 12-month-old group (Appendix 1 – figure 4B). Especially the LSM generated smaller prediction errors during the acquisition trials than in the LCM, suggesting that the LSM might be better at explaining the behaviors of acquisition (Appendix 1 – figure 4A and 4B; but see the reply for comment #34). While the LSM generated smaller prediction errors than the LCM in test 2 of the control group, it failed to replicate the observed DIs, a critical behavioral phenotype difference between control and *App^{NL-G-F}* mice (Appendix 1 – figure 6A to 6C; cf. Figure 2F to 2H, Figure 3 – figure supplement 1A to 1C).

Thus, although each model could capture different aspects of reinstatement, standing on the LCM to explain the reinstatement better aligns with our purpose. It should also be noted that we did not explore all parameter spaces of the LSM, hence we cannot rule out the possibility that alternative parameter sets could provide a better fit and explain the memory modification process well. A more comprehensive parameter search in the LSM may be a valuable direction for future research.

[#5] One concern here is also parameter overfitting. Did the authors try leaving out some data (trials/mice) and predicting their responses based on the fit derived from the training data?

Following the reviewer's suggestion, we confirmed if overfitting occurred using all trials to estimate parameters. Estimating parameters while actually leaving out trials would disorder the time lapse across trials, and thereby the prior of latent causes in each trial. Instead, we removed the constraint of prediction error by setting the error threshold to 1 for certain trials to virtually leave these trials out. We treated these trials as a virtual “training” dataset, while the rest of the trials were a “test” dataset. For the median CR data of each group (Figure 3), we estimated parameters under 6 conditions with unique training and test trials, then evaluated the prediction error for the training and test trials. Note that training and test trials were arbitrarily decided. Also, the error threshold for the acquisition trial was set to 1 as described in Materials and Methods, which we have further discussed the reason in the reply to comment #34 and treated acquisition trials separately from the test trials. We expect that the contribution of the data from the acquisition and test trials for parameter estimation could be discounted compared to those from the training trials with the constraint, and if overfitting occurred, the prediction error in the test data would be worse than that in the training trials.

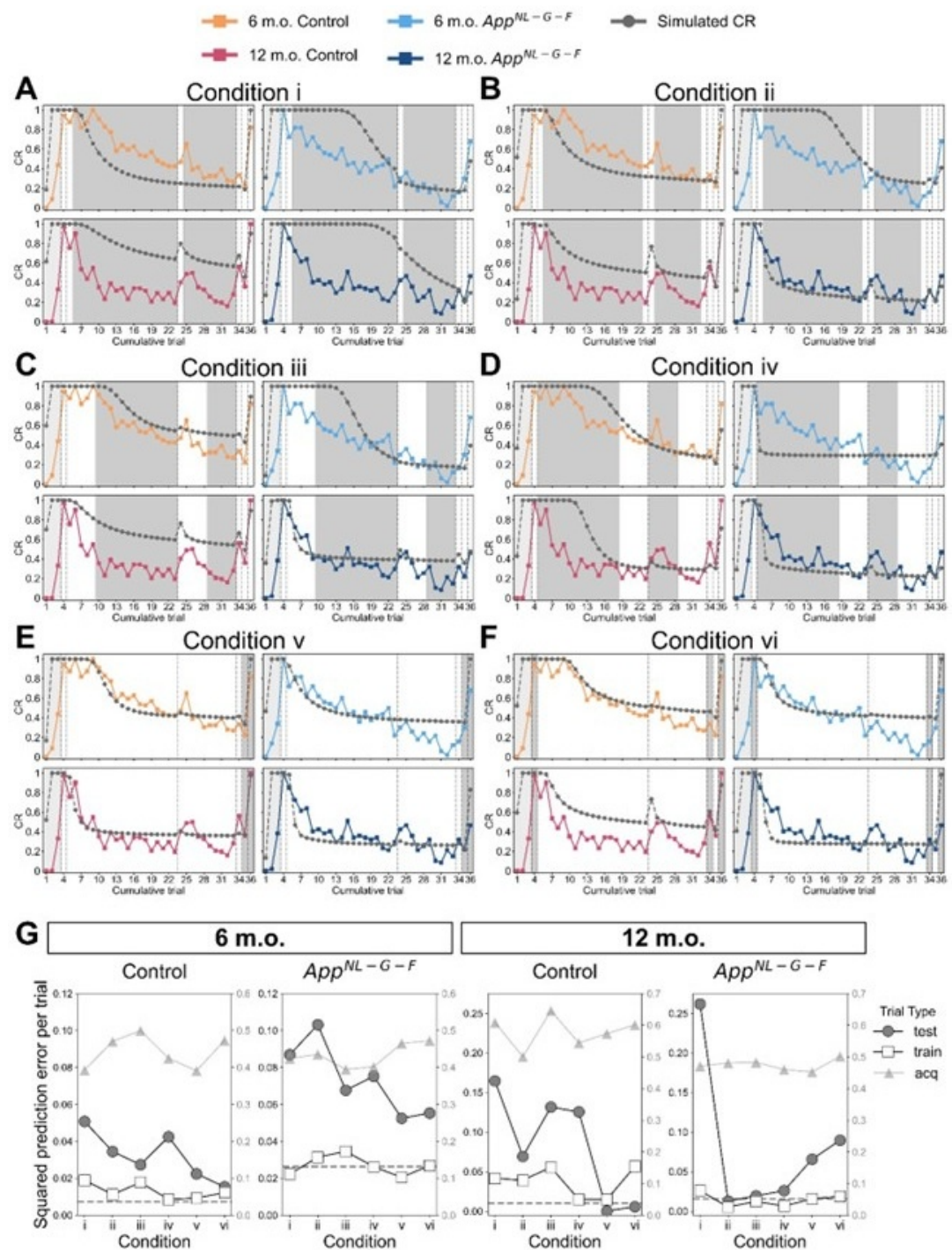
Author response image 1A to 1F showed the simulated and observed CR under each condition, where acquisition trials were in light-shaded areas, test trials were in dark-shaded areas, and the rest of the trials were training trials. Author response image 1G showed mean squared prediction error across the acquisition, training and test trials under each condition. The dashed gray line showed the mean squared prediction error of training trials in Figure 3 as a baseline.

In conditions i and ii, where two or four trials in the extinction were used for training (Author response image 1A and 1B), the prediction error was generally higher in test trials than in training trials. In conditions iii and iv where ten trials in the extinction were used for

training (Author response image 1C and 1D), the difference in prediction error between testing and training trials became smaller. These results suggest that providing more extinction trial data would reduce overfitting. In condition v (Author response image 1E), the results showed that using trials until extinction can predict reinstatement in control mice but not *App^{NL-G-F}* mice. Similarly, in condition vi (Author response image 1F), where test phase trials were left out, the prediction error differences were greater in *App^{NL-G-F}* mice. These results suggest that the test trials should be used for the parameter estimation to minimize prediction error for all groups. Overall, this analysis suggests that using all trials would reduce prediction error with few overfitting.

Author response image 1.

Leaving trials out in parameter estimation in the latent cause model. (A – F) The observed CR (colored line) is the median freezing rate during the CS presentation over the mice within each group, which is the same as that in Figure 3. The colors indicate different groups: orange represents 6-month-old control, light blue represents 6-month-old *App^{NL-G-F}* mice, pink represents 12-month-old control, and dark blue represents 12-month-old *App^{NL-G-F}* mice. Under six different leave-out conditions (i – vi), parameters were estimated and used for generating simulated CR (gray line). In each condition, trials were categorized as acquisition (light-shaded area), training data (white area), and test data (dark-shaded area) based on the error threshold during parameter estimation. Only the error threshold of the test data trial was different from the original method (see Material and Method) and set to 1. In conditions i to vi, the number of test data trials is 27, 25, 19, and 19 in extinction phases. In condition v, the number of test data trials is 2 (trials 35 and 36). In condition vi, test data trials were the 3 test phases (trials 4, 34, and 36). (G) Each subplot shows the mean squared prediction error for the test data trial (gray circles), training data trial (white squares), and acquisition trial (gray triangles) in each group. The left y-axis corresponds to data from test and training trials, and the right y-axis corresponds to data from acquisition trials. The dashed line indicates the results calculated from Figure 3 as a baseline.



Reviewer #1 (Recommendations for the authors):

Minor:

[#6] (1) I would like the authors to further clarify why 'explaining' the reinstatement deficit in the AD mouse model is important in working towards the understanding of AD i.e., which aspect of AD this could explain etc.

In this study, we utilized the reinstatement paradigm with the latent cause model as an internal model to illustrate how estimating internal states can improve understanding of

cognitive alteration associated with extensive A β accumulation in the brain. Our findings suggest that misclassification in the memory modification process, manifesting as overgeneralization and overdifferentiation, underlies the memory deficit in the *App^{NL-G-F}* knock-in model mice.

The parameters in the internal model associated with AD pathology (e.g., α and σ_x^2 in this study) can be viewed as computational phenotypes, filling the explanatory gap between neurobiological abnormalities and cognitive dysfunction in AD. This would advance the understanding of cognitive symptoms in the early stages of AD beyond conventional behavioral endpoints alone.

We further propose that altered internal states in *App^{NL-G-F}* knock-in mice may underlie a wide range of memory-related symptoms in AD as we observed that *App^{NL-G-F}* knock-in mice failed to retain competing memories in the reversal Barnes maze task. We speculate on how overgeneralization and overdifferentiation may explain some AD symptoms in the manuscript:

- Line 565-569: overgeneralization may explain deficits in discriminating highly similar visual stimuli reported in early-stage AD patients as they misclassify the lure as previously learned object
- Line 576-579: overdifferentiation may explain impaired ability to transfer previously learned association rules in early-stage AD patients as they misclassify them as separated knowledge.
- Line 579-582: overdifferentiation may explain delusions in AD patients as an extended latent cause model could simulate the emergence of delusional thinking

We provide one more example here that overgeneralization may explain that early-stage AD patients are more susceptible to proactive interference than cognitively normal elders in semantic memory tests (Curiel Cid et al., 2024; Loewenstein et al., 2015, 2016; Valles-Salgado et al., 2024), as they are more likely to infer previously learned material. Lastly, we expect that explaining memory-related symptoms within a unified framework may facilitate future hypothesis generation and contribute to the development of strategies for detecting the earliest cognitive alteration in AD.

[#7] (2) The authors state in the abstract/introduction that such computational modelling could be most beneficial for the early detection of memory disorders. The deficits observed here are pronounced in the older animals. It will help to further clarify if these older animals model the early stages of the disease. Do the authors expect severe deficits in this mouse model at even later time points?

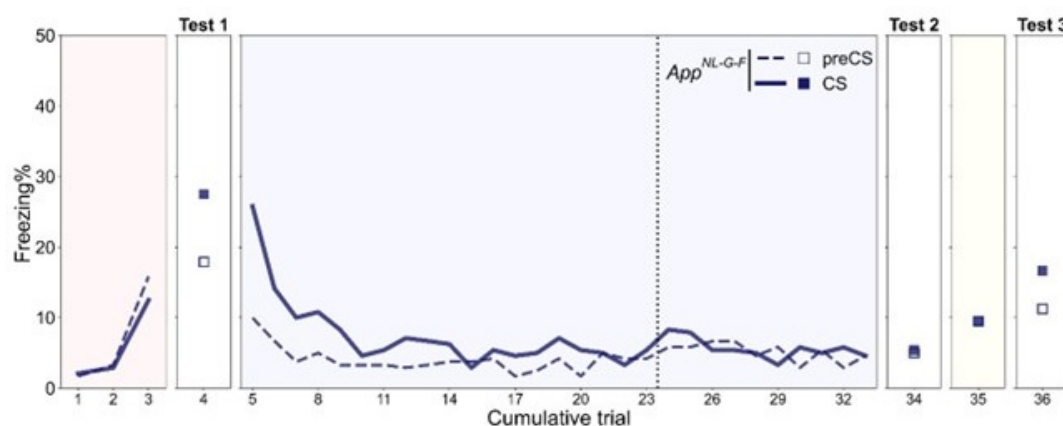
The early stage of the disease is marked by abnormal biomarkers associated with A β accumulation and neuroinflammation, while cognitive symptoms are mild or absent. This stage can persist for several years during which the level of A β may reach a plateau. As the disease progresses, tau pathology and neurodegeneration emerge and drive the transition into the late stage and the onset of dementia. The *App^{NL-G-F}* knock-in mice recapitulate the features present in the early stage (Saito et al., 2014), where extensive A β accumulation and neuroinflammation worsen along with ages (Figure 2 – figure supplement 1). Since *App^{NL-G-F}* knock-in mice are central to A β pathology without tauopathy and neurodegeneration, it should be noted that it does not represent the full spectrum of the disease even at advanced ages. Therefore, older animals still model the early stages of the diseases and are suitable to study the long-term effect of A β accumulation and neuroinflammation.

The age tested in previous reports using *App^{NL-G-F}* mice spanned a wide range from 2 months old to 24 months old. Different behavioral tasks have varied sensitivity but overall suggest the

dysfunction worsens with aging (Bellio et al., 2024; Mehla et al., 2019; Sakakibara et al., 2018). We have tested the reinstatement experiment with 17-month-old App^{NL-G-F} mice before (Author response image 2). They showed more advanced deficits with the same trends observed in 12-month-old App^{NL-G-F} mice, but their freezing rates were overall at a lower level. There is a concern that possible hearing loss may affect the results and interpretation, therefore we decided to focus on 12-month-old data.

Author response image 2.

Freezing rate across reinstatement paradigm in the 17-month-old App^{NL-G-F} mice. Dashed and solid lines indicate the median freezing rate over 34 mice before (preCS) and during (CS) tone presentation, respectively. Red, blue, and yellow backgrounds represent acquisition, extinction, and unsignaled shock in Figure 2A. The dashed vertical line separates the extinction 1 and extinction 2 phases.



[#8] (3) There are quite a few 'marginal' p -values in the paper at $p > 0.05$ but near it. Should we accept them all as statistically significant? The authors need to clarify if all the experimental groups are sufficiently powered.

For our study, we decided *a priori* that $p < 0.05$ would be considered statistically significant, as described in the Materials and Methods. Therefore, in our Results, we did not consider these marginal values as statistically significant but reported the trend, as they may indicate substantive significance.

We described our power analysis method in the manuscript Line 897-898 and have provided the results in Tables S21 and S22.

[#9] (4) The authors emphasize here that such computational modelling enables us to study the underlying 'reasoning' of the patient (in the abstract and introduction), I do not see how this is the case. The model states that there is a latent i.e. another underlying variable that was not previously considered.

Our use of the term “reasoning” was to distinguish the internal model, which describes how an agent makes sense of the world, from other generative models implemented for biomarker and disease progression prediction. However, we agree that using “reasoning” may be misleading and imprecise, so to reduce ambiguity we have removed this word in our manuscript Line 27: Nonetheless, internal models of the patient remain underexplored in AD; Line 85: However, previous approaches did not suppose an internal model of the world to predict future from current observation given prior knowledge.

[#10] (5) *The authors combine knock-in mice with controls to compute correlations of parameters of the model with behavior of animals (e.g. Figure 4B and Figure 5B). They run the risk of spurious correlations due to differences across groups, which they have indeed shown to exist (Figure 4A and 5A). It would help to show within-group correlations between DI and parameter fit, at least for the control group (which has a large spread of data).*

We agree that genotype (control, App^{NL-G-F}) could be a confounder between the estimated parameters and DI, thereby generating spurious correlations. To address this concern, we have provided within-group correlation in Figure 4 – figure supplement 2 for the 12-month-old group and Figure 5 – figure supplement 2 for the 6-month-old group.

In the 12-month-old group, the significant positive correlation between σx^2 and DI remained in both control and App^{NL-G-F} mice even if we adjusted the genotype effect, suggesting that it is very unlikely that the correlations in Figure 4B are due to the genotype-related confounding. On the other hand, the positive correlation between α and DI was found to be significant in the control mice but not in the App^{NL-G-F} mice. Most of α were distributed around the lower bound in App^{NL-G-F} mice, which possibly reduced the variance and correlation coefficient. These results support our original conclusion that α and σx^2 are parameters associated with a lower magnitude of reinstatement in aged App^{NL-G-F} mice.

In the 6-month-old group, the correlations shown in Figure 5B were not preserved within subgroups, suggesting genotype would be a confounder for α , σx^2 , and DI. We recognized that significant correlations in Figure 5B may arise from group differences, increased sample size, or greater variance after combining control and App^{NL-G-F} mice.

Therefore, we concluded that α and σx^2 are associated with the magnitude of reinstatement but modulated by the genotype effect depending on the age.

We have added interpretations of within-group correlation in the manuscript Line 307-308, 375-378.

[#11] (6) *It is unclear to me why overgeneralization of internal states will lead to the animals having trouble recalling a memory. Would this not lead to overgeneralization of memory recall instead?*

We assume that the reviewer is referring to “overgeneralization of internal states,” a case in which the animal’s internal state remained the same regardless of the observation, thereby leading to “overgeneralization of memory recall.” We agree that this could be one possible situation and appears less problematic than the case in which this memory is no longer retrievable.

However, in our manuscript, we did not deal with the case of “overgeneralization of internal states”. Rather, our findings illustrated how the memory modification process falls into overgeneralization or overdifferentiation and how it adversely affects the retention of competing memories, thereby causing App^{NL-G-F} mice to have trouble recalling the same memory as the control mice.

According to the latent cause model, retrieval failure is explained by a mismatch of internal states, namely when an agent perceives that the current cue does not match a previously experienced one, the old latent cause is less likely to be inferred due to its low likelihood (Gershman et al., 2017). For example, if a mouse exhibited higher CR in test 2, it would be interpreted as a successful fear memory retrieval due to overgeneralization of the fear memory. However, it reflects a failure of extinction memory retrieval due to the mismatch

between the internal states at extinction and test 2. This is an example that overgeneralization of memory induces the failure of memory retrieval.

On the other hand, *App^{NL-G-F}* mice exhibited higher CR in test 1, which is conventionally interpreted as a successful fear memory retrieval. When estimating their internal states, they would infer that their observation in test 1 well matches those under the acquisition latent causes, that is the overgeneralization of fear memory as shown by a higher posterior probability in acquisition latent causes in test 1 (Figure 4 – figure supplement 3). This is an example that over-generalization of memory does not always induce retrieval failure as we explained in the reply to comment #1.

Reviewer #2 (Public review):

Summary:

This manuscript proposes that the use of a latent cause model for the assessment of memory-based tasks may provide improved early detection of Alzheimer's Disease as well as more differentiated mapping of behavior to underlying causes. To test the validity of this model, the authors use a previously described knock-in mouse model of AD and subject the mice to several behaviors to determine whether the latent cause model may provide informative predictions regarding changes in the observed behaviors. They include a well-established fear learning paradigm in which distinct memories are believed to compete for control of behavior. More specifically, it's been observed that animals undergoing fear learning and subsequent fear extinction develop two separate memories for the acquisition phase and the extinction phase, such that the extinction does not simply 'erase' the previously acquired memory. Many models of learning require the addition of a separate context or state to be added during the extinction phase and are typically modeled by assuming the existence of a new state at the time of extinction. The Niv research group, Gershman et al. 2017, have shown that the use of a latent cause model applied to this behavior can elegantly predict the formation of latent states based on a Bayesian approach, and that these latent states can facilitate the persistence of the acquisition and extinction memory independently. The authors of this manuscript leverage this approach to test whether deficits in the production of the internal states, or the inference and learning of those states, may be disrupted in knock-in mice that show both a build-up of amyloid-beta plaques and a deterioration in memory as the mice age.

Strengths:

I think the authors' proposal to leverage the latent cause model and test whether it can lead to improved assessments in an animal model of AD is a promising approach for bridging the gap between clinical and basic research. The authors use a promising mouse model and apply this to a paradigm in which the behavior and neurobiology are relatively well understood - an ideal situation for assessing how a disease state may impact both the neurobiology and behavior. The latent cause model has the potential to better connect observed behavior to underlying causes and may pave a road for improved mapping of changes in behavior to neurobiological mechanisms in diseases such as AD.

Weaknesses:

I have several substantial concerns which I've detailed below. These include important details on how the behavior was analyzed, how the model was used to assess the behavior, and the interpretations that have been made based on the model.

[#12] (1) There is substantial data to suggest that during fear learning in mice separate memories develop for the acquisition and extinction phases, with the acquisition memory

becoming more strongly retrieved during spontaneous recovery and reinstatement. The Gershman paper, cited by the authors, shows how the latent causal model can predict this shift in latent states by allowing for the priors to decay over time, thereby increasing the posterior of the acquisition memory at the time of spontaneous recovery. In this manuscript, the authors suggest a similar mechanism of action for reinstatement, yet the model does not appear to return to the acquisition memory state after reinstatement, at least based on the examples shown in Figures 1 and 3. Rather, the model appears to mainly modify the weights in the most recent state, putatively the 'extinction state', during reinstatement. Of course, the authors must rely on how the model fits the data, but this seems problematic based on prior research indicating that reinstatement is most likely due to the reactivation of the acquisition memory. This may call into question whether the model is successfully modeling the underlying processes or states that lead to behavior and whether this is a valid approach for AD.

We thank the reviewer for insightful comments.

We agree that, as demonstrated in Gershman et al. (2017), the latent cause model accounts for spontaneous recovery via the inference of new latent causes during extinction and the temporal compression property provided by the prior. Moreover, it was also demonstrated that even a relatively low posterior can drive behavioral expression if the weight in the acquisition latent cause is preserved. For example, when the interval between retrieval and extinction was long enough that acquisition latent cause was not dominant during extinction, spontaneous recovery was observed despite the posterior probability of acquisition latent cause (C1) remaining below 0.1 in Figure 11D of Gershman et al. (2017).

In our study, a high response in test 3 (reinstatement) is explained by both acquisition and extinction latent cause. The former preserves the associative weight of the initial fear memory, while the latter has w_{context} learned in the unsignaled shock phase. These positive w were weighted by their posterior probability and together contributed to increased expected shock in test 3. Though the posterior probability of acquisition latent cause was lower than extinction latent cause in test 3 due to time passage, this would be a parallel instance mentioned above. To clarify their contributions to reinstatement, we have conducted additional simulations and the discussion in reply to the reviewer's next comment (see the reply to comment #13).

We recognize that our results might appear to deviate from the notion that reinstatement results from the strong reactivation of acquisition memory, where one would expect a high posterior probability of the acquisition latent cause. However, we would like to emphasize that the return of fear emerges from the interplay of competing memories. Previous studies have shown that contextual or cued fear reinstatement involves a neural activity switch back to fear state in the medial prefrontal cortex (mPFC), including the prelimbic cortex and infralimbic cortex, and the amygdala, including ventral intercalated amygdala neurons (ITCv), medial subdivision of central nucleus of the amygdala (CeM), and the basolateral amygdala (BLA) (Giustino et al., 2019; Hitora-Imamura et al., 2015; Zaki et al., 2022). We speculate that such transition is parallel to the internal states change in the latent cause model in terms of posterior probability and associative weight change.

Optogenetic manipulation experiments have further revealed how fear and extinction engrams contribute to extinction retrieval and reinstatement. For instance, Gu et al. (2022) used a cued fear conditioning paradigm and found that inhibition of extinction engrams in the BLA, ventral hippocampus (vHPC), and mPFC after extinction learning artificially increased freezing to the tone cue. Similar results were observed in contextual fear conditioning, where silencing extinction engrams in the hippocampus dentate gyrus (DG) impaired extinction retrieval (Lacagnina et al., 2019). These results suggest that the weakening extinction memory can induce a return of fear response even without a reminder

shock. On the other hand, Zaki et al. (2022) showed that inhibition of fear engrams in the BLA, DG, or hippocampus CA1 attenuated contextual fear reinstatement. However, they also reported that stimulation of these fear engrams was not sufficient to induce reinstatement, suggesting these fear engram only partially account for reinstatement.

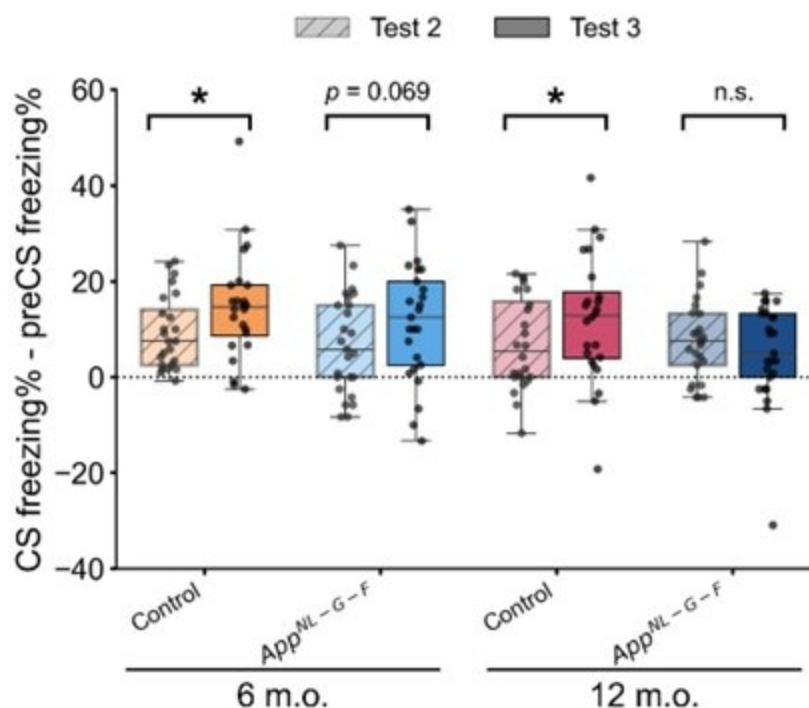
In summary, reinstatement likely results from bidirectional changes in the fear and extinction circuits, supporting our interpretation that both acquisition and extinction latent causes contribute to the reinstatement. Although it remains unclear whether these memory engrams represent latent causes, one possible interpretation is that w_{context} update in extinction latent causes during unsignaled shock indicates weakening of the extinction memory, while preservation of w in acquisition latent causes and their posterior probability suggests reactivation of previous fear memory.

[#13] (2) As stated by the authors in the introduction, the advantage of the fear learning approach is that the memory is modified across the acquisition-extinction-reinstatement phases. Although perhaps not explicitly stated by the authors, the post-reinstatement test (test 3) is the crucial test for whether there is reactivation of a previously stored memory, with the general argument being that the reinvigorated response to the CS can't simply be explained by relearning the CS-US pairing, because re-exposure the US alone leads to increase response to the CS at test. Of course there are several explanations for why this may occur, particularly when also considering the context as a stimulus. This is what I understood to be the justification for the use of a model, such as the latent cause model, that may better capture and compare these possibilities within a single framework. As such, it is critical to look at the level of responding to both the context alone and to the CS. It appears that the authors only look at the percent freezing during the CS, and it is not clear whether this is due to the contextual US learning during the US re-exposure or to increased response to the CS - presumably caused by reactivation of the acquisition memory. For example, the instance of the model shown in Figure 1 indicates that the 'extinction state', or state z6, develops a strong weight for the context during the reinstatement phase of presenting the shock alone. This state then leads to increased freezing during the final CS probe test as shown in the figure. By not comparing the difference in the evoked freezing CR at the test (ITI vs CS period), the purpose of the reinstatement test is lost in the sense of whether a previous memory was reactivated - was the response to the CS restored above and beyond the freezing to the context? I think the authors must somehow incorporate these different phases (CS vs ITI) into their model, particularly since this type of memory retrieval that depends on assessing latent states is specifically why the authors justified using the latent causal model.

To clarify the contribution of context, we have provided preCS freezing rate across trials in Figure 2 – figure supplement 2. As the reviewer pointed out, the preCS freezing rate did not remain at the same level across trials, especially within the 12-month-old control and App^{NL-GF} group (Figure 2 – figure supplement 2A and 2B), suggesting the effect context. A paired samples t -test comparing preCS freezing (Figure 2 – figure supplement 2E) and CS freezing (Figure 2E) in test 3 revealed significant differences in all groups: 6-month-old control, $t(23) = -6.344$, $p < 0.001$, $d = -1.295$; 6-month-old App^{NL-GF} , $t(24) = -4.679$, $p < 0.001$, $d = -0.936$; 12-month-old control, $t(23) = -4.512$, $p < 0.001$, $d = 0.921$; 12-month-old App^{NL-GF} , $t(24) = -2.408$, $p = 0.024$, $d = -0.482$. These results indicate that the response to CS was above and beyond the response to context only. We also compared the change in freezing rate (CS freezing rate minus preCS freezing rate) in test 2 and test 3 to examine the net response to the tone. The significant difference was found in the control group, but not in the App^{NL-GF} group (Author response image 3). The increased net response to the tone in the control group suggested that the reinstatement was partially driven by reactivation of acquisition memory, not solely by the contextual US learning during the unsignaled shock phase. We have added these results and discussion in the manuscript Line 220-231.

Author response image 3.

Net freezing rate in test 2 and test 3. Net freezing rate is defined as the CS freezing rate (i.e., freezing rate during 1 min CS presentation) minus the preCS freezing rate (i.e., 1 min before CS presentation). The dashed horizontal line indicates no freezing rate change from the preCS period to the CS presentation. * $p < 0.05$ by paired-sample Student's t -test, and the alternative hypothesis specifies that test 2 freezing rate change is less than test 3. Colors indicate different groups: orange represents 6-month-old control ($n = 24$), light blue represents 6-month-old App^{NL-G-F} mice ($n = 25$), pink represents 12-month-old control ($n = 24$), and dark blue represents 12-month-old App^{NL-G-F} mice ($n = 25$). Each black dot represents one animal. Statistical results were as follows: $t(23) = -1.927$, $p = 0.033$, Cohen's $d = -0.393$ in 6-month-old control; $t(24) = -1.534$, $p = 0.069$, Cohen's $d = -0.307$ in 6-month-old App^{NL-G-F} ; $t(23) = -1.775$, $p = 0.045$, Cohen's $d = -0.362$ in 12-month-old control; $t(24) = 0.86$, $p = 0.801$, Cohen's $d = 0.172$ in 12-month-old App^{NL-G-F} .

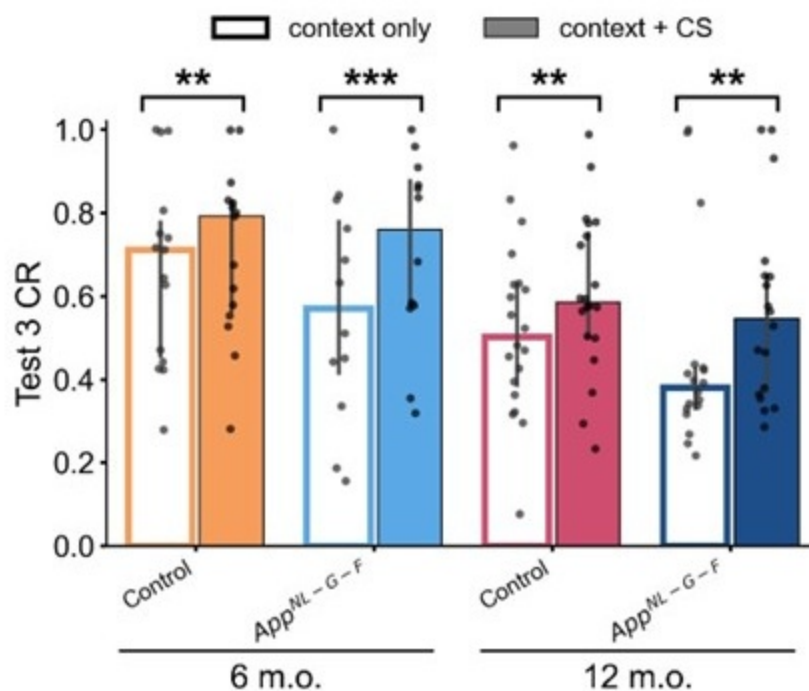


According to the latent cause model, if the reinstatement is merely induced by an association between the context and the US in the unsignaled shock phase, the CR given context only and that given context and CS in test 3 should be equal. However, the simulation conducted for each mouse using their estimated parameters confirmed that this was not the case in this study. The results showed that simulated CR was significantly higher in the context+CS condition than in the context only condition (Author response image 4). This trend is consistent with the behavioral results we mentioned above.

Author response image 4.

Simulation of context effect in test 3. Estimated parameter sets of each sample were used to run the simulation that only context or context with CS was present in test 3 (trial 36). The data are shown as median with interquartile range, where white bars with colored lines represent CR for context only and colored bars represent CR for context with CS. Colors indicate different groups: orange represents 6-month-old control ($n = 15$), light blue represents 6-month-old App^{NL-G-F} mice ($n = 12$), pink represents 12-month-old control ($n = 20$), and dark blue represents 12-month-old App^{NL-G-F} mice ($n = 12$).

and dark blue represents 12-month-old App^{NL-G-F} mice ($n = 18$). Each black dot represents one animal. $**p < 0.01$, and $***p < 0.001$ by Wilcoxon signed-rank test comparing context only and context + CS in each group, and the alternative hypothesis specifies that CR in context is not equal to CR in context with CS. Statistical results were as follows: $W = 15$, $p = 0.008$, effect size $r = -0.66$ in 6-month-old control; $W = 0$, $p < 0.001$, effect size $r = -0.88$ in 6-month-old App^{NL-G-F} ; $W = 25$, $p = 0.002$, effect size $r = -0.67$ in 12-month-old control; $W = 9$, $p = 0.002$, effect size $r = -0.75$ in 12-month-old App^{NL-G-F} .



We thank the reviewer for the suggestion about fitting and fixing certain parameters in different phases.

However, this strategy may not be optimal for our study for the following scientific reasons.

Our primary purpose is to explore internal states in the memory modification process that are associated with the deficit found in *App^{NL-G-F}* mice in the reinstatement paradigm. We did not restrict the question to memory retrieval, nor did we have a particular hypothesis such that only a few parameters of interest account for the impaired associative learning or structure learning in *App^{NL-G-F}* mice while all other parameters are comparable between groups. We are concerned that restricting questions to memory retrieval at the test is too parsimonious and might lead to misinterpretation of the results. As we explain in reply to comment #5, removing trials in extinction during parameter estimation reduces the model fit performance and runs the risk of overfitting within the individual. Therefore, we estimated all parameters for each animal, with the assumption that the estimated parameter set represents individual internal state (i.e., learning and memory characteristics) and should be fixed within the animal across all trials.

Figure 3 is the parameter estimation and simulation results using the median data of each group as an individual. The estimated parameter value is one of the possible cases in that group to demonstrate how a typical learning curve fits the latent cause model. The reviewer mentioned “20-fold reduction in learning rate” is the comparison of two data points, not the actual comparison between groups. The comparison between control and *App^{NL-G-F}* mice in the 12-month-old group for all parameters was provided in Table S7. The Mann-Whitney *U* test did not reveal a significant difference in learning rate (η): 12-month-old control (Mdn = 0.09, IQR=0.23) vs. 12-month-old *App^{NL-G-F}* (Mdn = 0.12, IQR=0.23), $U = 199$, $p = 0.587$.

We agree that lower learning rate could bias the learning toward inferring a new latent cause. However, this tendency may depend on the value of other parameters and varied in different phases in the reinstatement paradigm. Here, we used α as an example and demonstrate their interaction in Appendix 2 – table 2 with relatively extreme values: $\alpha = \{1, 3\}$ and $\eta = \{0.01, 0.5\}$ while the rest of the parameters fixed at the initial guess value.

When $\alpha = 1$, the number of latent causes across phases (K_{acq} , K_{ext} , K_{rem}) remain unchanged and their posterior probability in test 3 were comparable even if η increased from 0.01 to 0.5. This is an example that lower η does not lead to inferring new latent causes because of low α . The effect of low learning rate manifests in test 3 CR due to low $w_{context, acq}$ and $w_{context, ext}$.

When $\alpha = 3$, the number of acquisition latent causes (K_{acq}) was higher in the case of $\eta = 0.01$ than that of $\eta = 0.5$, showing the effect mentioned by the reviewer. However, test 1 CR is much lower when $\eta = 0.01$, indicating unsuccessful learning even after inferring a new latent cause. This is none of the cases observed in this study. During extinction phases, the effect of η is surpassed by the effect of high α , where the number of extinction latent causes (K_{ext}) is high and not affected by η . After the extinction phases, the effect of K kicks in as the total number of latent causes reaches its value ($K = 33$ in this example), especially in the case of $\eta = 0.01$. A new latent cause is inferred after extinction in the condition of $\eta = 0.5$, but the CR 3 is still high as the $w_{context, acq}$ and $w_{context, ext}$ are high. This is an example that a new latent cause is inferred in spite of higher η .

Overall, the learning rate would not have a prominent effect alone throughout the reinstatement paradigm, and it has a joint effect with other parameters. Note that the example here did not cover our estimated results, as the estimated learning rate was not significantly different between control and *App^{NL-G-F}* mice (see above). Please refer to the reply to comment #31 for more discussion about the interaction among parameters when the learning rate is fixed. We hope this clarifies the reviewer’s concern.

[#15] (4) Why didn't the authors use the latent causal model on the Barnes maze task? The authors mention in the discussion that different cognitive processes may be at play across the two tasks, yet reversal tasks have been suggested to be solved using latent states to be able to flip between the two different task states. In this way, it seems very fitting to use the latent cause model. Indeed, it may even be a better way to assess changes in α as there are presumably 12 observable stimuli/locations.

Please refer to our provisional response about the application of the latent cause model to the reversal Barnes maze task. Briefly, it would be difficult to directly apply the latent cause model to the Barnes maze data because this task involves operant learning, and thereby almost all conditions in the latent cause model are not satisfied. Please also see our reply to comment #24 for the discussion of the link between the latent cause model and Barnes maze task.

Reviewer #2 (Recommendations for the authors):

[#16] (1) I had a bit of difficulty finding all the details of the model. First, I had to mainly rely on the Gershman 2017 paper to understand the model. Even then, there were certain aspects of the model that were not clear. For instance, it's not quite clear to me when the new internal states are created and how the maximum number of states is determined. After reading the authors' methods and the Gershman paper, it seems that a new internal state is generated at each time point, aka z_t , and that the prior for that state decays onwards from α . Yet because most 'new' internal states don't ever take on much of a portion of the posterior, most of these states can be ignored. Is that a correct understanding? To state this another way, I interpret the equation on line 129 to indicate that the prior is determined by the power law for all existing internal states and that each new state starts with a value of α , yet I don't see the rule for creating a new state, or for iterating k other than that k iterates at each timestep. Yet this seems to not be consistent with the fact that the max number of states K is also a parameter fit. Please clarify this, or point me to where this is better defined.

I find this to be an important question for the current paper as it is unclear to me when the states were created. Most notably, in Figure 3, it's important to understand why there's an increase in the posterior of z_5 in the AD 12-month mice at test. Is state z_5 generated at trial 5? If so, the prior would be extremely small by trial 36, making it even more perplexing why z_5 has such a high posterior. If its weights are similar to z_3 and z_4 and they have been much more active recently, why would z_5 come into play?

We assume that the "new internal state" the reviewer is referring to is the "new latent cause." We would like to clarify that "internal state" in our study refers to all the latent causes at a given time point and observation. As this manuscript is submitted as a Research Advance article in *eLife*, we did not rephrase all the model details. Here, we explain when a new latent cause is created (i.e., the prior probability of a new latent cause is greater than 0) with the example of the 12-month-old group (Figure 3C and 3D).

Suppose that before the start of each trial, an agent inferred the most likely latent cause with maximum posterior, and it inferred k latent causes so far. A new latent cause can be inferred at the computation of the prior of latent causes at the beginning of each trial.

In the latent cause model, it follows a distance-dependent Chinese Restaurant Process (CRP; Blei and Frazier, 2011). The prior of each old latent cause is its posterior probability, which is the final count of the EM update before the current. In addition, the prior of old latent causes is sensitive to the time passage so that it exponentially decreases as a forgetting function modulated by g (see Figure 2 in Gershman et al., 2017). Simultaneously, the prior of a new

cause is assigned α . The new latent cause is inferred at this moment. Hence, the prior of latent causes is jointly determined by α , g and its posterior probability. The maximum number of latent causes K is set *a priori* and does not affect the prior while $k < K$ (see also reply to comment #30 for the discussion of boundary set for K and comment #31 for the discussion of the interaction between α and K). Note that only one new latent cause can be inferred in each trial, and $(k+1)^{\text{th}}$ latent cause, which has never been inferred so far, is chosen as the new latent cause.

In our manuscript, the subscript number in z_k denotes the order in which they were inferred, not the trial number. In Figures 3C and 3D, z_3 and z_4 were inferred in trials 5 and 6 during extinction; z_5 is a new latent cause inferred in trial 36. Therefore, the prior of z_5 is not extremely small compared to z_4 and z_3 .

In both control and App^{NL-G-F} mice in the 12-month-old (Figures 3C and 3D), z_3 is dominant until trial 35. The unsignaled shock at trial 35 generates a large prediction error as only context is presented and followed by the US. This prediction error reduces posterior of z_3 , while increasing the posterior of z_4 and w_{context} in z_3 and z_4 . This decrease of posterior of z_3 is more obvious in the App^{NL-G-F} than in the control group, prompting them to infer a new latent cause z_5 (Figure 3C and 3D). Although Figure 3C and 3D are illustrative examples as we explained in the reply to comment #14, this interpretation would be plausible as the App^{NL-G-F} group inferred a significantly larger number of latent causes after the extinction with slightly higher posteriors of them than those in the control group (Figure 4E).

[#17] (2) Related to the above, Are the states z_A and z_B defined by the authors to help the reader group the states into acquisition and extinction states, or are they somehow grouped by the model? If the latter is true, I don't understand how this would occur based on the model. If the former, could the authors state that these states were grouped together by the author?

We used z_A and z_B annotations to assist with the explanation, so this is not grouped by the model. We have stated this in the manuscript Line 181-182.

[#18] (3) This expands on the third point above. In Figure 3D, internal states z_3 , z_4 , and z_5 appear to be pretty much identical in weights in the App group. It's not clear to me why then the posterior of z_5 would all of a sudden jump up. If I understand correctly, the posterior is the likelihood of the observations given the internal state (presumably this should be similar across z_3 , z_4 , and z_5), multiplied by the prior of the state. z_3 and z_4 are the dominant inferred states up to state 36. Why would z_5 become more likely if there doesn't appear to be any error? I'm inferring no error because there are little or no changes in weights on trial 36, most prominently no changes in z_3 which is the dominant internal state in step 36. If there's little change in weights, or no errors, shouldn't the prior dominate the calculation of the posterior which would lead to z_3 and z_4 being most prominent at trial 36?

We have explained how z_5 of the 12-month-old App^{NL-G-F} was inferred in the reply to comment #16. Here, we explain the process underlying the rapid changes of the posterior of z_3 , z_4 , and z_5 from trial 35 to 36.

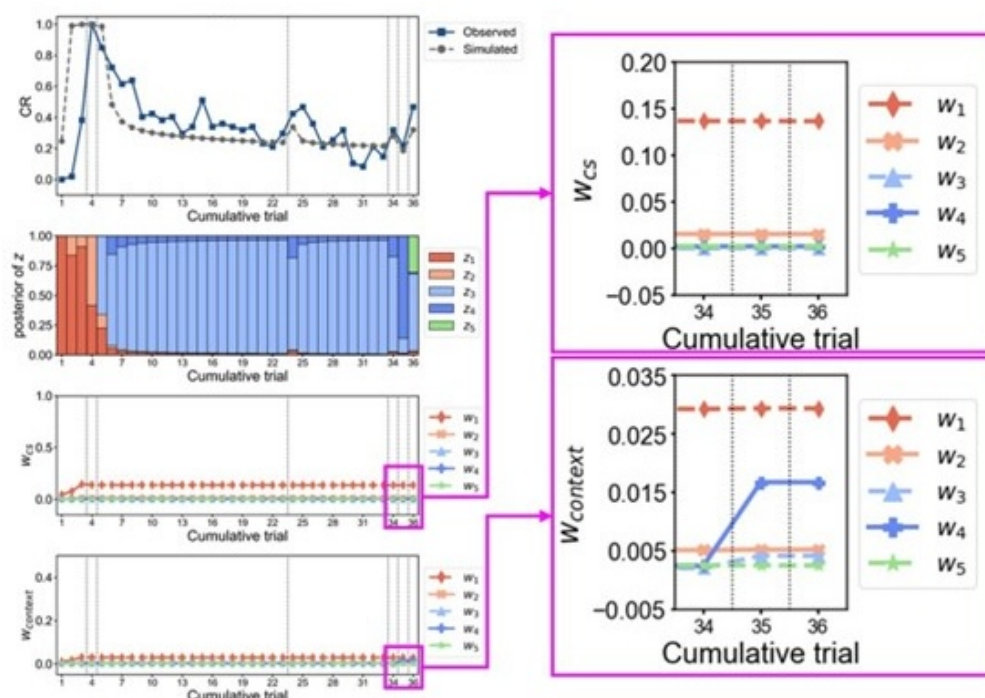
During the extinction, the mice inferred z_3 given the CS and the context in the absence of US. In trial 35, they observed the context and the unsignaled shock in the absence of the CS. This reduced the likelihood for the CS under z_3 and thereby the posterior of z_3 , while relatively increasing the posterior of z_4 . The associative weight between the context and the US, w_{context} , indeed increased in both z_3 and z_4 , but w_{context} of z_4 was updated more than that of z_3 due to its higher posterior probability. At the beginning of trial 36, a new latent cause z_5 was inferred with a certain prior (see also the reply for comment #16), and $w_5 = w_0$, where

w_0 is the initial value of weight. After normalizing the prior over latent causes, the emergence of z_5 reduced the prior probability of other latent causes compared to the case where the prior of z_5 is 0. Since the CS was presented while the US was absent in trial 36, the likelihood of the CS and that of the US under z_3 , and especially z_4 , given the cues and w became lower than the case in which z_5 has not been inferred yet. Consequently, the posterior of z_5 became salient (Figure 3D).

To maintain consistency across panels, we used a uniform y-axis range. However, we acknowledge that this may make it harder to notice the changes of associative weights in Figure 3D. We have provided the subpanel in Figure 3D with a smaller y-axis limit to reveal the weight changes at trial 35 in Author response image 5.

Author response image 5.

Magnified view of $w_{context}$ and w_{CS} in the last 3 trials in Figure 3D. The graph format is the same as in Figure 3D. The weight for CS (w_{CS}) and that for context ($w_{context}$) in each latent cause across trial 34 (test 2), 35 (unsigned shock), and 36 (test 3) in 12-month-old App^{NL-G-F} in Figure 3D was magnified in the upper and lower magenta box, respectively.



[#19] (8) In Figure 4B - The figure legend didn't appear to indicate at which time points the DIs are plotted.

We have amended the figure legend to indicate that DI between test 3 and test 1 is plotted.

[#20] (9) Lines 301-303 state that the posterior probabilities of the acquisition internal states in the 12month AD mice were much higher at test 1 and that this resulted in different levels of CR across the control and 12-month App group. This is shown in the Figure 4A supplement, but this is not apparent in Figure 3 panels C and D. Is the example shown in panel D not representative of the group? The CRs across the two examples in Figure 3 C and D look extremely similar at test 1. Furthermore, the posteriors of the internal states look pretty similar across the two groups for the first 4 trials. Both the App

and control have substantial posterior probabilities for the acquisition period, I don't see any additional states at test 1. The pattern of states during acquisition looks strikingly similar across the two groups, whereas the weights of the stimuli are considerably different. I think it would help the authors to use an example that better represents what the authors are referring to, or provide data to illustrate the difference. Figure 4C partly shows this, but it's not very clear how strong the posteriors are for the 3rd state in the controls.

Figure 3 serves as an example to explain the internal states in each group (see also the third paragraph in the reply to comment #14). Figure 4C to H showed the results from each sample for between-group comparison in selected features. Therefore, the results of direct comparisons of the parameter values and internal states between genotypes in Figure 3 are not necessarily the same as those in Figure 4. Both examples in Figure 3C and 3D inferred 2 latent causes during the acquisition. In terms of posterior till test 1 (trial 4), the two could be the same. However, such examples were not rare, as the proportion of the mice that inferred 2 latent causes during the acquisition was slightly lower than 50% in the control, and around 90% in the *App^{NL-G-F}* mice (Figure 4C). The posterior probability of acquisition latent cause in test 1 showed a similar pattern (Figure 4 – figure supplement 3), with values near 1 in around 50% of the control mice and around 90% of the *App^{NL-G-F}* mice.

[#21] (10) Line 320: This is a confusing sentence. I think the authors are saying that because the App group inferred a new state during test 3, this would protect the weights of the 'extinction' state as compared to the controls since the strength of the weight updates depends on the probability of the posterior.

In order to address this, we have revised this sentence to “Such internal states in *App^{NL-G-F}* mice would diverge the associative weight update from those in the control mice after extinction.” in the manuscript Line 349-351.

[#22] (11) In lines 517-519 the authors address the difference in generalizing the occurrence of stimuli across the App and control groups. It states that App mice with lower alpha generalized observations to an old cause rather than attributing it as a new state. Going back to statement 3 above, I think it's important to show that the model fit of a reduction in alpha does not go hand-in-hand with a reduction in the learning rates and hence the weights. Again, if the likelihoods are diminished due to the low weights, then the fit of alpha might be reduced as well. To reiterate my point above, if the observations in changes in generalization and differentiation occur because of a reduction in the learning rate, the modeling may not be providing a particularly insightful understanding of AD, other than that poor learning leads to ineffectual generalization and differentiation. Do these findings hold up if the learning rates are more comparable across the control and App group?

These findings were explained on the basis of comparable learning rates between control and *App^{NL-G-F}* mice in the 12-month-old group (see the reply to comment #14). In addition, we have conducted simulation for different α and σ_x^2 values under the condition of the fixed learning rate, where overgeneralization and overdifferentiation still occurred (see the reply to comment #26).

[#23] (12) Lines 391 - 393. This is a confusing sentence. "These results suggest that App NL-G-F mice could successfully form a spatial memory of the target hole, while the memory was less likely to be retrieved by a novel observation such as the absence of the escape box under the target hole at the probe test 1." The App mice show improved behavior across days of approaching the correct hole. Is this statement suggesting that

once they've approached the target hole, the lack of the escape box leads to a reduction in the retention of that memory?

We speculated that when the mice observed the absence of the escape box, a certain prediction error would be generated, which may have driven the memory modification. In App^{NL-G-F} mice, such modification, either overgeneralization or overdifferentiation, could render the memory of the target hole vulnerable; if overgeneralization occurred, the memory would be quickly overwritten as the goal no longer exists in this position in this maze, while if overdifferentiation occurred, a novel memory such that the goal does not exist in the maze different from previous one would be formed. In either case of misclassification, the probability of retrieving the goal position would be reduced. To reduce ambiguity in this sentence, we have revised the description in the manuscript Line 432-434 as follows: "These results suggest that App^{NL-G-F} mice could successfully form a spatial memory of the target hole, while they did not retrieve the spatial memory of the target hole as strongly as control mice when they observed the absence of the escape box during the probe test."

[#24] (13) The connection between the results of Barnes maze and the fear learning paradigm is weak. How can changes in overgeneralization due to a reduction in the creation of inferred states and differentiation due to a reduced α lead to the observations in the Barnes maze experiment?

We extrapolated our interpretation in the reinstatement modeling to behaviors in a different behavioral task, to explore the explanatory power of the latent cause framework formalizing mechanisms of associative learning and memory modification. Here, we explain the results of the reversal Barnes maze paradigm in terms of the latent cause model, while conferring the reinstatement paradigm.

Whilst we acknowledge that fear conditioning and spatial learning are not fully comparable, the reversal Barnes maze paradigm used in our study shares several key learning components with the reinstatement paradigm.

First, associative learning is fundamental in spatial learning (Leising & Blaisdell, 2009; Pearce, 2009). Although we did not make any specific assumptions of what kind of associations were learned in the Barnes maze, performance improvements in learning phases likely reflect trial-and-error updates of these associations involving sensory preconditioning or secondary conditioning. Second, the reversal training phases could resemble the extinction phase in the reinstatement paradigm, challenge previously established memory. In terms of the latent cause model, both the reversal learning phase in the reversal Barnes maze paradigm and the extinction phase in the reinstatement paradigm induce a mismatch of the internal state. This process likely introduces large prediction errors, triggering memory modification to reconcile competing memories.

Under the latent cause framework, we posit that the mice would either infer new memories or modify existing memories for the unexpected observations in the Barnes maze (e.g., changed location or absence of escape box) as in the reinstatement paradigm, but learn a larger number of association rules between stimuli in the maze compared to those in the reinstatement. In the reversal Barnes maze paradigm, the animals would infer that a latent cause generates the stimuli in the maze at certain associative weights in each trial, and would adjust behavior by retaining competing memories.

Both overgeneralization and overdifferentiation could explain the lower exploration time of the target hole in the App^{NL-G-F} mice in probe test 1. In the case of overgeneralization, the mice would overwrite the existing spatial memory of the target hole with a memory that the escape box is absent. In the case of overdifferentiation, the mice would infer a new memory such that the goal does not exist in the novel field, in addition to the old memory where the

goal exists in the previous field. In both cases, the App^{NL-G-F} mice would not infer that the location of the goal is fixed at a particular point and failed to retain competing spatial memories of the goal, leading to relying on a less precise, non-spatial strategy to solve the task.

Since there is no established way to formalize the Barnes maze learning in the latent cause model, we did not directly apply the latent cause model to the Barnes maze data. Instead, we used the view above to explore common processes in memory modification between the reinstatement and the Barnes maze paradigm.

The above description was added to the manuscript on page 13 (Line 410-414) and page 19-20 (Line 600-602, 626-639).

[#25] (14) In the fear conditioning task, it may be valuable to separate responding to the context and the cue at the time of the final test. The mice can learn about the context during the reinstatement, but there must be an inference to the cue as it's not present during the reinstatement phase. This would provide an opportunity for the model to perhaps access a prior state that was formed during acquisition. This would be more in line with the original proposal by Gershman et al. 2017 with spontaneous recovery.

Please refer to the reply to comment #13 regarding separating the response to context in test 3.

Reviewer #3 (Public review):

Summary:

This paper seeks to identify underlying mechanisms contributing to memory deficits observed in Alzheimer's disease (AD) mouse models. By understanding these mechanisms, they hope to uncover insights into subtle cognitive changes early in AD to inform interventions for early-stage decline.

Strengths:

The paper provides a comprehensive exploration of memory deficits in an AD mouse model, covering the early and late stages of the disease. The experimental design was robust, confirming age-dependent increases in A β plaque accumulation in the AD model mice and using multiple behavior tasks that collectively highlighted difficulties in maintaining multiple competing memory cues, with deficits most pronounced in older mice.

In the fear acquisition, extinction, and reinstatement task, AD model mice exhibited a significantly higher fear response after acquisition compared to controls, as well as a greater drop in fear response during reinstatement. These findings suggest that AD mice struggle to retain the fear memory associated with the conditioned stimulus, with the group differences being more pronounced in the older mice.

In the reversal Barnes maze task, the AD model mice displayed a tendency to explore the maze perimeter rather than the two potential target holes, indicating a failure to integrate multiple memory cues into their strategy. This contrasted with the control mice, which used the more confirmatory strategy of focusing on the two target holes. Despite this, the AD mice were quicker to reach the target hole, suggesting that their impairments were specific to memory retrieval rather than basic task performance.

The authors strengthened their findings by analyzing their data with a leading computational model, which describes how animals balance competing memories. They found that AD mice showed somewhat of a contradiction: a tendency to both treat trials

as more alike than they are (lower α) and similar stimuli as more distinct than they are (lower σ_x) compared to controls.

Weaknesses:

While conceptually solid, the model struggles to fit the data and to support the key hypothesis about AD mice's ability to retain competing memories. These issues are evident in Figure 3:

[#26] (1) The model misses key trends in the data, including the gradual learning of fear in all groups during acquisition, the absence of a fear response at the start of the experiment, the increase in fear at the start of day 2 of extinction (especially in controls), and the more rapid reinstatement of fear observed in older controls compared to acquisition.

We acknowledge these limitations and explained why they arise in the latent cause model as follows.

a. Absence of a fear response at the start of the experiment and the gradual learning of fear during acquisition

In the latent cause model, the CR is derived from a sigmoidal transformation from the predicted outcome with the assumption that its mapping to behavioral response may be nonlinear (see Equation 10 and section “Conditioned responding” in Gershman et al., 2017).

The magnitude of the unconditioned response (trial 1) is determined by w_0 , θ , and λ . An example was given in Appendix 2 – table 3. In general, a higher w_0 and a lower θ produce a higher trial 1 CR when other parameters are fixed. During the acquisition phase, once the expected shock exceeds θ , CR rapidly approaches 1, and further increases in expected shock produce few changes in CR. This rapid increase was also evident in the spontaneous recovery simulation (Figure 11) in Gershman et al. (2017). The steepness of this rapid increase is modulated by λ such that a higher value produces a shallower slope. This is a characteristic of the latent cause model, assuming CR follows a sigmoid function of expected shock, while the ordinal relationship over CRs is maintained with or without the sigmoid function, as Gershman et al. (2017) mentioned. If one assumes that the CR should be proportional to the expected shock, the model can reproduce the gradual response as a linear combination of w and posteriors of latent causes while omitting the sigmoid transformation (Figure 3).

b. Increase in fear at the start of day 2 extinction

This point is partially reproduced by the latent cause model. As shown in Figure 3, trial 24 (the first trial of day 2 extinction) showed an increase in both posterior probability of latent cause retaining fear memory and the simulated CRs in all groups except the 6-month-old control group, though the increase in CR was small due to the sigmoid transformation (see above). This can be explained by the latent cause model as 24 h time lapse between extinction 1 and 2 decreases the prior of the previously inferred latent cause, leading to an increase of those of other latent causes.

Unlike other groups, the 6-month-old control did not exhibit increased observed CR at trial 24 but at trial 25 (Figure 3A). The latent cause model failed to reproduce it, as there was no increase in posterior probability in trial 24 (Figure 3A). This could be partially explained by the low value of g , which counteracts the effect of the time interval between days: lower g keeps prior of the latent causes at the same level as those in the previous trial. Despite some failures in capturing this effect, our fitting policy was set to optimize prediction among the test trials given our primary purpose of explaining reinstatement.

c. more rapid reinstatement of fear observed in older controls compared to acquisition

We would like to point out that this was replicated by the latent cause model as shown in Figure 3 – figure supplement 1C. The DI between test 3 and test 1 calculated from the simulated CR was significantly higher in 12-month-old control than in *App^{NL-G-F}* mice (cf. Figure 2C to E).

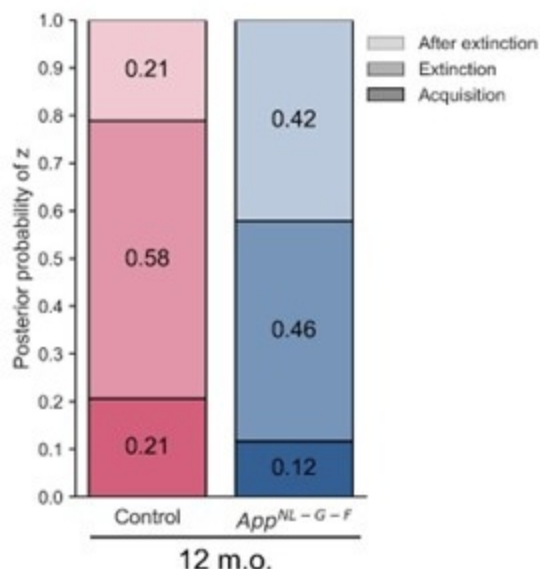
[#27] (2) *The model attributes the higher fear response in controls during reinstatement to a stronger association with the context from the unsignaled shock phase, rather than to any memory of the conditioned stimulus from acquisition. These issues lead to potential overinterpretation of the model parameters. The differences in α and σ are being used to make claims about cognitive processes (e.g., overgeneralization vs. overdifferentiation), but the model itself does not appear to capture these processes accurately. The authors could benefit from a model that better matches the data and that can capture the retention and recollection of a fear memory across phases.*

First, we would like to clarify that the latent cause model explains the reinstatement not only by the extinction latent cause with increased w_{context} but also the acquisition latent cause with preserved w_{CS} and w_{context} (see also reply to comment #13). Second, the latent cause model primarily attributes the higher fear reinstatement in control to a lower number of latent causes inferred after extinction (Figure 4E) and higher w_{context} in extinction latent cause (Figure 4G). We noted that there was a trend toward significance in the posterior probability of latent causes inferred after extinction (Figure 4E), which in turn influences those of acquisition latent causes. Although the posterior probability of acquisition latent cause appeared trivial and no significance was detected between control and *App^{NL-G-F}* mice (Figure 4C), it was suppressed by new latent causes in *App^{NL-G-F}* mice (Author response image 6).

This indicates that *App^{NL-G-F}* mice retrieved acquisition memory less strongly than control mice. Therefore, we argue that the latent cause model attributed a higher fear response in control during reinstatement not solely to the stronger association with the context but also to CS fear memory from acquisition. Although we tested whether additional models fit the reinstatement data in individual mice, these models did not satisfy our fitting criteria for many mice compared to the latent cause model (see also reply to comment #4 and #28).

Author response image 6.

Posterior probability of acquisition, extinction, and after extinction latent causes in test 3. The values within each bar indicate the mean posterior probability of acquisition latent cause (darkest shade), extinction latent cause (medium shade), and latent causes inferred after extinction (lightest shade) in test 3 over mice within genotype. Source data are the same as those used in Figure 4C–E (posterior of z).



Conclusion:

Overall, the data support the authors' hypothesis that AD model mice struggle to retain competing memories, with the effect becoming more pronounced with age. While I believe the right computational model could highlight these differences, the current model falls short in doing so.

Reviewer #3 (Recommendations for the authors):

[#28] Other computational models may better capture the data. Ideally, I'd look for a model that can capture the gradual learning during acquisition, and, in some mice, the inferring of a new latent cause during extinction, allowing the fear memory to be retained and referenced at the start of day 2 extinction and during later tests.

We have further evaluated another computational model, the latent state model, and compared it with the latent cause model. The simulation of reinstatement and parameter estimation method of the latent state model were described in the Appendix.

The latent state model proposed by Cochran and Cisler (2019) shares several concepts with the latent cause model, and well replicates empirical data under certain conditions. We expect that it can also explain the reinstatement.

Following the same analysis flow for the latent cause model, we estimated the parameters and simulated reinstatement in the latent state model from individual CRs and median of them. In the median freezing rate data of the 12-month-old control mice, the simulated CR replicated the observed CR well and exhibited the ideal features that the reviewer looked for: gradual learning during acquisition and an increased fear at the start of the second-day extinction (Appendix 1 – figure 1G). However, a lot of samples did not fit well to the latent state model. The number of anomalies was generally higher than that in the latent cause model (Appendix 1 – figure 2). Within the accepted samples, the sum of squared prediction error in all trials was significantly lower in the latent state model, which resulted from lower prediction error in the acquisition trials (Appendix 1 – figure 4A and 4B). In the three test trials, the squared prediction error was comparable between the latent state model and the latent cause model except for the test 2 trials in the control group (Appendix 1 – figure 4A and 4B, rightmost panel). On the other hand, almost all accepted samples continued to infer the

acquisition latent states during extinction without inferring new states (Appendix 1 – figure 5B and 5E, left panel), which differed from the ideal internal states the reviewer expected. While the latent state model fit performance seems to be better than the latent cause model, the accepted samples cannot reproduce the lower DI between test 3 and test 1 in aged *App^{NL}-G-F* mice (Appendix 1 – figure 6C). These results make the latent state model less suitable for our purpose and therefore we decided to stay with the latent cause model. It should also be noted that we did not explore all parameter spaces of the latent state model hence we cannot rule out the possibility that alternative parameter sets could provide a better fit and explain the memory modification process well. A more comprehensive parameter search in the LSM may be a valuable direction for future research.

If you decide not to go with a new model, my preference would be to drop the current modeling. However, if you wish to stay with the current model, I'd like to see justification or acknowledgment of the following:

[#29] (1) Lower bound on alpha of 1: This forces the model to infer new latent causes, but it seems that some mice, especially younger AD mice, might rely more on classical associative learning (e.g., Rescorla-Wagner) rather than inferring new causes.

We acknowledge that the default value set in Gershman et al. (2017) is 0.1, and the constraint we set is a much higher value. However, $\alpha = 1$ does not always force the model to infer new latent causes.

In the standard form Chinese restaurant process (CRP), the prior that n^{th} observation is assigned to a new cluster is given by $\alpha / (n - 1 + \alpha)$ (Blei & Gershman, 2012). When $\alpha = 1$, the prior of the new cluster for the 2nd observation will be 0.5; when $\alpha = 3$, this prior increases to 0.75. Thus, when $\alpha > 1$, the prior of the new cluster is above chance early in the sequence, which may relate to the reviewer's concern. However, this effect diminishes as the number of observations increases. For instance, the prior of the new cluster drops to 0.1 and 0.25 for the 10th observation when $\alpha = 1$ and 3, respectively. Furthermore, the prior in the latent cause model is governed by not only α but also g , a scaling parameter for the temporal difference between successive observations (see Results in the manuscript) following “distance-dependent” CRP, then normalized over all latent causes including a new latent cause. Thus, it does not necessarily imply that α greater than 1 forces agents to infer a new latent cause. As shown in Appendix 2 – table 4, the number of latent causes does not inflate in each trial when $\alpha = 1$. On the other hand, the high number of latent causes due to $\alpha = 2$ can be suppressed when $g = 0.01$. More importantly, the driving force is the prediction error generated in each trial (see also comment #31 about the interaction between α and σ_x^2). Raising the value of α *per se* can be viewed as increasing the probability to infer a new latent cause, not forcing the model to do so by higher α alone.

During parameter exploration using the median behavioral data under a wider range of α with a lower boundary at 0.1, the estimated value eventually exceeded 1. Therefore, we set the lower bound of α to be 1 to reduce inefficient sampling.

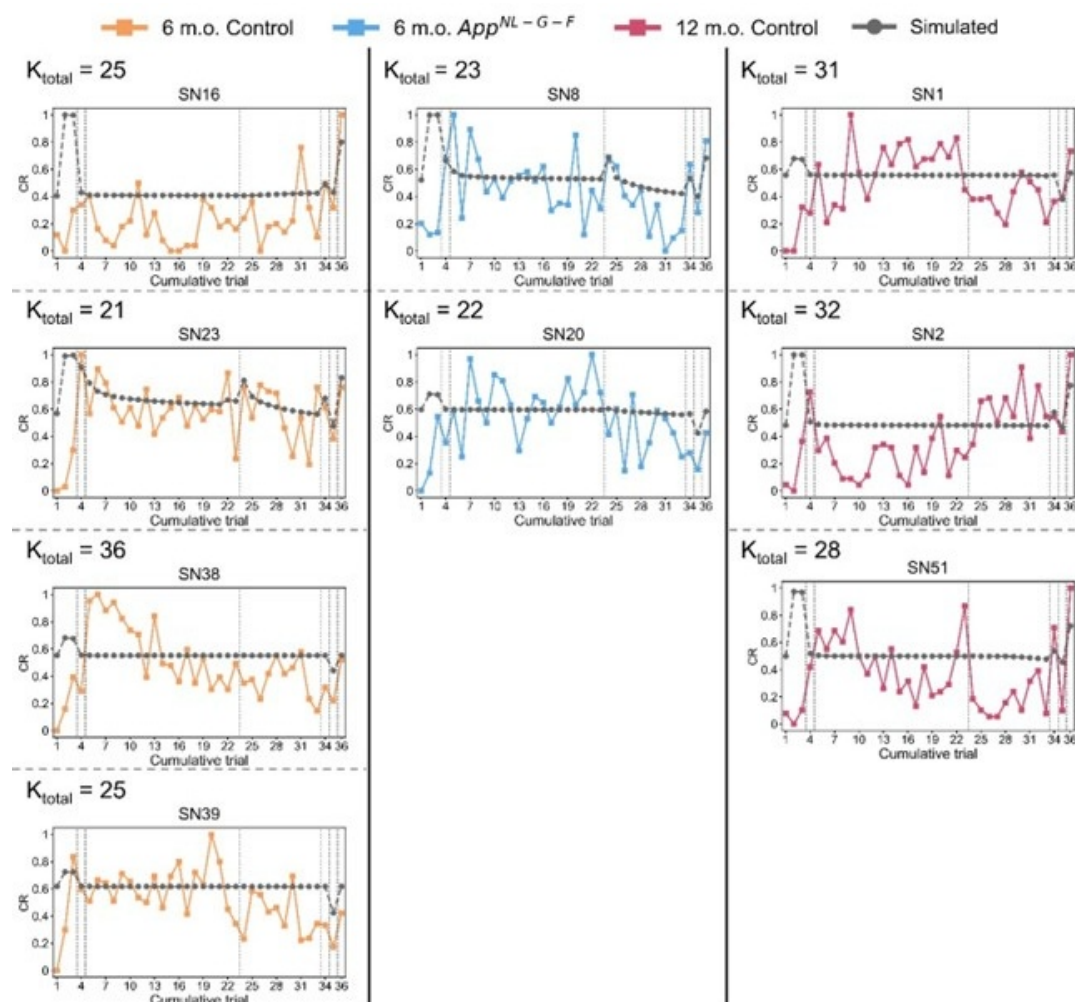
[#30] (2) Number of latent causes: Some mice infer nearly as many latent causes as trials, which seems unrealistic.

We set the upper boundary for the maximum number of latent causes (K) to be 36 to align with the infinite features of CRP. This allowed some mice to infer more than 20 latent causes in total. When we checked the learning curves in these mice, we found that they largely fluctuated or did not show clear decreases during the extinction (Author response image 7, colored lines). The simulated learning curves were almost flat in these trials (Author response image 7, gray lines). It might be difficult to estimate the internal states of such atypical mice if the sampling process tried to fit them by increasing the number of latent causes.

Nevertheless, most of the samples have a reasonable total number of latent causes: 12-month-old control mice, Mdn = 5, IQR = 4; 12-month-old *App^{NL-G-F}* mice, Mdn = 5, IQR = 1.75; 6-month-old control mice, Mdn = 7, IQR = 12.5; 6-month-old *App^{NL-G-F}* mice, Mdn = 5, IQR = 5.25. These data were provided in Tables S9 and S12.

Author response image 7.

Samples with a high number of latent causes. Observed CR (colored line) and simulated CR (gray line) for individual samples with a total number of inferred latent causes exceeding 20.



[#31] (3) Parameter estimation: With 10 parameters fitting one-dimensional curves, many parameters (e.g., α and σ_x) are likely highly correlated and poorly identified. Consider presenting scatter plots of the parameters (e.g., α vs σ_x) in the Supplement.

We have provided the scatter plots with a correlation matrix in Figure 4 – figure supplement 1 for the 12-month-old group and Figure 5 – figure supplement 1 for the 6-month-old group. As pointed out by the reviewer, there are significant rank correlations between parameters including α and σ_x^2 in both the 6 and 12-month-old groups. However, we also noted that there are no obvious linear relationships between the parameters.

The correlation above raises a potential problem of non-identifiability among parameters. First, we computed the variance inflation index (VIF) for all parameters to examine the risk of multicollinearity, though we did not consider a linear regression between parameters and

DI in this study. All VIF values were below the conventional threshold 10 (Appendix 2 – table 5), suggesting that severe multicollinearity is unlikely to bias our conclusions. Second, we have conducted the simulation with different combinations of α , σ_x^2 , and K to clarify their contribution to overgeneralization and overdifferentiation observed in the 12-month-old group.

In Appendix 2 – table 6, the values of α and σ_x^2 were either their upper or lower boundary set in parameter estimation, while the value K was selected heuristically to demonstrate its effect. Given the observed positive correlation between α and σ_x^2 , and their negative correlation with K (Figure 4 - figure supplement 1), we consider the product of $K = \{4, 35\}$, $\alpha = \{1, 3\}$ and $\sigma_x^2 = \{0.01, 3\}$. Among these combinations, the representative condition for the control group is $\alpha = 3$, $\sigma_x^2 = 3$, and that for the App^{NL-G-F} group is $\alpha = 1$, $\sigma_x^2 = 0.01$. In the latter condition, overgeneralization and overdifferentiation, which showed higher test 1 CR, lower number of acquisition latent causes (K_{acq}), lower test 3 CR, lower DI between test 3 and test 1, and higher number of latent causes after extinction (K_{rem}), was extremely induced.

We found conditions that fall outside of empirical correlation, such as $\alpha = 3$, $\sigma_x^2 = 0.01$, also reproduced overgeneralization and overdifferentiation. Similarly, the combination, $\alpha = 1$, $\sigma_x^2 = 3$, exhibited control-like behavior when $K = 4$ but shifted toward App^{NL-G-F} -like behavior when $K = 36$. The effect of K was also evident when $\alpha = 3$ and $\sigma_x^2 = 3$, where $K = 36$ led to over-differentiation. We note that these conditions were artificially set and likely not representative of biologically plausible. These results underscore the non-identifiability concern raised by the reviewer. Therefore, we acknowledge that merely attributing overgeneralization to lower α or overdifferentiation to lower σ_x^2 may be overly reductive. Instead, these patterns likely arise from the joint effect of α , σ_x^2 , and K . We have revised the manuscript accordingly in Results and Discussion (page 11-13, 18-19).

[#32] (4) Data normalization: Normalizing the data between 0 and 1 removes the interpretability of % freezing, making mice with large changes in freezing indistinguishable seem similar to mice with small changes.

As we describe in our reply to comment #26, the conditioned response in the latent cause model was scaled between 0 and 1, and we assume 0 and 1 mean the minimal and maximal CR within each mouse, respectively. Furthermore, although we initially tried to fit simulated CRs to raw CRs, we found that the fitting level was low due to the individual difference in the degree of behavioral expression: some mice exhibited a larger range of CR, while others showed a narrower one. Thus, we decided to normalize the data. We agree that this processing will make the mice with high changes in freezing% indistinguishable from those with low changes. However, the freezing% changes within the mouse were preserved and did not affect the discrimination index.

[#33] (5) Overlooking parameter differences: Differences in parameters, like w_0 that didn't fit the hypothesis may have been ignored.

Our initial hypothesis is that internal states were altered in App^{NL-G-F} mice, and we did not have a specific hypothesis on which parameter would contribute to such a state. We mainly focus on the parameters (1) that are significantly different between control and App^{NL-G-F} mice and (2) that are significantly correlated to the empirical behavioral data, DI between test 3 and test 1.

In the 12-month-old group, besides α and σ_x^2 , w_0 and K showed marginal p -value in Mann-Whitney U test (Table S7) and moderate correlation with the DI (Table S8). While differences in K were already discussed in the manuscript, we did miss the point that w_0 could contribute to the differences in w between control and App^{NL-G-F} (Figure 4G) in the previous manuscript. We explain the contribution of w_0 on the reinstatement results here. When other parameters

are fixed, higher w_0 would lead to higher CR in test 3, because higher w_0 would allow increasing w_{context} by the unsignaled shock, leading to reinstatement (Appendix 2 – table 7). It is likely that higher w_0 would be sampled through the parameter estimation in the 12-month-old control but not App^{NL-G-F} . On the other hand, the number of latent causes is not sensitive to w_0 when other parameters were fixed at the initial guess value (Appendix 2 – table 1), suggesting w_0 has a small contribution to memory modification process.

Thus, we speculate that although the difference in w_0 between control and App^{NL-G-F} mice may arise from the sampling process, resulting in a positive correlation with DI between test 3 and test 1, its contribution to diverged internal states would be smaller relative to α or σ_x^2 as a wide range of w_0 has no effect on the number of latent causes (Appendix 2 – table 7). We have added the discussion of differences in w_0 in the 12-month-old group in manuscript Line 357-359.

In the 6-month-old group, besides α and σ_x^2 , θ is significantly higher in the AD mice group (Table S10) but not correlated with the DI (Table S11). We have already discussed this point in the manuscript.

[#34] (6) Initial response: Higher initial responses in the model at the start of the experiment may reflect poor model fit.

Please refer to our reply to comment #26 for our explanation of what contributes to high initial responses in the latent cause model.

In addition, achieving a good fit for the acquisition CRs was not our primary purpose, as the response measured in the acquisition phase includes not only a conditioned response to the CS and context but also an unconditioned response to the novel stimuli (CS and US). This mixed response presumably increased the variance of the measured freezing rate over individuals, therefore we did not cover the results in the discussion.

Rather, we favor models at least replicating the establishment of conditioning, extinction and reinstatement of fear memory in order to explain the memory modification process. As we mentioned in the reply for comment #4, alternative models, the latent state model and the Rescorla-Wagner model, failed to replicate the observation (cf. Figure 3 – figure supplement 1A-1C). Thus, we chose to stand on the latent cause model as it aligns better with the purpose of this study.

[#35] In addition, please be transparent if data is excluded, either during the fitting procedure or when performing one-way ANCOVA. Avoid discarding data when possible, but if necessary, provide clarity on the nature of excluded data (e.g., how many, why were they excluded, which group, etc?).

We clarify the information of excluded data as follows. We had 25 mice for the 6-month-old control group, 26 mice for the 6-month-old App^{NL-G-F} group, 29 mice for the 12-month-old control group, and 26 mice for the 12-month-old App^{NL-G-F} group (Table S1).

Our first exclusion procedure was applied to the freezing rate data in the test phase. If the mouse had a freezing rate outside of the 1.5 IQR in any of the test phases, it is regarded as an outlier and removed from the analysis (see Statistical analysis in Materials and Methods). One mouse in the 6-month-old control group, one mouse in the 6-month-old App^{NL-G-F} group, five mice in the 12-month-old control group, and two mice in the 12-month-old App^{NL-G-F} group were excluded.

Our second exclusion procedure was applied during the fitting and parameter estimation (see parameter estimation in Materials and Methods). We have provided the number of anomaly samples during parameter estimation in Appendix 1 – figure 2.

Lastly, we would like to state that all the sample sizes written in the figure legends do not include outliers detected through the exclusion procedure mentioned above.

[#36] Finally, since several statistical tests were used and the differences are small, I suggest noting that multiple comparisons were not controlled for, so p-values should be interpreted cautiously.

We have provided power analyses in Tables S21 and S22 with methods described in the manuscript (Line 897-898) and added a note that not all of the multiple comparisons were corrected for in the manuscript (Line 898-899).

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